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Lee et al.

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(54) **INTRA BLOCK COPY-BASED
ENCODING/DECODING METHOD, DEVICE,
AND BITSTREAM STORAGE MEDIUM**

(58) **Field of Classification Search**
None
See application file for complete search history.

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Mar. 19-27, 2019.

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PC

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H04N 19/593 (2014.01)
H04N 19/105 (2014.01)
H04N 19/176 (2014.01)
H04N 19/513 (2014.01)

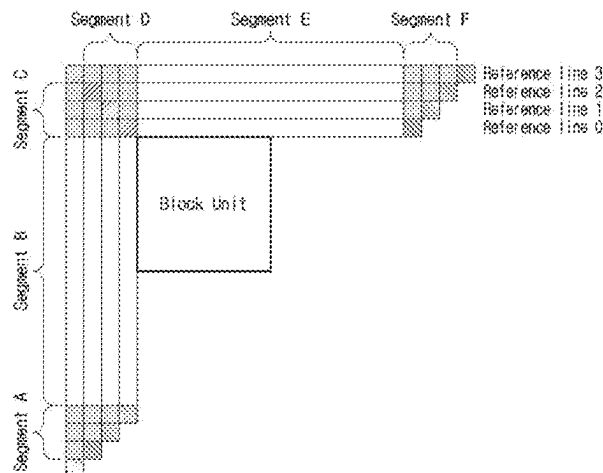
(57) **ABSTRACT**

An image encoding/decoding method and apparatus are
disclosed. The image decoding method includes acquiring
prediction mode information of a current block from a
bitstream, decoding an intra block copy prediction mode of
the current block using the prediction mode information of
the current block, and reconstructing the current block based
on the intra block copy prediction mode. The intra block
copy prediction mode is at least one of a block copy based
SKIP mode, a block copy based MERGE mode or a block
copy based AMVP mode.

(52) **U.S. Cl.**

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19/513 (2014.11)

9 Claims, 18 Drawing Sheets



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FIG. 1

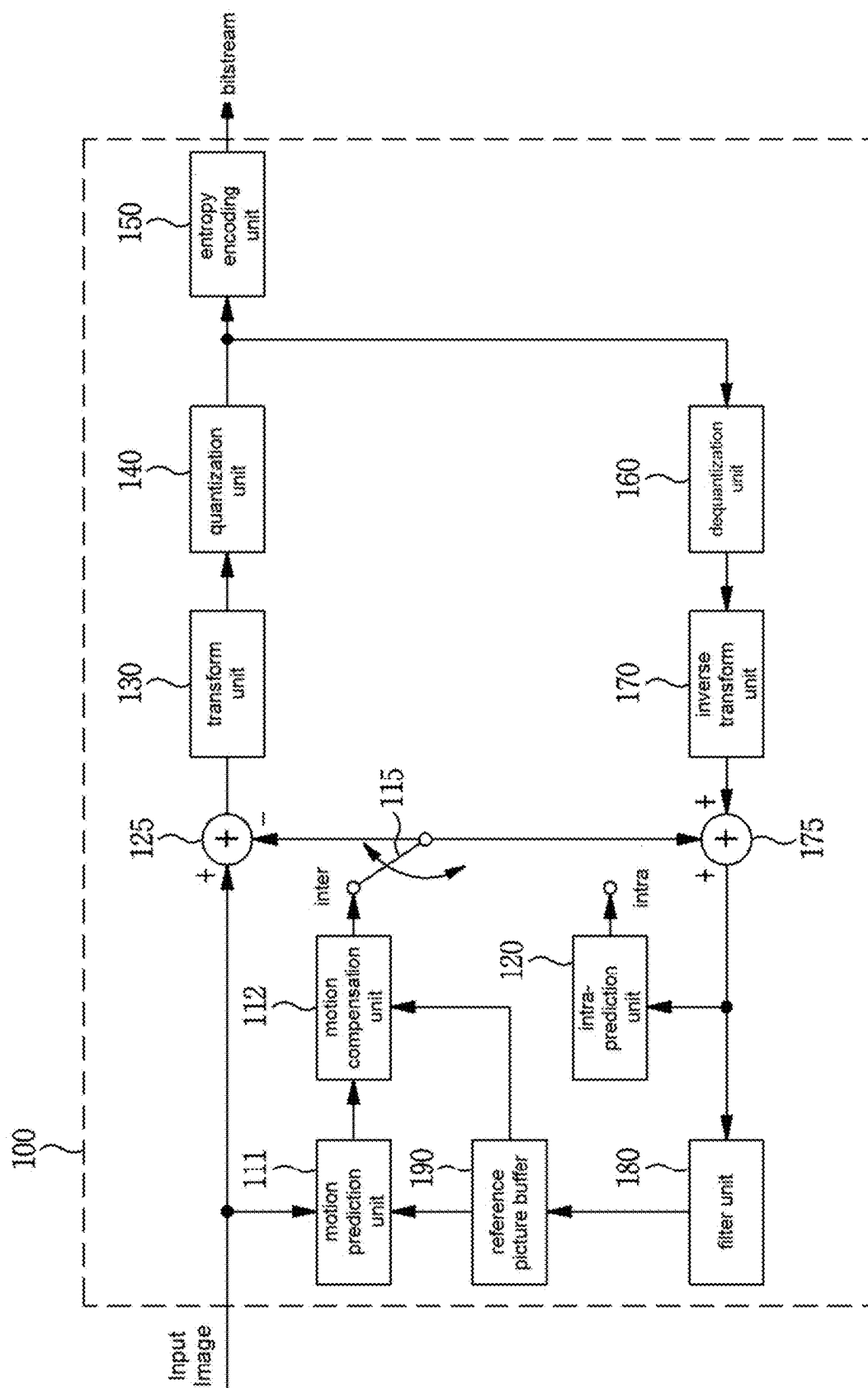


FIG. 2

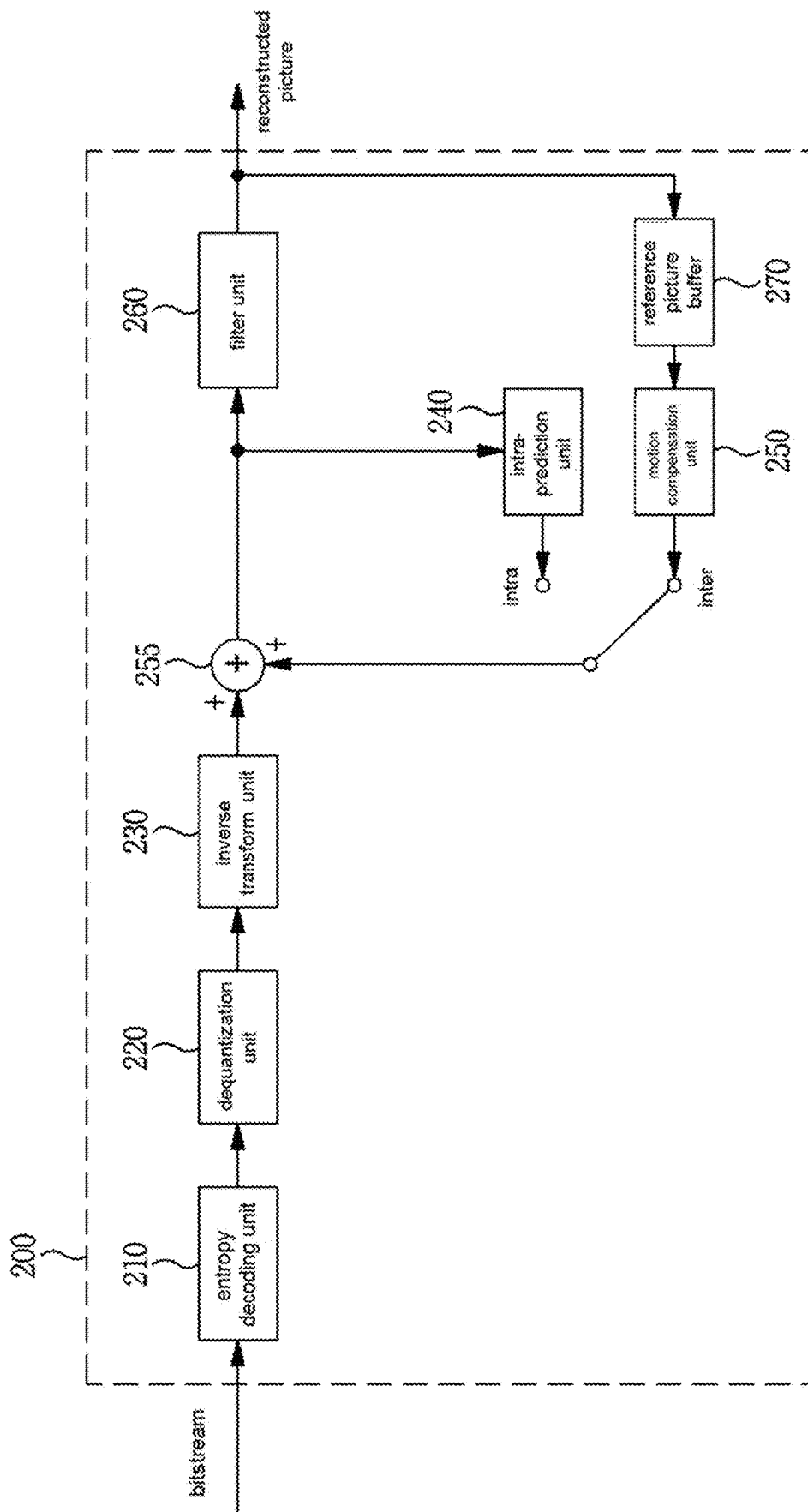


FIG. 3

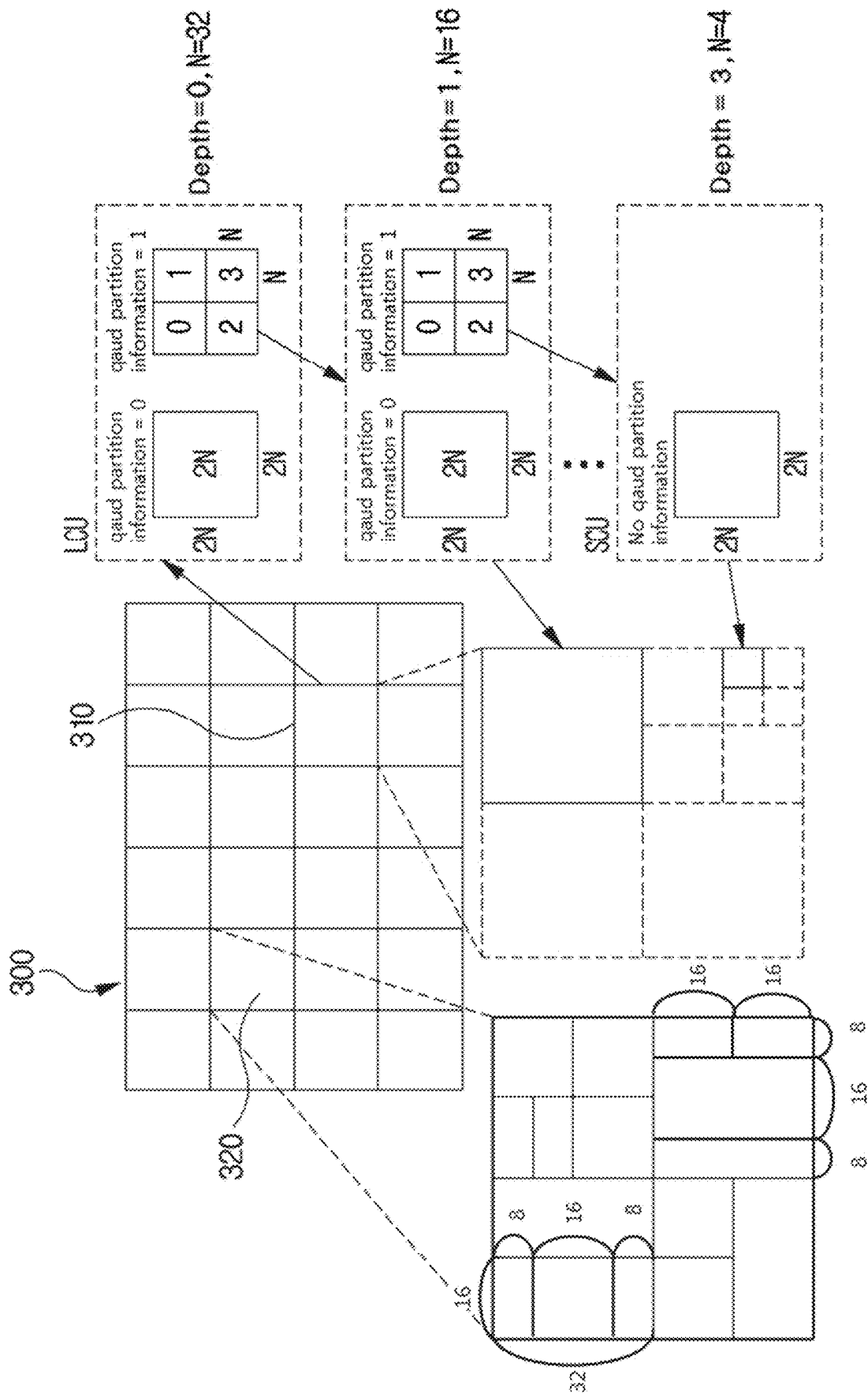


FIG. 4

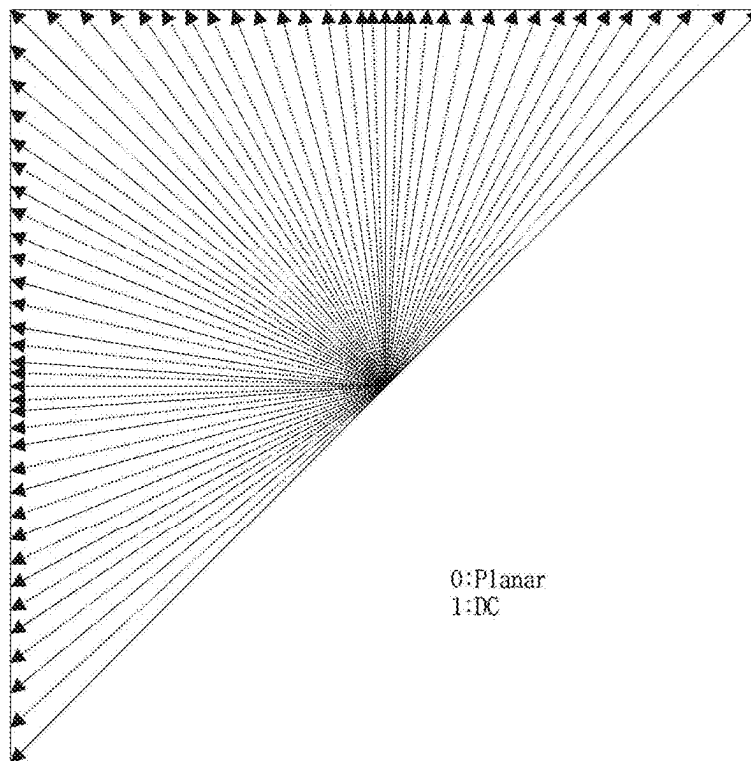


FIG. 5

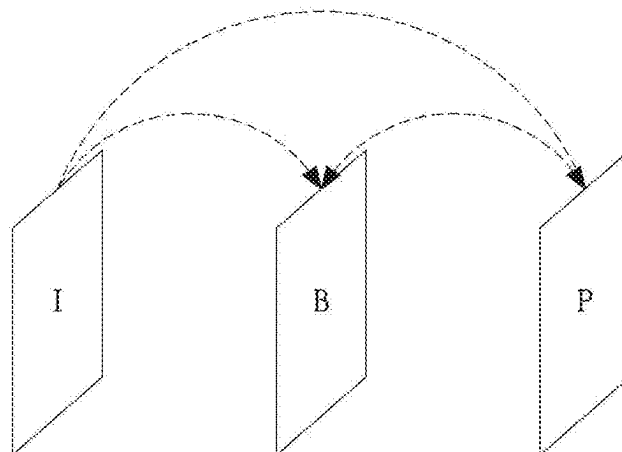


FIG. 6

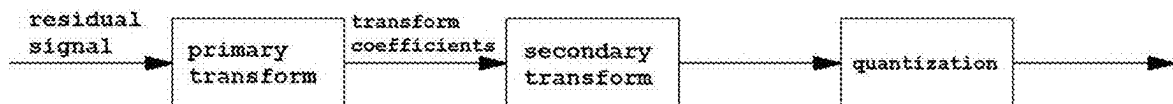


FIG. 7

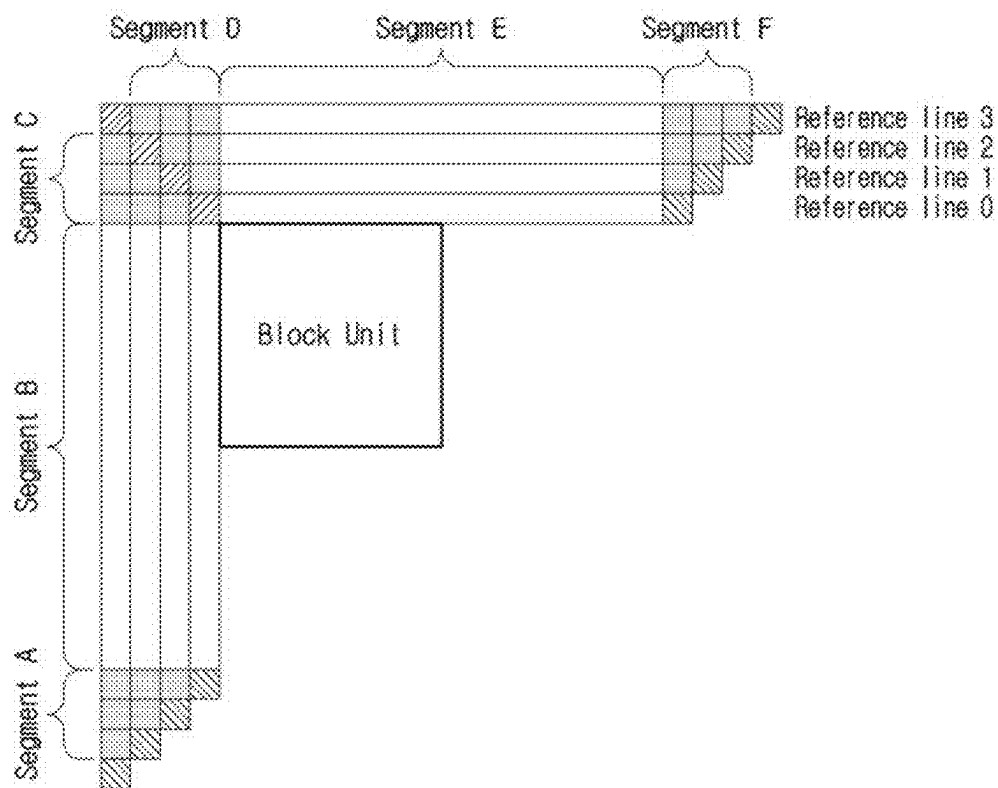


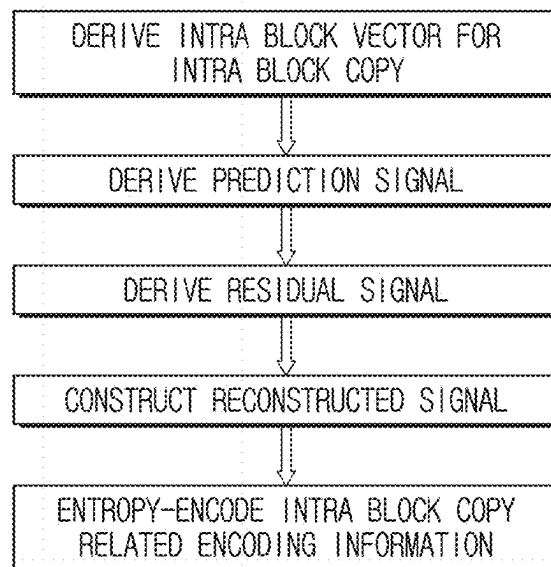
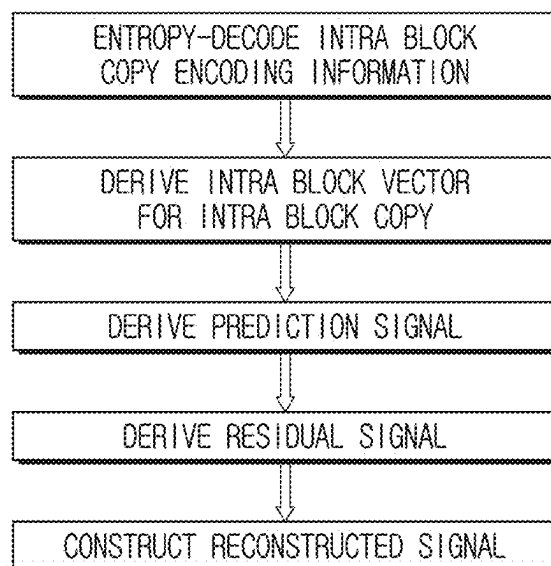
FIG. 8A**FIG. 8B**

FIG. 9

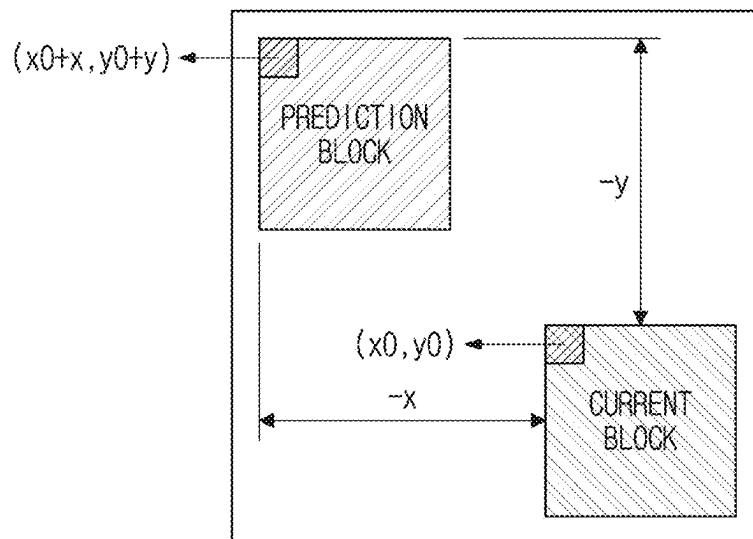


FIG. 10

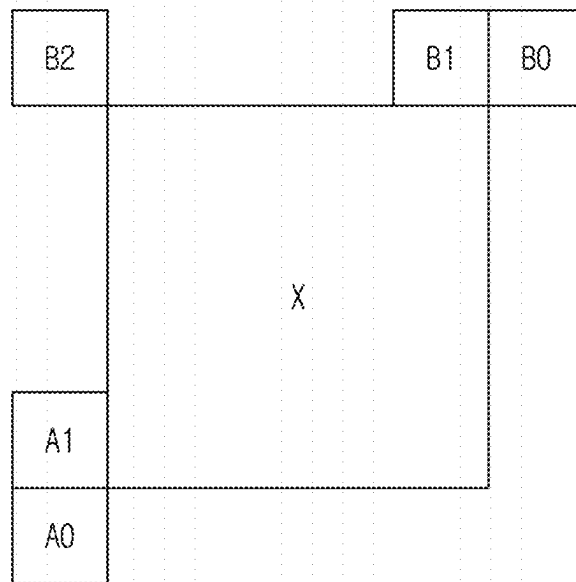


FIG. 11

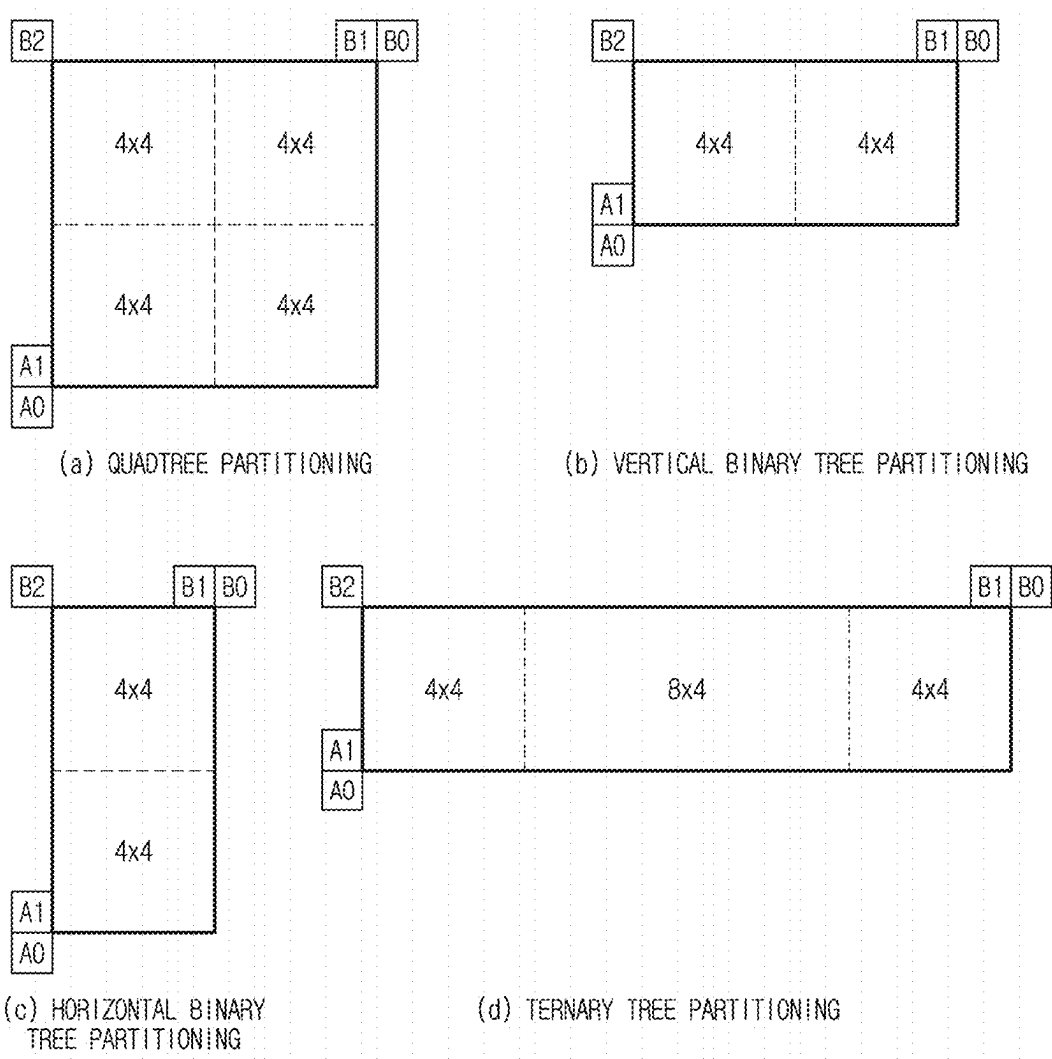


FIG. 12

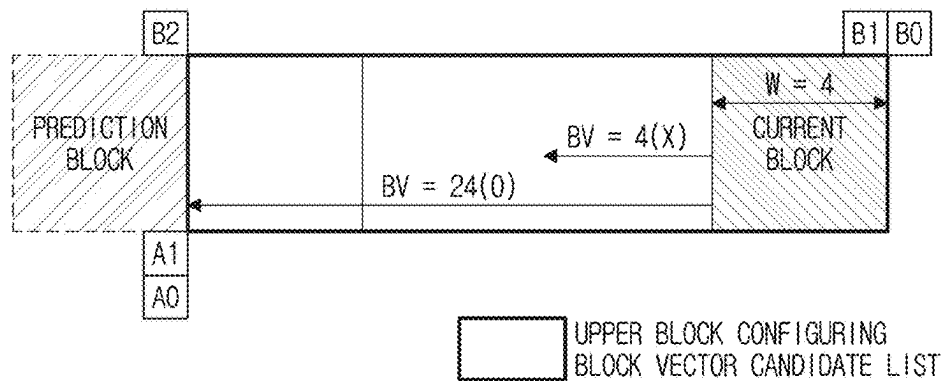


FIG. 13

pic parameter set rbsp() {	
...	
constant slice header params enabled flag	u(1)
if(constant slice header params enabled flag) {	
pps dep quant enabled idc	u(2)
for(i = 0; i < 2; i++)	
pps ref pic list sps idc[i]	u(2)
pps mvd ll zero idc	u(2)
pps collocated from 0 idc	u(2)
pps six minus max num merge cand plus1	ue(v)
pps six minus max num ibc merge cand plus1	ue(v)
pps max num merge cand minus max num triangle cand plus1	ue(v)
}	
picture header rbsp() {	
...	
if (sps ibc enabled flag && pps six minus max num ibc merge cand plus1)	
pic six minus max num ibc merge cand	ue(v)
...	

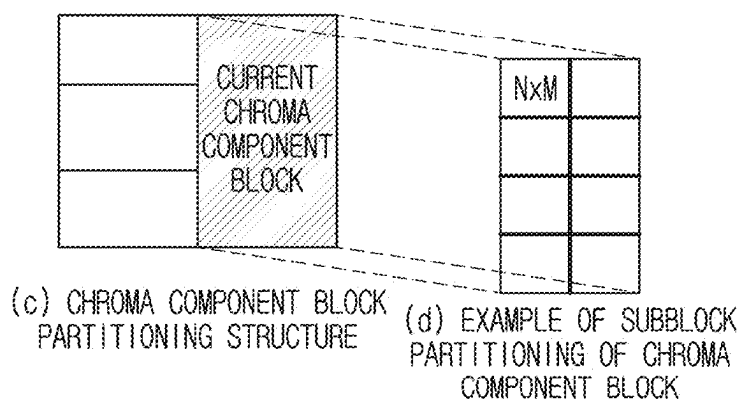
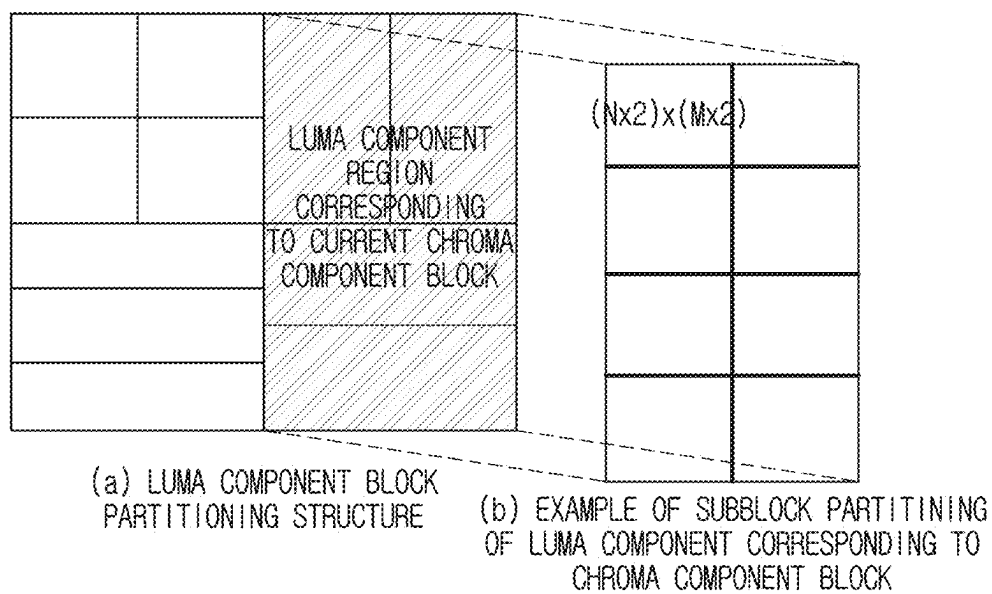
FIG. 14

} else if(CuPredMode[x0][y0] == MODE_IBC) {	
mvd_coding(x0, y0, 0, 0)	
if(MaxNumMergeCand > 1)	
mvp_l0_flag[x0][y0]	ae(v)
if(sps_amvr_enabled_flag &&	
(MvdL0[x0][y0][0] != 0 MvdL0[x0][y0][1] != 0)) {	
amvr_precision_flag[x0][y0]	ae(v)
}	

FIG. 15

if(CuPredMode[x0][y0] == MODE_IBC) {	
merge_idx[x0][y0]	ae(v)
} else {	

FIG. 16



———— BLOCK PARTITIONING
—— SUBBLOCK PARTITIONING

FIG. 17

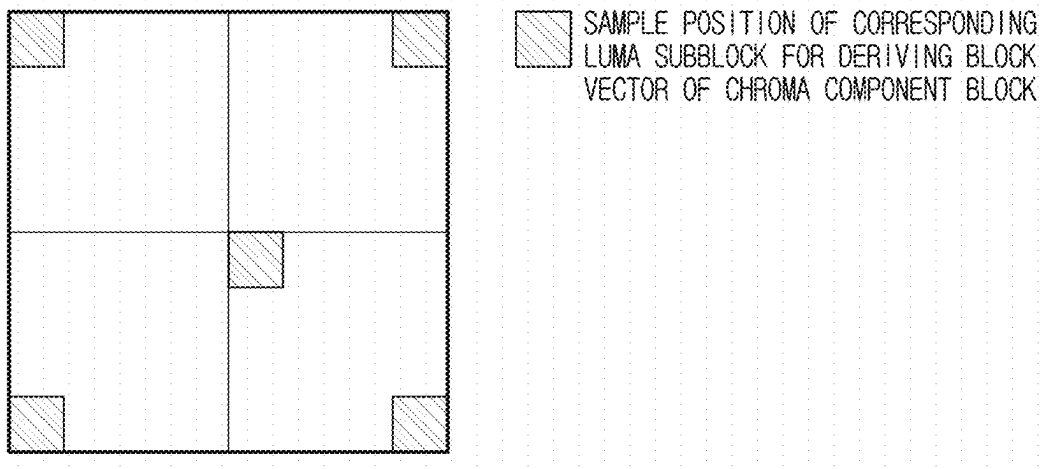


FIG. 18

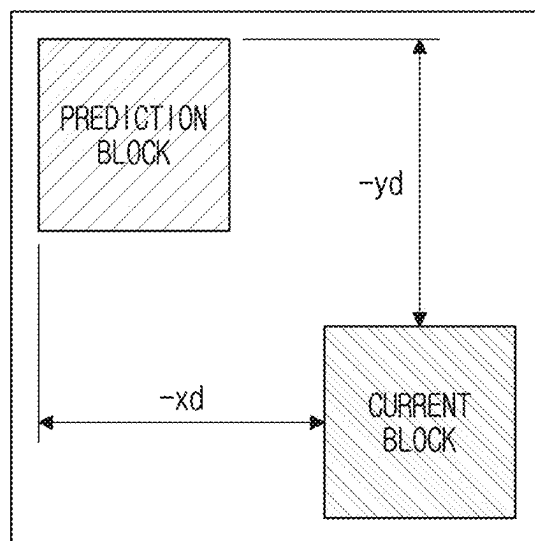


FIG. 19

INTRA BLOCK COPY SKIP MODE	INTRA BLOCK COPY SKIP MODE
INTRA BLOCK COPY SKIP MODE	INTRA BLOCK COPY SKIP MODE
INTRA BLOCK COPY SKIP MODE	INTRA BLOCK COPY SKIP MODE
INTRA BLOCK COPY SKIP MODE	INTRA BLOCK COPY SKIP MODE

FIG. 20

INTRA BLOCK COPY SKIP MODE	INTRA BLOCK COPY MERGE MODE
INTRA BLOCK COPY SKIP MODE	INTRA BLOCK COPY AMVP MODE
INTRA BLOCK COPY MERGE MODE	INTRA BLOCK COPY AMVP MODE
INTRA BLOCK COPY SKIP MODE	INTRA BLOCK COPY SKIP MODE

FIG. 21A

Descriptor	
	coding_unit(x0, y0, cbWidth, cbHeight, treeType) {
	if(tile_group_type != 1 sps_ibc_enabled_flag) {
	if(treeType != DUAL_TREE_CHROMA)
ac(v)	cu_skip_flag[x0][y0]
	if(cu_skip_flag[x0][y0] == 0 && tile_group_type != 1)
	pred_mode_flag
ac(v)	if(((tile_group_type == 1 && cu_skip_flag[x0][y0] == 0) (tile_group_type != 1 && CuPredMode[x0][y0] != MODE_INTRA))
	&& sps_ibc_enabled_flag)
	pred_mode_ibc_flag
ac(v)	}
	if(CuPredMode[x0][y0] == MODE_INTRA) {
	
	} else if(treeType != DUAL_TREE_CHROMA) { /* MODE_INTER or MODE_IBC */
	if(cu_skip_flag[x0][y0] == 0)
ac(v)	merge_flag[x0][y0]
	if(merge_flag[x0][y0]) {
	merge_data(x0, y0, cbWidth, cbHeight)
	} else if(CuPredMode[x0][y0] == MODE_IBC) {
	mvd_coding(x0, y0, 0, 0)
ac(v)	mvp_l0_flag[x0][y0]
	if(sps_mvnr_enabled_flag && (MvdL0[x0][y0][0] != 0 MvdL0[x0][y0][1] != 0)) {
ac(v)	amvr_precision_flag[x0][y0]
	}
	} else {
	
	}
	}
	if(pcm_flag[x0][y0]) {
	if(CuPredMode[x0][y0] != MODE_INTRA && merge_flag[x0][y0] == 0)
ac(v)	cu_cbf
	if(cu_cbf) {
	
	transform_tree(x0, y0, cbWidth, cbHeight, treeType)
	}
	}
	}

FIG. 21B

transform	unit(x0, y0, tbWidth, tbHeight, treeType, subTulIndex) {	Descriptor
if (treeType == SINGLE_TREE treeType == DUAL_TREE_LUMA) {		
if (IntraSubPartitionsSplitType == ISP_NO_SPLIT && !(cu_sbt_flag && (subTulIndex == 0 && cu_sbt_pos_flag)		
(subTulIndex == 1 && !cu_sbt_pos_flag))) (IntraSubPartitionsSplitType != ISP_NO_SPLIT &&		
(subTulIndex < NumIntraSubPartitions - 1 !InferTuCbfLuma)))		
tu_cbf_luma[x0 y0]		ae(v)
if (IntraSubPartitionsSplitType != ISP_NO_SPLIT)		
InferTuCbfLuma = InferTuCbfLuma && !tu_cbf_luma[x0 y0]		
}		
if (treeType == SINGLE_TREE treeType == DUAL_TREE_CHROMA) {		
if (IntraSubPartitionsSplitType == ISP_NO_SPLIT && !(cu_sbt_flag && (subTulIndex == 0 && cu_sbt_pos_flag)		
(subTulIndex == 1 && !cu_sbt_pos_flag))) (IntraSubPartitionsSplitType != ISP_NO_SPLIT &&		
(subTulIndex == NumIntraSubPartitions - 1))) {		
tu_cbf_cbf[x0 y0]		ae(v)
tu_cbf_cr[x0 y0]		ae(v)
}		
}		
...		
if (tu_cbf_luma[x0 y0])		
residual_coding(x0, y0, Log2(tbWidth), Log2(tbHeight), 0)		
if (tu_cbf_cbf[x0 y0])		
residual_coding(xC, yC, Log2(wC), Log2(hC), 1)		
if (tu_cbf_cr[x0 y0])		
residual_coding(xC, yC, Log2(wC), Log2(hC), 2)		
}		

FIG. 22

	Descriptor
transform_unit(x0, y0, tbWidth, tbHeight, treeType, subTuIndex) {	
if (treeType == SINGLE_TREE (treeType == DUAL_TREE_LUMA && CurPredModel[x0][y0] == MODE_INTRA)) {	
if (IntraSubPartitionsSplitType == ISP_NO_SPLIT && !(cu_sbt_flag && ((subTuIndex == 0 && cu_sbt_pos_flag)	
(subTuIndex == 1 && !cu_sbt_pos_flag)))) (IntraSubPartitionsSplitType != ISP_NO_SPLIT &&	
(subTuIndex < NumIntraSubPartitions - 1 InferTuCbfLuma)))	
tu_cbf_luma[x0][y0]	ae(v)
if (IntraSubPartitionsSplitType != ISP_NO_SPLIT)	
InferTuCbfLuma = InferTuCbfLuma && !tu_cbf_luma[x0][y0]	
}	
if (treeType == SINGLE_TREE treeType == DUAL_TREE_CHROMA) {	
if (IntraSubPartitionsSplitType == ISP_NO_SPLIT && !(cu_sbt_flag && ((subTuIndex == 0 && cu_sbt_pos_flag)	
(subTuIndex == 1 && !cu_sbt_pos_flag)))) (IntraSubPartitionsSplitType != ISP_NO_SPLIT &&	
(subTuIndex == NumIntraSubPartitions - 1))) {	
tu_cbf_cb[x0][y0]	ae(v)
tu_cbf_cr[x0][y0]	ae(v)
}	
}	
...	
}	

FIG. 23

	Descriptor
transform_unit[x0, y0, tbWidth, tbHeight, treeType, subTuIndex] {	
if(treeType == SINGLE_TREE (treeType == DUAL_TREE LUMA && !CurPredMode[x0][y0] == MODE_IBC)) {	
if(IntraSubPartitionsSplitType == ISP_NO_SPLIT && !(cu_sbt_flag && ((subTuIndex == 0 && cu_sbt_pos_flag)	
(subTuIndex == 1 && !cu_sbt_pos_flag))) (IntraSubPartitionsSplitType != ISP_NO_SPLIT &&	
(subTuIndex < NumIntraSubPartitions - 1 !InferTuCbfLuma))) {	
tu_cbf_luma[x0][y0]	ae(v)
if(IntraSubPartitionsSplitType != ISP_NO_SPLIT)	
InferTuCbfLuma = InferTuCbfLuma && !tu_cbf_luma[x0][y0]	
}	
if(treeType == SINGLE_TREE treeType == DUAL_TREE_CHROMA) {	
if(IntraSubPartitionsSplitType == ISP_NO_SPLIT && !(cu_sbt_flag && ((subTuIndex == 0 && cu_sbt_pos_flag)	
(subTuIndex == 1 && !cu_sbt_pos_flag))) (IntraSubPartitionsSplitType != ISP_NO_SPLIT &&	
(subTuIndex == NumIntraSubPartitions - 1))) {	
tu_cbf_cb[x0][y0]	ae(v)
tu_cbf_cr[x0][y0]	ae(v)
}	
}	
...	
}	

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INTRA BLOCK COPY-BASED ENCODING/DECODING METHOD, DEVICE, AND BITSTREAM STORAGE MEDIUM

CROSS-REFERENCE TO RELATED APPLICATION

This application is a division of U.S. application Ser. No. 17/613,395, filed on Nov. 22, 2021, which is a National Stage Entry of PCT International Application No. PCT/KR2020/007842, filed on Jun. 17, 2020, which claims priority to Korean Patent Application No. 10-2019-0071282, filed on Jun. 17, 2019, and Korean Patent Application No. 10-2019-0174309, filed on Dec. 24, 2019, the entire contents of which are hereby incorporated by references in its entirety.

TECHNICAL FIELD

The present invention relates to an image encoding/decoding method and apparatus, and a recording medium for storing a bitstream. More particularly, the present invention relates to a method and apparatus for encoding/decoding an image based on intra block copy, and a recording medium for storing a bitstream generated by the image encoding method or apparatus of the present invention.

BACKGROUND ART

Recently, the demand for high resolution and quality images such as high definition (HD) or ultra-high definition (UHD) images has increased in various applications. As the resolution and quality of images are improved, the amount of data correspondingly increases. This is one of the causes of increase in transmission cost and storage cost when transmitting image data through existing transmission media such as wired or wireless broadband channels or when storing image data. In order to solve such problems with high resolution and quality image data, a high efficiency image encoding/decoding technique is required.

There are various video compression techniques such as an inter prediction technique of predicting the values of pixels within a current picture from the values of pixels within a preceding picture or a subsequent picture, an intra prediction technique of predicting the values of pixels within a region of a current picture from the values of pixels within another region of the current picture, a transform and quantization technique of compressing the energy of a residual signal, and an entropy coding technique of allocating frequently occurring pixel values with shorter codes and less occurring pixel values with longer codes.

Conventional image encoding/decoding methods and apparatuses use limited intra block copy based encoding/decoding technology and thus have limitations in encoding efficiency improvement.

DISCLOSURE

Technical Problem

An object of the present invention is to provide an image encoding/decoding method and apparatus with improved compression efficiency.

Another object of the present invention is to provide an image encoding/decoding method and apparatus based on intra block copy with improved compression efficiency.

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Another object of the present invention is to provide a recording medium for storing a bitstream generated by an image encoding/decoding method or apparatus according to the present invention.

5 Another object of the present invention is to provide an image encoding/decoding method and apparatus based on a configuration of a block vector candidate list.

Technical Solution

10 According to the present invention, a method of decoding an image includes acquiring prediction mode information of a current block from a bitstream, decoding an intra block copy prediction mode of the current block using the prediction mode information of the current block, and reconstructing the current block based on the intra block copy prediction mode. The intra block copy prediction mode is at least one of a block copy based SKIP mode, a block copy based MERGE mode or a block copy based AMVP mode.

20 The reconstructing of the current block may include constructing a block vector candidate list for the current block, deriving a block vector for the current block from a candidate block vector included in the block vector candidate list, and reconstructing the current block using the derived block vector.

25 The constructing of the block vector candidate list for the current block may include acquiring, from a bitstream, first information indicating a maximum number of allowable candidate block vectors in the block vector candidate list.

30 The first information may be signaled in a sequence parameter set.

The first information may be entropy-decoded based on second information indicating whether an intra block copy prediction mode is used.

35 A maximum number of allowable candidate block vectors in the block vector candidate list may be derived by a difference between a predetermined positive integer and a value of the first information.

The predetermined positive integer may be 6.

40 When the intra block copy prediction mode of the current block is in a block copy based AMVP mode and the maximum number of allowable candidate block vectors in the block vector candidate list is greater than a predetermined value, information indicating an index of an L0 motion vector predictor may be decoded.

45 When the number of candidate block vectors included in the block vector candidate list is less than the maximum number of allowable candidate vectors in the block vector candidate list, the constructing of the block vector candidate list for the current block may include adding history-based block vector candidates to the block vector candidate list until the number of candidate block vectors becomes equal to the maximum number of allowable candidate vectors in the block vector candidate list.

50 The method may further include determining whether a predetermined candidate of the history-based block vector candidates overlaps a candidate block vector included in the block vector candidate list.

55 When the current block has a size less than a predetermined size, an update process of a history-based block vector candidate list of the current block may not be performed.

60 According to the present invention, a method of encoding an image includes encoding a current block based on an intra block copy prediction mode of the current block, and encoding the intra block copy prediction mode of the current block using prediction mode information of the current

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block. The intra block copy prediction mode is at least one of a block copy based SKIP mode, a block copy based MERGE mode or a block copy based AMVP mode.

The encoding of the current block may include constructing a block vector candidate list for the current block, deriving a block vector for the current block from a candidate block vector included in the block vector candidate list, and encoding the current block using the derived block vector.

The method may further include encoding first information indicating a maximum number of allowable candidate block vectors in the block vector candidate list into a bitstream.

The first information may be signaled in a sequence parameter set.

The first information may be encoded based on second information indicating whether an intra block copy prediction mode is used.

A maximum number of allowable candidate block vectors in the block vector candidate list may be encoded by a difference between a predetermined positive integer and a value of the first information.

The predetermined positive integer may be 6.

When the intra block copy prediction mode of the current block is in a block copy based AMVP mode and the maximum number of allowable candidate block vectors in the block vector candidate list is greater than a predetermined value, information indicating an index of an L0 motion vector predictor may be encoded.

When the number of candidate block vectors included in the block vector candidate list is less than the maximum number of allowable candidate vectors in the block vector candidate list, the constructing of the block vector candidate list for the current block may include adding history-based block vector candidates to the block vector candidate list until the number of candidate block vectors becomes equal to the maximum number of allowable candidate vectors in the block vector candidate list.

When the current block has a size less than a predetermined size, an update process of a history-based block vector candidate list of the current block may not be performed.

A recording medium according to the present invention may store a bitstream generated by the image encoding method according to the present invention.

Advantageous Effects

According to the present invention, it is possible to provide an image encoding/decoding method and apparatus with improved compression efficiency.

According to the present invention, it is possible to provide a symmetric/rotation image encoding/decoding method and apparatus based on intra block copy with improved compression efficiency.

According to the present invention, it is possible to provide a recording medium for storing a bitstream generated by an image encoding/decoding method or apparatus according to the present invention.

DESCRIPTION OF DRAWINGS

FIG. 1 is a block diagram showing a configuration of an encoding apparatus according to an embodiment to which the present invention is applied.

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FIG. 2 is a block diagram showing a configuration of a decoding apparatus according to an embodiment and to which the present invention is applied.

FIG. 3 is a view schematically showing a partition structure of an image when encoding and decoding the image.

FIG. 4 is a view showing an intra-prediction process.

FIG. 5 is a diagram illustrating an embodiment of an inter-picture prediction process.

FIG. 6 is a diagram illustrating a transform and quantization process.

FIG. 7 is a diagram illustrating reference samples capable of being used for intra prediction.

FIG. 8A is a flowchart illustrating an image encoding method according to an embodiment of the present invention. FIG. 8B is a flowchart illustrating an image decoding method according to an embodiment of the present invention.

FIG. 9 is a view illustrating a relation between a current block and a prediction block according to an embodiment of the present invention.

FIG. 10 is a view illustrating neighbor blocks adjacent to a current block according to an embodiment of the present invention.

FIG. 11 is a view illustrating a current block partitioned when a predetermined threshold is 32 according to an embodiment of the present invention.

FIG. 12 is a view illustrating a process of sharing and using a block vector candidate list constructed in an upper block according to an embodiment of the present invention.

FIG. 13 is a view illustrating a process of deriving information indicating an allowable maximum number in a block vector candidate list according to an embodiment of the present invention.

FIG. 14 is a view illustrating a process of entropy-encoding/decoding an L0 motion vector prediction flag according to an embodiment of the present invention.

FIG. 15 is a view illustrating a process of entropy-encoding/decoding merge index information according to an embodiment of the present invention.

FIG. 16 is a view illustrating a correspondence relation between a chroma component block and a luma component region according to another embodiment of the present invention.

FIG. 17 is a view illustrating a luma component subblock according to another embodiment of the present invention.

FIG. 18 is a view illustrating a relation between a current block and a prediction block according to another embodiment of the present invention.

FIG. 19 is a view illustrating when prediction encoding modes of luma component subblocks corresponding to chroma component blocks according to an embodiment of the present invention are the same.

FIG. 20 is a view illustrating when prediction encoding modes of luma component subblocks corresponding to chroma component blocks according to an embodiment of the present invention are different.

FIGS. 21A, 21B, 22 and 23 are views illustrating encoding information transmitted in association with intra block partitioning according to an embodiment of the present invention.

MODE FOR INVENTION

A variety of modifications may be made to the present invention and there are various embodiments of the present invention, examples of which will now be provided with reference to drawings and described in detail. However, the

present invention is not limited thereto, although the exemplary embodiments can be construed as including all modifications, equivalents, or substitutes in a technical concept and a technical scope of the present invention. The similar reference numerals refer to the same or similar functions in various aspects. In the drawings, the shapes and dimensions of elements may be exaggerated for clarity. In the following detailed description of the present invention, references are made to the accompanying drawings that show, by way of illustration, specific embodiments in which the invention may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to implement the present disclosure. It should be understood that various embodiments of the present disclosure, although different, are not necessarily mutually exclusive. For example, specific features, structures, and characteristics described herein, in connection with one embodiment, may be implemented within other embodiments without departing from the spirit and scope of the present disclosure. In addition, it should be understood that the location or arrangement of individual elements within each disclosed embodiment may be modified without departing from the spirit and scope of the present disclosure. The following detailed description is, therefore, not to be taken in a limiting sense, and the scope of the present disclosure is defined only by the appended claims, appropriately interpreted, along with the full range of equivalents to what the claims claim.

Terms used in the specification, 'first', 'second', etc. can be used to describe various components, but the components are not to be construed as being limited to the terms. The terms are only used to differentiate one component from other components. For example, the 'first' component may be named the 'second' component without departing from the scope of the present invention, and the 'second' component may also be similarly named the 'first' component. The term 'and/or' includes a combination of a plurality of items or any one of a plurality of terms.

It will be understood that when an element is simply referred to as being 'connected to' or 'coupled to' another element without being 'directly connected to' or 'directly coupled to' another element in the present description, it may be 'directly connected to' or 'directly coupled to' another element or be connected to or coupled to another element, having the other element intervening therebetween. In contrast, it should be understood that when an element is referred to as being "directly coupled" or "directly connected" to another element, there are no intervening elements present.

Furthermore, constitutional parts shown in the embodiments of the present invention are independently shown so as to represent characteristic functions different from each other. Thus, it does not mean that each constitutional part is constituted in a constitutional unit of separated hardware or software. In other words, each constitutional part includes each of enumerated constitutional parts for convenience. Thus, at least two constitutional parts of each constitutional part may be combined to form one constitutional part or one constitutional part may be divided into a plurality of constitutional parts to perform each function. The embodiment where each constitutional part is combined and the embodiment where one constitutional part is divided are also included in the scope of the present invention, if not departing from the essence of the present invention.

The terms used in the present specification are merely used to describe particular embodiments, and are not intended to limit the present invention. An expression used in the singular encompasses the expression of the plural,

unless it has a clearly different meaning in the context. In the present specification, it is to be understood that terms such as "including", "having", etc. are intended to indicate the existence of the features, numbers, steps, actions, elements, parts, or combinations thereof disclosed in the specification, and are not intended to preclude the possibility that one or more other features, numbers, steps, actions, elements, parts, or combinations thereof may exist or may be added. In other words, when a specific element is referred to as being "included", elements other than the corresponding element are not excluded, but additional elements may be included in embodiments of the present invention or the scope of the present invention.

In addition, some of constituents may not be indispensable constituents performing essential functions of the present invention but be selective constituents improving only performance thereof. The present invention may be implemented by including only the indispensable constitutional parts for implementing the essence of the present invention except the constituents used in improving performance. The structure including only the indispensable constituents except the selective constituents used in improving only performance is also included in the scope of the present invention.

Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings. In describing exemplary embodiments of the present invention, well-known functions or constructions will not be described in detail since they may unnecessarily obscure the understanding of the present invention. The same constituent elements in the drawings are denoted by the same reference numerals, and a repeated description of the same elements will be omitted.

Hereinafter, an image may mean a picture configuring a video, or may mean the video itself. For example, "encoding or decoding or both of an image" may mean "encoding or decoding or both of a moving picture", and may mean "encoding or decoding or both of one image among images of a moving picture."

Hereinafter, terms "moving picture" and "video" may be used as the same meaning and be replaced with each other.

Hereinafter, a target image may be an encoding target image which is a target of encoding and/or a decoding target image which is a target of decoding. Also, a target image may be an input image inputted to an encoding apparatus, and an input image inputted to a decoding apparatus. Here, a target image may have the same meaning with the current image.

Hereinafter, terms "image", "picture", "frame" and "screen" may be used as the same meaning and be replaced with each other.

Hereinafter, a target block may be an encoding target block which is a target of encoding and/or a decoding target block which is a target of decoding. Also, a target block may be the current block which is a target of current encoding and/or decoding. For example, terms "target block" and "current block" may be used as the same meaning and be replaced with each other.

Hereinafter, terms "block" and "unit" may be used as the same meaning and be replaced with each other. Or a "block" may represent a specific unit.

Hereinafter, terms "region" and "segment" may be replaced with each other.

Hereinafter, a specific signal may be a signal representing a specific block. For example, an original signal may be a signal representing a target block. A prediction signal may

be a signal representing a prediction block. A residual signal may be a signal representing a residual block.

In embodiments, each of specific information, data, flag, index, element and attribute, etc. may have a value. A value of information, data, flag, index, element and attribute equal to "0" may represent a logical false or the first predefined value. In other words, a value "0", a false, a logical false and the first predefined value may be replaced with each other. A value of information, data, flag, index, element and attribute equal to "1" may represent a logical true or the second predefined value. In other words, a value "1", a true, a logical true and the second predefined value may be replaced with each other.

When a variable i or j is used for representing a column, a row or an index, a value of i may be an integer equal to or greater than 0, or equal to or greater than 1. That is, the column, the row, the index, etc. may be counted from 0 or may be counted from 1.

DESCRIPTION OF TERMS

Encoder: means an apparatus performing encoding. That is, means an encoding apparatus.

Decoder: means an apparatus performing decoding. That is, means a decoding apparatus.

Block: is an $M \times N$ array of a sample. Herein, M and N may mean positive integers, and the block may mean a sample array of a two-dimensional form. The block may refer to a unit. A current block may mean an encoding target block that becomes a target when encoding, or a decoding target block that becomes a target when decoding. In addition, the current block may be at least one of an encode block, a prediction block, a residual block, and a transform block.

Sample: is a basic unit constituting a block. It may be expressed as a value from 0 to $2^{B_d}-1$ according to a bit depth (B_d). In the present invention, the sample may be used as a meaning of a pixel. That is, a sample, a pel, a pixel may have the same meaning with each other.

Unit: may refer to an encoding and decoding unit. When encoding and decoding an image, the unit may be a region generated by partitioning a single image. In addition, the unit may mean a subdivided unit when a single image is partitioned into subdivided units during encoding or decoding. That is, an image may be partitioned into a plurality of units. When encoding and decoding an image, a predetermined process for each unit may be performed. A single unit may be partitioned into sub-units that have sizes smaller than the size of the unit. Depending on functions, the unit may mean a block, a macroblock, a coding tree unit, a code tree block, a coding unit, a coding block, a prediction unit, a prediction block, a residual unit, a residual block, a transform unit, a transform block, etc. In addition, in order to distinguish a unit from a block, the unit may include a luma component block, a chroma component block associated with the luma component block, and a syntax element of each color component block. The unit may have various sizes and forms, and particularly, the form of the unit may be a two-dimensional geometrical figure such as a square shape, a rectangular shape, a trapezoid shape, a triangular shape, a pentagonal shape, etc. In addition, unit information may include at least one of a unit type indicating the coding unit, the prediction unit, the transform unit, etc., and a unit size, a unit depth, a sequence of encoding and decoding of a unit, etc.

Coding Tree Unit: is configured with a single coding tree block of a luma component Y, and two coding tree blocks related to chroma components Cb and Cr. In addition, it may

mean that including the blocks and a syntax element of each block. Each coding tree unit may be partitioned by using at least one of a quad-tree partitioning method, a binary-tree partitioning method and ternary-tree partitioning method to configure a lower unit such as coding unit, prediction unit, transform unit, etc. It may be used as a term for designating a sample block that becomes a process unit when encoding/decoding an image as an input image. Here, the quad-tree may mean a quaternary-tree.

When the size of the coding block is within a predetermined range, the division is possible using only quad-tree partitioning. Here, the predetermined range may be defined as at least one of a maximum size and a minimum size of a coding block in which the division is possible using only quad-tree partitioning. Information indicating a maximum/minimum size of a coding block in which quad-tree partitioning is allowed may be signaled through a bitstream, and the information may be signaled in at least one unit of a sequence, a picture parameter, a tile group, or a slice (segment). Alternatively, the maximum/minimum size of the coding block may be a fixed size predetermined in the coder/decoder. For example, when the size of the coding block corresponds to 256×256 to 64×64 , the division is possible only using quad-tree partitioning. Alternatively, when the size of the coding block is larger than the size of the maximum conversion block, the division is possible only using quad-tree partitioning. Herein, the block to be divided may be at least one of a coding block and a transform block. In this case, information indicating the division of the coded block (for example, *split_flag*) may be a flag indicating whether or not to perform the quad-tree partitioning. When the size of the coding block falls within a predetermined range, the division is possible only using binary tree or ternary tree partitioning. In this case, the above description of the quad-tree partitioning may be applied to binary tree partitioning or ternary tree partitioning in the same manner.

Coding Tree Block: may be used as a term for designating any one of a Y coding tree block, Cb coding tree block, and Cr coding tree block.

Neighbor Block: may mean a block adjacent to a current block. The block adjacent to the current block may mean a block that comes into contact with a boundary of the current block, or a block positioned within a predetermined distance from the current block. The neighbor block may mean a block adjacent to a vertex of the current block. Herein, the block adjacent to the vertex of the current block may mean a block vertically adjacent to a neighbor block that is horizontally adjacent to the current block, or a block horizontally adjacent to a neighbor block that is vertically adjacent to the current block.

Reconstructed Neighbor block: may mean a neighbor block adjacent to a current block and which has been already spatially/temporally encoded or decoded. Herein, the reconstructed neighbor block may mean a reconstructed neighbor unit. A reconstructed spatial neighbor block may be a block within a current picture and which has been already reconstructed through encoding or decoding or both. A reconstructed temporal neighbor block is a block at a corresponding position as the current block of the current picture within a reference image, or a neighbor block thereof.

Unit Depth: may mean a partitioned degree of a unit. In a tree structure, the highest node (Root Node) may correspond to the first unit which is not partitioned. Also, the highest node may have the least depth value. In this case, the highest node may have a depth of level 0. A node having a depth of level 1 may represent a unit generated by partitioning once the first unit. A node having a depth of level 2

may represent a unit generated by partitioning twice the first unit. A node having a depth of level n may represent a unit generated by partitioning n -times the first unit. A Leaf Node may be the lowest node and a node which cannot be partitioned further. A depth of a Leaf Node may be the maximum level. For example, a predefined value of the maximum level may be 3. A depth of a root node may be the lowest and a depth of a leaf node may be the deepest. In addition, when a unit is expressed as a tree structure, a level in which a unit is present may mean a unit depth.

Bitstream: may mean a bitstream including encoding image information.

Parameter Set: corresponds to header information among a configuration within a bitstream. At least one of a video parameter set, a sequence parameter set, a picture parameter set, and an adaptation parameter set may be included in a parameter set. In addition, a parameter set may include a slice header, a tile group header, and tile header information. The term "tile group" means a group of tiles and has the same meaning as a slice.

An adaptation parameter set may mean a parameter set that can be shared by being referred to in different pictures, subpictures, slices, tile groups, tiles, or bricks. In addition, information in an adaptation parameter set may be used by referring to different adaptation parameter sets for a subpicture, a slice, a tile group, a tile, or a brick inside a picture.

In addition, regarding the adaptation parameter set, different adaptation parameter sets may be referred to by using identifiers of different adaptation parameter sets for a subpicture, a slice, a tile group, a tile, or a brick inside a picture.

In addition, regarding the adaptation parameter set, different adaptation parameter sets may be referred to by using identifiers of different adaptation parameter sets for a slice, a tile group, a tile, or a brick inside a subpicture.

In addition, regarding the adaptation parameter set, different adaptation parameter sets may be referred to by using identifiers of different adaptation parameter sets for a tile or a brick inside a slice.

In addition, regarding the adaptation parameter set, different adaptation parameter sets may be referred to by using identifiers of different adaptation parameter sets for a brick inside a tile.

Information on an adaptation parameter set identifier may be included in a parameter set or a header of the subpicture, and an adaptation parameter set corresponding to the adaptation parameter set identifier may be used for the subpicture.

The information on the adaptation parameter set identifier may be included in a parameter set or a header of the tile, and an adaptation parameter set corresponding to the adaptation parameter set identifier may be used for the tile.

The information on the adaptation parameter set identifier may be included in a header of the brick, and an adaptation parameter set corresponding to the adaptation parameter set identifier may be used for the brick.

The picture may be partitioned into one or more tile rows and one or more tile columns.

The subpicture may be partitioned into one or more tile rows and one or more tile columns within a picture. The subpicture may be a region having the form of a rectangle/square within a picture and may include one or more CTUs. In addition, at least one or more tiles/bricks/slices may be included within one subpicture.

The tile may be a region having the form of a rectangle/square within a picture and may include one or more CTUs. In addition, the tile may be partitioned into one or more bricks.

The brick may mean one or more CTU rows within a tile. The tile may be partitioned into one or more bricks, and each brick may have at least one or more CTU rows. A tile that is not partitioned into two or more may mean a brick.

The slice may include one or more tiles within a picture and may include one or more bricks within a tile.

Parsing: may mean determination of a value of a syntax element by performing entropy decoding, or may mean the entropy decoding itself.

Symbol: may mean at least one of a syntax element, a coding parameter, and a transform coefficient value of an encoding/decoding target unit. In addition, the symbol may mean an entropy encoding target or an entropy decoding result.

Prediction Mode: may be information indicating a mode encoded/decoded with intra prediction or a mode encoded/decoded with inter prediction.

Prediction Unit: may mean a basic unit when performing prediction such as inter-prediction, intra-prediction, inter-compensation, intra-compensation, and motion compensation. A single prediction unit may be partitioned into a plurality of partitions having a smaller size, or may be partitioned into a plurality of lower prediction units. A plurality of partitions may be a basic unit in performing prediction or compensation. A partition which is generated by dividing a prediction unit may also be a prediction unit.

Prediction Unit Partition: may mean a form obtained by partitioning a prediction unit.

Reference picture list may refer to a list including one or more reference pictures used for inter prediction or motion compensation. There are several types of usable reference picture lists, including LC (List combined), L0 (List 0), L1 (List 1), L2 (List 2), L3 (List 3).

Inter prediction indicator may refer to a direction of inter prediction (unidirectional prediction, bidirectional prediction, etc.) of a current block. Alternatively, it may refer to the number of reference pictures used to generate a prediction block of a current block. Alternatively, it may refer to the number of prediction blocks used at the time of performing inter prediction or motion compensation on a current block.

Prediction list utilization flag indicates whether a prediction block is generated using at least one reference picture in a specific reference picture list. An inter prediction indicator can be derived using a prediction list utilization flag, and conversely, a prediction list utilization flag can be derived using an inter prediction indicator. For example, when the prediction list utilization flag has a first value of zero (0), it means that a reference picture in a reference picture list is not used to generate a prediction block. On the other hand, when the prediction list utilization flag has a second value of one (1), it means that a reference picture list is used to generate a prediction block.

Reference picture index may refer to an index indicating a specific reference picture in a reference picture list.

Reference picture may mean a reference picture which is referred to by a specific block for the purposes of inter prediction or motion compensation of the specific block. Alternatively, the reference picture may be a picture including a reference block referred to by a current block for inter prediction or motion compensation. Hereinafter, the terms "reference picture" and "reference image" have the same meaning and can be interchangeably.

Motion vector may be a two-dimensional vector used for inter prediction or motion compensation. The motion vector may mean an offset between an encoding/decoding target block and a reference block. For example, (mvX, mvY) may

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represent a motion vector. Here, mvX may represent a horizontal component and mvY may represent a vertical component.

Search range may be a two-dimensional region which is searched to retrieve a motion vector during inter prediction. For example, the size of the search range may be MxN. Here, M and N are both integers.

Motion vector candidate may refer to a prediction candidate block or a motion vector of the prediction candidate block when predicting a motion vector. In addition, a motion vector candidate may be included in a motion vector candidate list.

Motion vector candidate list may mean a list composed of one or more motion vector candidates.

Motion vector candidate index may mean an indicator indicating a motion vector candidate in a motion vector candidate list. Alternatively, it may be an index of a motion vector predictor.

Motion information may mean information including at least one of the items including a motion vector, a reference picture index, an inter prediction indicator, a prediction list utilization flag, reference picture list information, a reference picture, a motion vector candidate, a motion vector candidate index, a merge candidate, and a merge index.

Merge candidate list may mean a list composed of one or more merge candidates.

Merge candidate may mean a spatial merge candidate, a temporal merge candidate, a combined merge candidate, a combined bi-predictive merge candidate, or a zero merge candidate. The merge candidate may include motion information such as an inter prediction indicator, a reference picture index for each list, a motion vector, a prediction list utilization flag, and an inter prediction indicator.

Merge index may mean an indicator indicating a merge candidate in a merge candidate list. Alternatively, the merge index may indicate a block from which a merge candidate has been derived, among reconstructed blocks spatially/temporally adjacent to a current block. Alternatively, the merge index may indicate at least one piece of motion information of a merge candidate.

Transform Unit: may mean a basic unit when performing encoding/decoding such as transform, inverse-transform, quantization, dequantization, transform coefficient encoding/decoding of a residual signal. A single transform unit may be partitioned into a plurality of lower-level transform units having a smaller size. Here, transformation/inverse-transformation may comprise at least one among the first transformation/the first inverse-transformation and the second transformation/the second inverse-transformation.

Scaling: may mean a process of multiplying a quantized level by a factor. A transform coefficient may be generated by scaling a quantized level. The scaling also may be referred to as dequantization.

Quantization Parameter: may mean a value used when generating a quantized level using a transform coefficient during quantization. The quantization parameter also may mean a value used when generating a transform coefficient by scaling a quantized level during dequantization. The quantization parameter may be a value mapped on a quantization step size.

Delta Quantization Parameter: may mean a difference value between a predicted quantization parameter and a quantization parameter of an encoding/decoding target unit.

Scan: may mean a method of sequencing coefficients within a unit, a block or a matrix. For example, changing a two-dimensional matrix of coefficients into a one-dimensional matrix may be referred to as scanning, and changing

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a one-dimensional matrix of coefficients into a two-dimensional matrix may be referred to as scanning or inverse scanning.

Transform Coefficient: may mean a coefficient value generated after transform is performed in an encoder. It may mean a coefficient value generated after at least one of entropy decoding and dequantization is performed in a decoder. A quantized level obtained by quantizing a transform coefficient or a residual signal, or a quantized transform coefficient level also may fall within the meaning of the transform coefficient.

Quantized Level: may mean a value generated by quantizing a transform coefficient or a residual signal in an encoder. Alternatively, the quantized level may mean a value that is a dequantization target to undergo dequantization in a decoder. Similarly, a quantized transform coefficient level that is a result of transform and quantization also may fall within the meaning of the quantized level.

Non-zero Transform Coefficient: may mean a transform coefficient having a value other than zero, or a transform coefficient level or a quantized level having a value other than zero.

Quantization Matrix: may mean a matrix used in a quantization process or a dequantization process performed to improve subjective or objective image quality. The quantization matrix also may be referred to as a scaling list.

Quantization Matrix Coefficient: may mean each element within a quantization matrix. The quantization matrix coefficient also may be referred to as a matrix coefficient.

Default Matrix: may mean a predetermined quantization matrix preliminarily defined in an encoder or a decoder.

Non-default Matrix: may mean a quantization matrix that is not preliminarily defined in an encoder or a decoder but is signaled by a user.

Statistic Value: a statistic value for at least one among a variable, a coding parameter, a constant value, etc. which have a computable specific value may be one or more among an average value, a sum value, a weighted average value, a weighted sum value, the minimum value, the maximum value, the most frequent value, a median value, an interpolated value of the corresponding specific values.

FIG. 1 is a block diagram showing a configuration of an encoding apparatus according to an embodiment to which the present invention is applied.

An encoding apparatus **100** may be an encoder, a video encoding apparatus, or an image encoding apparatus. A video may include at least one image. The encoding apparatus **100** may sequentially encode at least one image.

Referring to FIG. 1, the encoding apparatus **100** may include a motion prediction unit **111**, a motion compensation unit **112**, an intra-prediction unit **120**, a switch **115**, a subtractor **125**, a transform unit **130**, a quantization unit **140**, an entropy encoding unit **150**, a dequantization unit **160**, an inverse-transform unit **170**, an adder **175**, a filter unit **180**, and a reference picture buffer **190**.

The encoding apparatus **100** may perform encoding of an input image by using an intra mode or an inter mode or both. In addition, encoding apparatus **100** may generate a bitstream including encoded information through encoding the input image, and output the generated bitstream. The generated bitstream may be stored in a computer readable recording medium, or may be streamed through a wired/wireless transmission medium. When an intra mode is used as a prediction mode, the switch **115** may be switched to an intra. Alternatively, when an inter mode is used as a prediction mode, the switch **115** may be switched to an inter mode. Herein, the intra mode may mean an intra-prediction mode,

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and the inter mode may mean an inter-prediction mode. The encoding apparatus **100** may generate a prediction block for an input block of the input image. In addition, the encoding apparatus **100** may encode a residual block using a residual of the input block and the prediction block after the prediction block being generated. The input image may be called as a current image that is a current encoding target. The input block may be called as a current block that is current encoding target, or as an encoding target block.

When a prediction mode is an intra mode, the intra-prediction unit **120** may use a sample of a block that has been already encoded/decoded and is adjacent to a current block as a reference sample. The intra-prediction unit **120** may perform spatial prediction for the current block by using a reference sample, or generate prediction samples of an input block by performing spatial prediction. Herein, the intra prediction may mean intra-prediction,

When a prediction mode is an inter mode, the motion prediction unit **111** may retrieve a region that best matches with an input block from a reference image when performing motion prediction, and deduce a motion vector by using the retrieved region. In this case, a search region may be used as the region. The reference image may be stored in the reference picture buffer **190**. Here, when encoding/decoding for the reference image is performed, it may be stored in the reference picture buffer **190**.

The motion compensation unit **112** may generate a prediction block by performing motion compensation for the current block using a motion vector. Herein, inter-prediction may mean inter-prediction or motion compensation.

When the value of the motion vector is not an integer, the motion prediction unit **111** and the motion compensation unit **112** may generate the prediction block by applying an interpolation filter to a partial region of the reference picture. In order to perform inter-picture prediction or motion compensation on a coding unit, it may be determined that which mode among a skip mode, a merge mode, an advanced motion vector prediction (AMVP) mode, and a current picture referring mode is used for motion prediction and motion compensation of a prediction unit included in the corresponding coding unit. Then, inter-picture prediction or motion compensation may be differently performed depending on the determined mode.

The subtractor **125** may generate a residual block by using a difference of an input block and a prediction block. The residual block may be called as a residual signal. The residual signal may mean a difference between an original signal and a prediction signal. In addition, the residual signal may be a signal generated by transforming or quantizing, or transforming and quantizing a difference between the original signal and the prediction signal. The residual block may be a residual signal of a block unit.

The transform unit **130** may generate a transform coefficient by performing transform of a residual block, and output the generated transform coefficient. Herein, the transform coefficient may be a coefficient value generated by performing transform of the residual block. When a transform skip mode is applied, the transform unit **130** may skip transform of the residual block.

A quantized level may be generated by applying quantization to the transform coefficient or to the residual signal. Hereinafter, the quantized level may be also called as a transform coefficient in embodiments.

The quantization unit **140** may generate a quantized level by quantizing the transform coefficient or the residual signal according to a parameter, and output the generated quantized

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level. Herein, the quantization unit **140** may quantize the transform coefficient by using a quantization matrix.

The entropy encoding unit **150** may generate a bitstream by performing entropy encoding according to a probability distribution on values calculated by the quantization unit **140** or on coding parameter values calculated when performing encoding, and output the generated bitstream. The entropy encoding unit **150** may perform entropy encoding of sample information of an image and information for decoding an image. For example, the information for decoding the image may include a syntax element.

When entropy encoding is applied, symbols are represented so that a smaller number of bits are assigned to a symbol having a high chance of being generated and a larger number of bits are assigned to a symbol having a low chance of being generated, and thus, the size of bit stream for symbols to be encoded may be decreased. The entropy encoding unit **150** may use an encoding method for entropy encoding such as exponential Golomb, context-adaptive variable length coding (CA VLC), context-adaptive binary arithmetic coding (CABAC), etc. For example, the entropy encoding unit **150** may perform entropy encoding by using a variable length coding/code (VLC) table. In addition, the entropy encoding unit **150** may deduce a binarization method of a target symbol and a probability model of a target symbol/bin, and perform arithmetic coding by using the deduced binarization method, and a context model.

In order to encode a transform coefficient level (quantized level), the entropy encoding unit **150** may change a two-dimensional block form coefficient into a one-dimensional vector form by using a transform coefficient scanning method.

A coding parameter may include information (flag, index, etc.) such as syntax element that is encoded in an encoder and signaled to a decoder, and information derived when performing encoding or decoding. The coding parameter may mean information required when encoding or decoding an image. For example, at least one value or a combination form of a unit/block size, a unit/block depth, unit/block partition information, unit/block shape, unit/block partition structure, whether to partition of a quad-tree form, whether to partition of a binary-tree form, a partition direction of a binary-tree form (horizontal direction or vertical direction), a partition form of a binary-tree form (symmetric partition or asymmetric partition), whether or not a current coding unit is partitioned by ternary tree partitioning, direction (horizontal or vertical direction) of the ternary tree partitioning, type (symmetric or asymmetric type) of the ternary tree partitioning, whether a current coding unit is partitioned by multi-type tree partitioning, direction (horizontal or vertical direction) of the multi-type tree partitioning, type (symmetric or asymmetric type) of the multi-type tree partitioning, and a tree (binary tree or ternary tree) structure of the multi-type tree partitioning, a prediction mode (intra prediction or inter prediction), a luma intra-prediction mode/direction, a chroma intra-prediction mode/direction, intra partition information, inter partition information, a coding block partition flag, a prediction block partition flag, a transform block partition flag, a reference sample filtering method, a reference sample filter tab, a reference sample filter coefficient, a prediction block filtering method, a prediction block filter tap, a prediction block filter coefficient, a prediction block boundary filtering method, a prediction block boundary filter tab, a prediction block boundary filter coefficient, an intra-prediction mode, an inter-prediction mode, motion information, a motion vector, a motion vector difference, a reference picture index, a inter-

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prediction angle, an inter-prediction indicator, a prediction list utilization flag, a reference picture list, a reference picture, a motion vector predictor index, a motion vector predictor candidate, a motion vector candidate list, whether to use a merge mode, a merge index, a merge candidate, a merge candidate list, whether to use a skip mode, an interpolation filter type, an interpolation filter tab, an interpolation filter coefficient, a motion vector size, a presentation accuracy of a motion vector, a transform type, a transform size, information of whether or not a primary (first) transform is used, information of whether or not a secondary transform is used, a primary transform index, a secondary transform index, information of whether or not a residual signal is present, a coded block pattern, a coded block flag (CBF), a quantization parameter, a quantization parameter residue, a quantization matrix, whether to apply an intra loop filter, an intra loop filter coefficient, an intra loop filter tab, an intra loop filter shape/form, whether to apply a deblocking filter, a deblocking filter coefficient, a deblocking filter tab, a deblocking filter strength, a deblocking filter shape/form, whether to apply an adaptive sample offset, an adaptive sample offset value, an adaptive sample offset category, an adaptive sample offset type, whether to apply an adaptive loop filter, an adaptive loop filter coefficient, an adaptive loop filter tab, an adaptive loop filter shape/form, a binarization/inverse-binarization method, a context model determining method, a context model updating method, whether to perform a regular mode, whether to perform a bypass mode, a context bin, a bypass bin, a significant coefficient flag, a last significant coefficient flag, a coded flag for a unit of a coefficient group, a position of the last significant coefficient, a flag for whether a value of a coefficient is larger than 1, a flag for whether a value of a coefficient is larger than 2, a flag for whether a value of a coefficient is larger than 3, information on a remaining coefficient value, a sign information, a reconstructed luma sample, a reconstructed chroma sample, a residual luma sample, a residual chroma sample, a luma transform coefficient, a chroma transform coefficient, a quantized luma level, a quantized chroma level, a transform coefficient level scanning method, a size of a motion vector search area at a decoder side, a shape of a motion vector search area at a decoder side, a number of time of a motion vector search at a decoder side, information on a CTU size, information on a minimum block size, information on a maximum block size, information on a maximum block depth, information on a minimum block depth, an image displaying/outputting sequence, slice identification information, a slice type, slice partition information, tile identification information, a tile type, tile partition information, tile group identification information, a tile group type, tile group partition information, a picture type, a bit depth of an input sample, a bit depth of a reconstruction sample, a bit depth of a residual sample, a bit depth of a transform coefficient, a bit depth of a quantized level, and information on a luma signal or information on a chroma signal may be included in the coding parameter.

Herein, signaling the flag or index may mean that a corresponding flag or index is entropy encoded and included in a bitstream by an encoder, and may mean that the corresponding flag or index is entropy decoded from a bitstream by a decoder.

When the encoding apparatus **100** performs encoding through inter-prediction, an encoded current image may be used as a reference image for another image that is processed afterwards. Accordingly, the encoding apparatus **100** may

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reconstruct or decode the encoded current image, or store the reconstructed or decoded image as a reference image in reference picture buffer **190**.

A quantized level may be dequantized in the dequantization unit **160**, or may be inverse-transformed in the inverse-transform unit **170**. A dequantized or inverse-transformed coefficient or both may be added with a prediction block by the adder **175**. By adding the dequantized or inverse-transformed coefficient or both with the prediction block, a reconstructed block may be generated. Herein, the dequantized or inverse-transformed coefficient or both may mean a coefficient on which at least one of dequantization and inverse-transform is performed, and may mean a reconstructed residual block.

A reconstructed block may pass through the filter unit **180**. The filter unit **180** may apply at least one of a deblocking filter, a sample adaptive offset (SAO), and an adaptive loop filter (ALF) to a reconstructed sample, a reconstructed block or a reconstructed image. The filter unit **180** may be called as an in-loop filter.

The deblocking filter may remove block distortion generated in boundaries between blocks. In order to determine whether or not to apply a deblocking filter, whether or not to apply a deblocking filter to a current block may be determined based samples included in several rows or columns which are included in the block. When a deblocking filter is applied to a block, another filter may be applied according to a required deblocking filtering strength.

In order to compensate an encoding error, a proper offset value may be added to a sample value by using a sample adaptive offset. The sample adaptive offset may correct an offset of a deblocked image from an original image by a sample unit. A method of partitioning samples of an image into a predetermined number of regions, determining a region to which an offset is applied, and applying the offset to the determined region, or a method of applying an offset in consideration of edge information on each sample may be used.

The adaptive loop filter may perform filtering based on a comparison result of the filtered reconstructed image and the original image. Samples included in an image may be partitioned into predetermined groups, a filter to be applied to each group may be determined, and differential filtering may be performed for each group. Information of whether or not to apply the ALF may be signaled by coding units (CUs), and a form and coefficient of the ALF to be applied to each block may vary.

The reconstructed block or the reconstructed image having passed through the filter unit **180** may be stored in the reference picture buffer **190**. A reconstructed block processed by the filter unit **180** may be a part of a reference image. That is, a reference image is a reconstructed image composed of reconstructed blocks processed by the filter unit **180**. The stored reference image may be used later in inter prediction or motion compensation.

FIG. 2 is a block diagram showing a configuration of a decoding apparatus according to an embodiment and to which the present invention is applied.

A decoding apparatus **200** may a decoder, a video decoding apparatus, or an image decoding apparatus.

Referring to FIG. 2, the decoding apparatus **200** may include an entropy decoding unit **210**, a dequantization unit **220**, an inverse-transform unit **230**, an intra-prediction unit **240**, a motion compensation unit **250**, an adder **225**, a filter unit **260**, and a reference picture buffer **270**.

The decoding apparatus **200** may receive a bitstream output from the encoding apparatus **100**. The decoding

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apparatus **200** may receive a bitstream stored in a computer readable recording medium, or may receive a bitstream that is streamed through a wired/wireless transmission medium. The decoding apparatus **200** may decode the bitstream by using an intra mode or an inter mode. In addition, the decoding apparatus **200** may generate a reconstructed image generated through decoding or a decoded image, and output the reconstructed image or decoded image.

When a prediction mode used when decoding is an intra mode, a switch may be switched to an intra. Alternatively, when a prediction mode used when decoding is an inter mode, a switch may be switched to an inter mode.

The decoding apparatus **200** may obtain a reconstructed residual block by decoding the input bitstream, and generate a prediction block. When the reconstructed residual block and the prediction block are obtained, the decoding apparatus **200** may generate a reconstructed block that becomes a decoding target by adding the reconstructed residual block with the prediction block. The decoding target block may be called a current block.

The entropy decoding unit **210** may generate symbols by entropy decoding the bitstream according to a probability distribution. The generated symbols may include a symbol of a quantized level form. Herein, an entropy decoding method may be an inverse-process of the entropy encoding method described above.

In order to decode a transform coefficient level (quantized level), the entropy decoding unit **210** may change a one-directional vector form coefficient into a two-dimensional block form by using a transform coefficient scanning method.

A quantized level may be dequantized in the dequantization unit **220**, or inverse-transformed in the inverse-transform unit **230**. The quantized level may be a result of dequantizing or inverse-transforming or both, and may be generated as a reconstructed residual block. Herein, the dequantization unit **220** may apply a quantization matrix to the quantized level.

When an intra mode is used, the intra-prediction unit **240** may generate a prediction block by performing, for the current block, spatial prediction that uses a sample value of a block adjacent to a decoding target block and which has been already decoded.

When an inter mode is used, the motion compensation unit **250** may generate a prediction block by performing, for the current block, motion compensation that uses a motion vector and a reference image stored in the reference picture buffer **270**.

The adder **225** may generate a reconstructed block by adding the reconstructed residual block with the prediction block. The filter unit **260** may apply at least one of a deblocking filter, a sample adaptive offset, and an adaptive loop filter to the reconstructed block or reconstructed image. The filter unit **260** may output the reconstructed image. The reconstructed block or reconstructed image may be stored in the reference picture buffer **270** and used when performing inter-prediction. A reconstructed block processed by the filter unit **260** may be a part of a reference image. That is, a reference image is a reconstructed image composed of reconstructed blocks processed by the filter unit **260**. The stored reference image may be used later in inter prediction or motion compensation.

FIG. 3 is a view schematically showing a partition structure of an image when encoding and decoding the image. FIG. 3 schematically shows an example of partitioning a single unit into a plurality of lower units.

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In order to efficiently partition an image, when encoding and decoding, a coding unit (CU) may be used. The coding unit may be used as a basic unit when encoding/decoding the image. In addition, the coding unit may be used as a unit for distinguishing an intra prediction mode and an inter prediction mode when encoding/decoding the image. The coding unit may be a basic unit used for prediction, transform, quantization, inverse-transform, dequantization, or an encoding/decoding process of a transform coefficient.

Referring to FIG. 3, an image **300** is sequentially partitioned in a largest coding unit (LCU), and a LCU unit is determined as a partition structure. Herein, the LCU may be used in the same meaning as a coding tree unit (CTU). A unit partitioning may mean partitioning a block associated with to the unit. In block partition information, information of a unit depth may be included. Depth information may represent a number of times or a degree or both in which a unit is partitioned. A single unit may be partitioned into a plurality of lower level units hierarchically associated with depth information based on a tree structure. In other words, a unit and a lower level unit generated by partitioning the unit may correspond to a node and a child node of the node, respectively. Each of partitioned lower unit may have depth information. Depth information may be information representing a size of a CU, and may be stored in each CU. Unit depth represents times and/or degrees related to partitioning a unit. Therefore, partitioning information of a lower-level unit may comprise information on a size of the lower-level unit.

A partition structure may mean a distribution of a coding unit (CU) within an LCU **310**. Such a distribution may be determined according to whether or not to partition a single CU into a plurality (positive integer equal to or greater than 2 including 2, 4, 8, 16, etc.) of CUs. A horizontal size and a vertical size of the CU generated by partitioning may respectively be half of a horizontal size and a vertical size of the CU before partitioning, or may respectively have sizes smaller than a horizontal size and a vertical size before partitioning according to a number of times of partitioning. The CU may be recursively partitioned into a plurality of CUs. By the recursive partitioning, at least one among a height and a width of a CU after partitioning may decrease comparing with at least one among a height and a width of a CU before partitioning. Partitioning of the CU may be recursively performed until to a predefined depth or predefined size. For example, a depth of an LCU may be 0, and a depth of a smallest coding unit (SCU) may be a predefined maximum depth. Herein, the LCU may be a coding unit having a maximum coding unit size, and the SCU may be a coding unit having a minimum coding unit size as described above. Partitioning is started from the LCU **310**, a CU depth increases by 1 as a horizontal size or a vertical size or both of the CU decreases by partitioning. For example, for each depth, a CU which is not partitioned may have a size of $2N \times 2N$. Also, in case of a CU which is partitioned, a CU with a size of $2N \times 2N$ may be partitioned into four CUs with a size of $N \times N$. A size of N may decrease to half as a depth increase by 1.

In addition, information whether or not the CU is partitioned may be represented by using partition information of the CU. The partition information may be 1-bit information. All CUs, except for a SCU, may include partition information. For example, when a value of partition information is a first value, the CU may not be partitioned, when a value of partition information is a second value, the CU may be partitioned

Referring to FIG. 3, an LCU having a depth 0 may be a 64×64 block. 0 may be a minimum depth. A SCU having a depth 3 may be an 8×8 block. 3 may be a maximum depth. A CU of a 32×32 block and a 16×16 block may be respectively represented as a depth 1 and a depth 2.

For example, when a single coding unit is partitioned into four coding units, a horizontal size and a vertical size of the four partitioned coding units may be a half size of a horizontal and vertical size of the CU before being partitioned. In one embodiment, when a coding unit having a 32×32 size is partitioned into four coding units, each of the four partitioned coding units may have a 16×16 size. When a single coding unit is partitioned into four coding units, it may be called that the coding unit may be partitioned into a quad-tree form.

For example, when one coding unit is partitioned into two sub-coding units, the horizontal or vertical size (width or height) of each of the two sub-coding units may be half the horizontal or vertical size of the original coding unit. For example, when a coding unit having a size of 32×32 is vertically partitioned into two sub-coding units, each of the two sub-coding units may have a size of 16×32. For example, when a coding unit having a size of 8×32 is horizontally partitioned into two sub-coding units, each of the two sub-coding units may have a size of 8×16. When one coding unit is partitioned into two sub-coding units, it can be said that the coding unit is binary-partitioned or is partitioned by a binary tree partition structure.

For example, when one coding unit is partitioned into three sub-coding units, the horizontal or vertical size of the coding unit can be partitioned with a ratio of 1:2:1, thereby producing three sub-coding units whose horizontal or vertical sizes are in a ratio of 1:2:1. For example, when a coding unit having a size of 16×32 is horizontally partitioned into three sub-coding units, the three sub-coding units may have sizes of 16×8, 16×16, and 16×8 respectively, in the order from the uppermost to the lowermost sub-coding unit. For example, when a coding unit having a size of 32×32 is vertically split into three sub-coding units, the three sub-coding units may have sizes of 8×32, 16×32, and 8×32, respectively in the order from the left to the right sub-coding unit. When one coding unit is partitioned into three sub-coding units, it can be said that the coding unit is ternary-partitioned or partitioned by a ternary tree partition structure.

In FIG. 3, a coding tree unit (CTU) 320 is an example of a CTU to which a quad tree partition structure, a binary tree partition structure, and a ternary tree partition structure are all applied.

As described above, in order to partition the CTU, at least one of a quad tree partition structure, a binary tree partition structure, and a ternary tree partition structure may be applied. Various tree partition structures may be sequentially applied to the CTU, according to a predetermined priority order. For example, the quad tree partition structure may be preferentially applied to the CTU. A coding unit that cannot be partitioned any longer using a quad tree partition structure may correspond to a leaf node of a quad tree. A coding unit corresponding to a leaf node of a quad tree may serve as a root node of a binary and/or ternary tree partition structure. That is, a coding unit corresponding to a leaf node of a quad tree may be further partitioned by a binary tree partition structure or a ternary tree partition structure, or may not be further partitioned. Therefore, by preventing a coding block that results from binary tree partitioning or ternary tree partitioning of a coding unit corresponding to a leaf node of

a quad tree from undergoing further quad tree partitioning, block partitioning and/or signaling of partition information can be effectively performed.

The fact that a coding unit corresponding to a node of a quad tree is partitioned may be signaled using quad partition information. The quad partition information having a first value (e.g., "1") may indicate that a current coding unit is partitioned by the quad tree partition structure. The quad partition information having a second value (e.g., "0") may indicate that a current coding unit is not partitioned by the quad tree partition structure. The quad partition information may be a flag having a predetermined length (e.g., one bit).

There may not be a priority between the binary tree partitioning and the ternary tree partitioning. That is, a coding unit corresponding to a leaf node of a quad tree may further undergo arbitrary partitioning among the binary tree partitioning and the ternary tree partitioning. In addition, a coding unit generated through the binary tree partitioning or the ternary tree partitioning may undergo a further binary tree partitioning or a further ternary tree partitioning, or may not be further partitioned.

A tree structure in which there is no priority among the binary tree partitioning and the ternary tree partitioning is referred to as a multi-type tree structure. A coding unit corresponding to a leaf node of a quad tree may serve as a root node of a multi-type tree. Whether to partition a coding unit which corresponds to a node of a multi-type tree may be signaled using at least one of multi-type tree partition indication information, partition direction information, and partition tree information. For partitioning of a coding unit corresponding to a node of a multi-type tree, the multi-type tree partition indication information, the partition direction, and the partition tree information may be sequentially signaled.

The multi-type tree partition indication information having a first value (e.g., "1") may indicate that a current coding unit is to undergo a multi-type tree partitioning. The multi-type tree partition indication information having a second value (e.g., "0") may indicate that a current coding unit is not to undergo a multi-type tree partitioning.

When a coding unit corresponding to a node of a multi-type tree is further partitioned by a multi-type tree partition structure, the coding unit may include partition direction information. The partition direction information may indicate in which direction a current coding unit is to be partitioned for the multi-type tree partitioning. The partition direction information having a first value (e.g., "1") may indicate that a current coding unit is to be vertically partitioned. The partition direction information having a second value (e.g., "0") may indicate that a current coding unit is to be horizontally partitioned.

When a coding unit corresponding to a node of a multi-type tree is further partitioned by a multi-type tree partition structure, the current coding unit may include partition tree information. The partition tree information may indicate a tree partition structure which is to be used for partitioning of a node of a multi-type tree. The partition tree information having a first value (e.g., "1") may indicate that a current coding unit is to be partitioned by a binary tree partition structure. The partition tree information having a second value (e.g., "0") may indicate that a current coding unit is to be partitioned by a ternary tree partition structure.

The partition indication information, the partition tree information, and the partition direction information may each be a flag having a predetermined length (e.g., one bit).

At least any one of the quadtree partition indication information, the multi-type tree partition indication infor-

mation, the partition direction information, and the partition tree information may be entropy encoded/decoded. For the entropy-encoding/decoding of those types of information, information on a neighboring coding unit adjacent to the current coding unit may be used. For example, there is a high probability that the partition type (the partitioned or non-partitioned, the partition tree, and/or the partition direction) of a left neighboring coding unit and/or an upper neighboring coding unit of a current coding unit is similar to that of the current coding unit. Therefore, context information for entropy encoding/decoding of the information on the current coding unit may be derived from the information on the neighboring coding units. The information on the neighboring coding units may include at least any one of quad partition information, multi-type tree partition indication information, partition direction information, and partition tree information.

As another example, among binary tree partitioning and ternary tree partitioning, binary tree partitioning may be preferentially performed. That is, a current coding unit may primarily undergo binary tree partitioning, and then a coding unit corresponding to a leaf node of a binary tree may be set as a root node for ternary tree partitioning. In this case, neither quad tree partitioning nor binary tree partitioning may not be performed on the coding unit corresponding to a node of a ternary tree.

A coding unit that cannot be partitioned by a quad tree partition structure, a binary tree partition structure, and/or a ternary tree partition structure becomes a basic unit for coding, prediction and/or transformation. That is, the coding unit cannot be further partitioned for prediction and/or transformation. Therefore, the partition structure information and the partition information used for partitioning a coding unit into prediction units and/or transformation units may not be present in a bit stream.

However, when the size of a coding unit (i.e., a basic unit for partitioning) is larger than the size of a maximum transformation block, the coding unit may be recursively partitioned until the size of the coding unit is reduced to be equal to or smaller than the size of the maximum transformation block. For example, when the size of a coding unit is 64×64 and when the size of a maximum transformation block is 32×32 , the coding unit may be partitioned into four 32×32 blocks for transformation. For example, when the size of a coding unit is 32×64 and the size of a maximum transformation block is 32×32 , the coding unit may be partitioned into two 32×32 blocks for the transformation. In this case, the partitioning of the coding unit for transformation is not signaled separately, and may be determined through comparison between the horizontal or vertical size of the coding unit and the horizontal or vertical size of the maximum transformation block. For example, when the horizontal size (width) of the coding unit is larger than the horizontal size (width) of the maximum transformation block, the coding unit may be vertically bisected. For example, when the vertical size (height) of the coding unit is larger than the vertical size (height) of the maximum transformation block, the coding unit may be horizontally bisected.

Information of the maximum and/or minimum size of the coding unit and information of the maximum and/or minimum size of the transformation block may be signaled or determined at an upper level of the coding unit. The upper level may be, for example, a sequence level, a picture level, a slice level, a tile group level, a tile level, or the like. For example, the minimum size of the coding unit may be determined to be 4×4 . For example, the maximum size of the

transformation block may be determined to be 64×64 . For example, the minimum size of the transformation block may be determined to be 4×4 .

Information of the minimum size (quad tree minimum size) of a coding unit corresponding to a leaf node of a quad tree and/or information of the maximum depth (the maximum tree depth of a multi-type tree) from a root node to a leaf node of the multi-type tree may be signaled or determined at an upper level of the coding unit. For example, the upper level may be a sequence level, a picture level, a slice level, a tile group level, a tile level, or the like. Information of the minimum size of a quad tree and/or information of the maximum depth of a multi-type tree may be signaled or determined for each of an intra-picture slice and an inter-picture slice.

Difference information between the size of a CTU and the maximum size of a transformation block may be signaled or determined at an upper level of the coding unit. For example, the upper level may be a sequence level, a picture level, a slice level, a tile group level, a tile level, or the like. Information of the maximum size of the coding units corresponding to the respective nodes of a binary tree (hereinafter, referred to as a maximum size of a binary tree) may be determined based on the size of the coding tree unit and the difference information. The maximum size of the coding units corresponding to the respective nodes of a ternary tree (hereinafter, referred to as a maximum size of a ternary tree) may vary depending on the type of slice. For example, for an intra-picture slice, the maximum size of a ternary tree may be 32×32 . For example, for an inter-picture slice, the maximum size of a ternary tree may be 128×128 . For example, the minimum size of the coding units corresponding to the respective nodes of a binary tree (hereinafter, referred to as a minimum size of a binary tree) and/or the minimum size of the coding units corresponding to the respective nodes of a ternary tree (hereinafter, referred to as a minimum size of a ternary tree) may be set as the minimum size of a coding block.

As another example, the maximum size of a binary tree and/or the maximum size of a ternary tree may be signaled or determined at the slice level. Alternatively, the minimum size of the binary tree and/or the minimum size of the ternary tree may be signaled or determined at the slice level.

Depending on size and depth information of the above-described various blocks, quad partition information, multi-type tree partition indication information, partition tree information and/or partition direction information may be included or may not be included in a bit stream.

For example, when the size of the coding unit is not larger than the minimum size of a quad tree, the coding unit does not contain quad partition information. Thus, the quad partition information may be deduced from a second value.

For example, when the sizes (horizontal and vertical sizes) of a coding unit corresponding to a node of a multi-type tree are larger than the maximum sizes (horizontal and vertical sizes) of a binary tree and/or the maximum sizes (horizontal and vertical sizes) of a ternary tree, the coding unit may not be binary-partitioned or ternary-partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but may be deduced from a second value.

Alternatively, when the sizes (horizontal and vertical sizes) of a coding unit corresponding to a node of a multi-type tree are the same as the maximum sizes (horizontal and vertical sizes) of a binary tree and/or are two times as large as the maximum sizes (horizontal and vertical sizes) of a ternary tree, the coding unit may not be further binary-

partitioned or ternary-partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but be derived from a second value. This is because when a coding unit is partitioned by a binary tree partition structure and/or a ternary tree partition structure, a coding unit smaller than the minimum size of a binary tree and/or the minimum size of a ternary tree is generated.

Alternatively, the binary tree partitioning or the ternary tree partitioning may be limited on the basis of the size of a virtual pipeline data unit (hereinafter, a pipeline buffer size). For example, when the coding unit is divided into sub-coding units which do not fit the pipeline buffer size by the binary tree partitioning or the ternary tree partitioning, the corresponding binary tree partitioning or ternary tree partitioning may be limited. The pipeline buffer size may be the size of the maximum transform block (e.g., 64×64). For example, when the pipeline buffer size is 64×64, the division below may be limited.

N×M (N and/or M is 128) Ternary tree partitioning for coding units

128×N (N≤64) Binary tree partitioning in horizontal direction for coding units

N×128 (N≤64) Binary tree partitioning in vertical direction for coding units

Alternatively, when the depth of a coding unit corresponding to a node of a multi-type tree is equal to the maximum depth of the multi-type tree, the coding unit may not be further binary-partitioned and/or ternary-partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but may be deduced from a second value.

Alternatively, only when at least one of vertical direction binary tree partitioning, horizontal direction binary tree partitioning, vertical direction ternary tree partitioning, and horizontal direction ternary tree partitioning is possible for a coding unit corresponding to a node of a multi-type tree, the multi-type tree partition indication information may be signaled. Otherwise, the coding unit may not be binary-partitioned and/or ternary-partitioned. Accordingly, the multi-type tree partition indication information may not be signaled but may be deduced from a second value.

Alternatively, only when both of the vertical direction binary tree partitioning and the horizontal direction binary tree partitioning or both of the vertical direction ternary tree partitioning and the horizontal direction ternary tree partitioning are possible for a coding unit corresponding to a node of a multi-type tree, the partition direction information may be signaled. Otherwise, the partition direction information may not be signaled but may be derived from a value indicating possible partitioning directions.

Alternatively, only when both of the vertical direction binary tree partitioning and the vertical direction ternary tree partitioning or both of the horizontal direction binary tree partitioning and the horizontal direction ternary tree partitioning are possible for a coding tree corresponding to a node of a multi-type tree, the partition tree information may be signaled. Otherwise, the partition tree information may not be signaled but be deduced from a value indicating a possible partitioning tree structure.

FIG. 4 is a view showing an intra-prediction process.

Arrows from center to outside in FIG. 4 may represent prediction directions of intra prediction modes.

Intra encoding and/or decoding may be performed by using a reference sample of a neighbor block of the current block. A neighbor block may be a reconstructed neighbor block. For example, intra encoding and/or decoding may be

performed by using a coding parameter or a value of a reference sample included in a reconstructed neighbor block.

A prediction block may mean a block generated by performing intra prediction. A prediction block may correspond to at least one among CU, PU and TU. A unit of a prediction block may have a size of one among CU, PU and TU. A prediction block may be a square block having a size of 2×2, 4×4, 16×16, 32×32 or 64×64 etc. or may be a rectangular block having a size of 2×8, 4×8, 2×16, 4×16 and 8×16 etc.

Intra prediction may be performed according to intra prediction mode for the current block. The number of intra prediction modes which the current block may have may be a fixed value and may be a value determined differently according to an attribute of a prediction block. For example, an attribute of a prediction block may comprise a size of a prediction block and a shape of a prediction block, etc.

The number of intra-prediction modes may be fixed to N regardless of a block size. Or, the number of intra prediction modes may be 3, 5, 9, 17, 34, 35, 36, 65, or 67 etc. Alternatively, the number of intra-prediction modes may vary according to a block size or a color component type or both. For example, the number of intra prediction modes may vary according to whether the color component is a luma signal or a chroma signal. For example, as a block size becomes large, a number of intra-prediction modes may increase. Alternatively, a number of intra-prediction modes of a luma component block may be larger than a number of intra-prediction modes of a chroma component block.

An intra-prediction mode may be a non-angular mode or an angular mode. The non-angular mode may be a DC mode or a planar mode, and the angular mode may be a prediction mode having a specific direction or angle. The intra-prediction mode may be expressed by at least one of a mode number, a mode value, a mode numeral, a mode angle, and mode direction. A number of intra-prediction modes may be M, which is larger than 1, including the non-angular and the angular mode. In order to intra-predict a current block, a step of determining whether or not samples included in a reconstructed neighbor block may be used as reference samples of the current block may be performed. When a sample that is not usable as a reference sample of the current block is present, a value obtained by duplicating or performing interpolation on at least one sample value among samples included in the reconstructed neighbor block or both may be used to replace with a non-usable sample value of a sample, thus the replaced sample value is used as a reference sample of the current block.

FIG. 7 is a diagram illustrating reference samples capable of being used for intra prediction.

As shown in FIG. 7, at least one of the reference sample line 0 to the reference sample line 3 may be used for intra prediction of the current block. In FIG. 7, the samples of a segment A and a segment F may be padded with the samples closest to a segment B and a segment E, respectively, instead of retrieving from the reconstructed neighboring block. Index information indicating the reference sample line to be used for intra prediction of the current block may be signaled. For example, in FIG. 7, reference sample line indicators 0, 1, and 2 may be signaled as index information indicating reference sample lines 0, 1 and 2. When the upper boundary of the current block is the boundary of the CTU, only the reference sample line 0 may be available. Therefore, in this case, the index information may not be signaled. When a reference sample line other than the reference

sample line 0 is used, filtering for a prediction block, which will be described later, may not be performed.

When intra-predicting, a filter may be applied to at least one of a reference sample and a prediction sample based on an intra-prediction mode and a current block size.

In case of a planar mode, when generating a prediction block of a current block, according to a position of a prediction target sample within a prediction block, a sample value of the prediction target sample may be generated by using a weighted sum of an upper and left side reference sample of a current sample, and a right upper side and left lower side reference sample of the current block. In addition, in case of a DC mode, when generating a prediction block of a current block, an average value of upper side and left side reference samples of the current block may be used. In addition, in case of an angular mode, a prediction block may be generated by using an upper side, a left side, a right upper side, and/or a left lower side reference sample of the current block. In order to generate a prediction sample value, interpolation of a real number unit may be performed.

In the case of intra prediction between color components, a prediction block for the current block of the second color component may be generated on the basis of the corresponding reconstructed block of the first color component. For example, the first color component may be a luma component, and the second color component may be a chroma component. For intra prediction between color components, the parameters of the linear model between the first color component and the second color component may be derived on the basis of the template. The template may include upper and/or left neighboring samples of the current block and upper and/or left neighboring samples of the reconstructed block of the first color component corresponding thereto. For example, the parameters of the linear model may be derived using a sample value of a first color component having a maximum value among samples in a template and a sample value of a second color component corresponding thereto, and a sample value of a first color component having a minimum value among samples in the template and a sample value of a second color component corresponding thereto. When the parameters of the linear model are derived, a corresponding reconstructed block may be applied to the linear model to generate a prediction block for the current block. According to a video format, subsampling may be performed on the neighboring samples of the reconstructed block of the first color component and the corresponding reconstructed block. For example, when one sample of the second color component corresponds to four samples of the first color component, four samples of the first color component may be sub-sampled to compute one corresponding sample. In this case, the parameter derivation of the linear model and intra prediction between color components may be performed on the basis of the corresponding sub-sampled samples. Whether or not to perform intra prediction between color components and/or the range of the template may be signaled as the intra prediction mode.

The current block may be partitioned into two or four sub-blocks in the horizontal or vertical direction. The partitioned sub-blocks may be sequentially reconstructed. That is, the intra prediction may be performed on the sub-block to generate the sub-prediction block. In addition, dequantization and/or inverse transform may be performed on the sub-blocks to generate sub-residual blocks. A reconstructed sub-block may be generated by adding the sub-prediction block to the sub-residual block. The reconstructed sub-block may be used as a reference sample for intra prediction of the sub-sub-blocks. The sub-block may be a block including a

predetermined number (for example, 16) or more samples. Accordingly, for example, when the current block is an 8×4 block or a 4×8 block, the current block may be partitioned into two sub-blocks. Also, when the current block is a 4×4 block, the current block may not be partitioned into sub-blocks. When the current block has other sizes, the current block may be partitioned into four sub-blocks. Information on whether or not to perform the intra prediction based on the sub-blocks and/or the partitioning direction (horizontal or vertical) may be signaled. The intra prediction based on the sub-blocks may be limited to be performed only when reference sample line 0 is used. When the intra prediction based on the sub-block is performed, filtering for the prediction block, which will be described later, may not be performed.

The final prediction block may be generated by performing filtering on the prediction block that is intra-predicted. The filtering may be performed by applying predetermined weights to the filtering target sample, the left reference sample, the upper reference sample, and/or the upper left reference sample. The weight and/or the reference sample (range, position, etc.) used for the filtering may be determined on the basis of at least one of a block size, an intra prediction mode, and a position of the filtering target sample in the prediction block. The filtering may be performed only in the case of a predetermined intra prediction mode (e.g., DC, planar, vertical, horizontal, diagonal, and/or adjacent diagonal modes). The adjacent diagonal mode may be a mode in which k is added to or subtracted from the diagonal mode. For example, k may be a positive integer of 8 or less.

An intra-prediction mode of a current block may be entropy encoded/decoded by predicting an intra-prediction mode of a block present adjacent to the current block. When intra-prediction modes of the current block and the neighbor block are identical, information that the intra-prediction modes of the current block and the neighbor block are identical may be signaled by using predetermined flag information. In addition, indicator information of an intra-prediction mode that is identical to the intra-prediction mode of the current block among intra-prediction modes of a plurality of neighbor blocks may be signaled. When intra-prediction modes of the current block and the neighbor block are different, intra-prediction mode information of the current block may be entropy encoded/decoded by performing entropy encoding/decoding based on the intra-prediction mode of the neighbor block.

FIG. 5 is a diagram illustrating an embodiment of an inter-picture prediction process.

In FIG. 5, a rectangle may represent a picture. In FIG. 5, an arrow represents a prediction direction. Pictures may be categorized into intra pictures (I pictures), predictive pictures (P pictures), and Bi-predictive pictures (B pictures) according to the encoding type thereof.

The I picture may be encoded through intra-prediction without requiring inter-picture prediction. The P picture may be encoded through inter-picture prediction by using a reference picture that is present in one direction (i.e., forward direction or backward direction) with respect to a current block. The B picture may be encoded through inter-picture prediction by using reference pictures that are preset in two directions (i.e., forward direction and backward direction) with respect to a current block. When the inter-picture prediction is used, the encoder may perform inter-picture prediction or motion compensation and the decoder may perform the corresponding motion compensation.

Hereinbelow, an embodiment of the inter-picture prediction will be described in detail.

The inter-picture prediction or motion compensation may be performed using a reference picture and motion information.

Motion information of a current block may be derived during inter-picture prediction by each of the encoding apparatus **100** and the decoding apparatus **200**. The motion information of the current block may be derived by using motion information of a reconstructed neighboring block, motion information of a collocated block (also referred to as a col block or a co-located block), and/or a block adjacent to the co-located block. The co-located block may mean a block that is located spatially at the same position as the current block, within a previously reconstructed collocated picture (also referred to as a col picture or a co-located picture). The co-located picture may be one picture among one or more reference pictures included in a reference picture list.

The derivation method of the motion information may be different depending on the prediction mode of the current block. For example, a prediction mode applied for inter prediction includes an AMVP mode, a merge mode, a skip mode, a merge mode with a motion vector difference, a subblock merge mode, a geometric partitioning mode, an combined inter intra prediction mode, affine mode, and the like. Herein, the merge mode may be referred to as a motion merge mode.

For example, when the AMVP is used as the prediction mode, at least one of motion vectors of the reconstructed neighboring blocks, motion vectors of the co-located blocks, motion vectors of blocks adjacent to the co-located blocks, and a (0, 0) motion vector may be determined as motion vector candidates for the current block, and a motion vector candidate list is generated by using the motion vector candidates. The motion vector candidate of the current block can be derived by using the generated motion vector candidate list. The motion information of the current block may be determined based on the derived motion vector candidate. The motion vectors of the collocated blocks or the motion vectors of the blocks adjacent to the collocated blocks may be referred to as temporal motion vector candidates, and the motion vectors of the reconstructed neighboring blocks may be referred to as spatial motion vector candidates.

The encoding apparatus **100** may calculate a motion vector difference (MVD) between the motion vector of the current block and the motion vector candidate and may perform entropy encoding on the motion vector difference (MVD). In addition, the encoding apparatus **100** may perform entropy encoding on a motion vector candidate index and generate a bitstream. The motion vector candidate index may indicate an optimum motion vector candidate among the motion vector candidates included in the motion vector candidate list. The decoding apparatus may perform entropy decoding on the motion vector candidate index included in the bitstream and may select a motion vector candidate of a decoding target block from among the motion vector candidates included in the motion vector candidate list by using the entropy-decoded motion vector candidate index. In addition, the decoding apparatus **200** may add the entropy-decoded MVD and the motion vector candidate extracted through the entropy decoding, thereby deriving the motion vector of the decoding target block.

Meanwhile, the coding apparatus **100** may perform entropy-coding on resolution information of the calculated

MVD. The decoding apparatus **200** may adjust the resolution of the entropy-decoded MVD using the MVD resolution information.

Meanwhile, the coding apparatus **100** calculates a motion vector difference (MVD) between a motion vector and a motion vector candidate in the current block on the basis of an affine model, and performs entropy-coding on the MVD. The decoding apparatus **200** derives a motion vector on a per sub-block basis by deriving an affine control motion vector of a decoding target block through the sum of the entropy-decoded MVD and an affine control motion vector candidate.

The bitstream may include a reference picture index indicating a reference picture. The reference picture index may be entropy-encoded by the encoding apparatus **100** and then signaled as a bitstream to the decoding apparatus **200**. The decoding apparatus **200** may generate a prediction block of the decoding target block based on the derived motion vector and the reference picture index information.

Another example of the method of deriving the motion information of the current may be the merge mode. The merge mode may mean a method of merging motion of a plurality of blocks. The merge mode may mean a mode of deriving the motion information of the current block from the motion information of the neighboring blocks. When the merge mode is applied, the merge candidate list may be generated using the motion information of the reconstructed neighboring blocks and/or the motion information of the collocated blocks. The motion information may include at least one of a motion vector, a reference picture index, and an inter-picture prediction indicator. The prediction indicator may indicate one-direction prediction (L0 prediction or L1 prediction) or two-direction predictions (L0 prediction and L1 prediction).

The merge candidate list may be a list of motion information stored. The motion information included in the merge candidate list may be at least one of motion information (spatial merge candidate) of a neighboring block adjacent to the current block, motion information (temporal merge candidate) of the collocated block of the current block in the reference picture, new motion information generated by a combination of the motion information existing in the merge candidate list, motion information (history-based merge candidate) of the block that is encoded/decoded before the current block, and zero merge candidate.

The encoding apparatus **100** may generate a bitstream by performing entropy encoding on at least one of a merge flag and a merge index and may signal the bitstream to the decoding apparatus **200**. The merge flag may be information indicating whether or not to perform the merge mode for each block, and the merge index may be information indicating that which neighboring block, among the neighboring blocks of the current block, is a merge target block. For example, the neighboring blocks of the current block may include a left neighboring block on the left of the current block, an upper neighboring block disposed above the current block, and a temporal neighboring block temporally adjacent to the current block.

Meanwhile, the coding apparatus **100** performs entropy-coding on the correction information for correcting the motion vector among the motion information of the merge candidate and signals the same to the decoding apparatus **200**. The decoding apparatus **200** can correct the motion vector of the merge candidate selected by the merge index on the basis of the correction information. Here, the correction information may include at least one of information on whether or not to perform the correction, correction direc-

tion information, and correction size information. As described above, the prediction mode that corrects the motion vector of the merge candidate on the basis of the signaled correction information may be referred to as a merge mode having the motion vector difference.

The skip mode may be a mode in which the motion information of the neighboring block is applied to the current block as it is. When the skip mode is applied, the encoding apparatus 100 may perform entropy encoding on information of the fact that the motion information of which block is to be used as the motion information of the current block to generate a bit stream, and may signal the bitstream to the decoding apparatus 200. The encoding apparatus 100 may not signal a syntax element regarding at least any one of the motion vector difference information, the encoding block flag, and the transform coefficient level to the decoding apparatus 200.

The subblock merge mode may mean a mode that derives the motion information in units of sub-blocks of a coding block (CU). When the subblock merge mode is applied, a subblock merge candidate list may be generated using motion information (sub-block based temporal merge candidate) of the sub-block collocated to the current sub-block in the reference image and/or an affine control point motion vector merge candidate.

The geometric partitioning mode may mean a mode that derives motion information by partitioning the current block into the predefined directions, derives each prediction sample using each of the derived motion information, and derives the prediction sample of the current block by weighting each of the derived prediction samples.

The inter-intra combined prediction mode may mean a mode that derives a prediction sample of the current block by weighting a prediction sample generated by inter prediction and a prediction sample generated by intra prediction.

The decoding apparatus 200 may correct the derived motion information by itself. The decoding apparatus 200 may search the predetermined region on the basis of the reference block indicated by the derived motion information and derive the motion information having the minimum SAD as the corrected motion information.

The decoding apparatus 200 may compensate a prediction sample derived via inter prediction using an optical flow.

FIG. 6 is a diagram illustrating a transform and quantization process.

As illustrated in FIG. 6, a transform and/or quantization process is performed on a residual signal to generate a quantized level signal. The residual signal is a difference between an original block and a prediction block (i.e., an intra prediction block or an inter prediction block). The prediction block is a block generated through intra prediction or inter prediction. The transform may be a primary transform, a secondary transform, or both. The primary transform of the residual signal results in transform coefficients, and the secondary transform of the transform coefficients results in secondary transform coefficients.

At least one scheme selected from among various transform schemes which are preliminarily defined is used to perform the primary transform. For example, examples of the predefined transform schemes include discrete cosine transform (DCT), discrete sine transform (DST), and Karhunen-Loève transform (KLT). The transform coefficients generated through the primary transform may undergo the secondary transform. The transform schemes used for the primary transform and/or the secondary transform may be determined according to coding parameters of the current block and/or neighboring blocks of the current block. Alter-

natively, transform information indicating the transform scheme may be signaled. The DCT-based transform may include, for example, DCT-2, DCT-8, and the like. The DST-based transform may include, for example, DST-7.

A quantized-level signal (quantization coefficients) may be generated by performing quantization on the residual signal or a result of performing the primary transform and/or the secondary transform. The quantized level signal may be scanned according to at least one of a diagonal up-right scan, a vertical scan, and a horizontal scan, depending on an intra prediction mode of a block or a block size/shape. For example, as the coefficients are scanned in a diagonal up-right scan, the coefficients in a block form change into a one-dimensional vector form. Aside from the diagonal up-right scan, the horizontal scan of horizontally scanning a two-dimensional block form of coefficients or the vertical scan of vertically scanning a two-dimensional block form of coefficients may be used depending on the intra prediction mode and/or the size of a transform block. The scanned quantized-level coefficients may be entropy-encoded to be inserted into a bitstream.

A decoder entropy-decodes the bitstream to obtain the quantized-level coefficients. The quantized-level coefficients may be arranged in a two-dimensional block form through inverse scanning. For the inverse scanning, at least one of a diagonal up-right scan, a vertical scan, and a horizontal scan may be used.

The quantized-level coefficients may then be dequantized, then be secondary-inverse-transformed as necessary, and finally be primary-inverse-transformed as necessary to generate a reconstructed residual signal.

Inverse mapping in a dynamic range may be performed for a luma component reconstructed through intra prediction or inter prediction before in-loop filtering. The dynamic range may be divided into 16 equal pieces and the mapping function for each piece may be signaled. The mapping function may be signaled at a slice level or a tile group level. An inverse mapping function for performing the inverse mapping may be derived on the basis of the mapping function. In-loop filtering, reference picture storage, and motion compensation are performed in an inverse mapped region, and a prediction block generated through inter prediction is converted into a mapped region via mapping using the mapping function, and then used for generating the reconstructed block. However, since the intra prediction is performed in the mapped region, the prediction block generated via the intra prediction may be used for generating the reconstructed block without mapping/inverse mapping.

When the current block is a residual block of a chroma component, the residual block may be converted into an inverse mapped region by performing scaling on the chroma component of the mapped region. The availability of the scaling may be signaled at the slice level or the tile group level. The scaling may be applied only when the mapping for the luma component is available and the division of the luma component and the division of the chroma component follow the same tree structure. The scaling may be performed on the basis of an average of sample values of a luma prediction block corresponding to the color difference block. In this case, when the current block uses inter prediction, the luma prediction block may mean a mapped luma prediction block. A value necessary for the scaling may be derived by referring to a lookup table using an index of a piece to which an average of sample values of a luma prediction block belongs. Finally, by scaling the residual block using the derived value, the residual block may be switched to the inverse mapped region. Then, chroma component block

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restoration, intra prediction, inter prediction, in-loop filtering, and reference picture storage may be performed in the inverse mapped area.

Information indicating whether the mapping/inverse mapping of the luma component and chroma component is available may be signaled through a set of sequence parameters.

The prediction block of the current block may be generated on the basis of a block vector indicating a displacement between the current block and the reference block in the current picture. In this way, a prediction mode for generating a prediction block with reference to the current picture is referred to as an intra block copy (IBC) mode. The IBC mode may be applied to $M \times N$ ($M \leq 64$, $N \leq 64$) coding units. The IBC mode may include a skip mode, a merge mode, an AMVP mode, and the like. In the case of a skip mode or a merge mode, a merge candidate list is constructed, and the merge index is signaled so that one merge candidate may be specified. The block vector of the specified merge candidate may be used as a block vector of the current block. The merge candidate list may include at least one of a spatial candidate, a history-based candidate, a candidate based on an average of two candidates, and a zero-merge candidate. In the case of an AMVP mode, the difference block vector may be signaled. In addition, the prediction block vector may be derived from the left neighboring block and the upper neighboring block of the current block. The index on which neighboring block to use may be signaled. The prediction block in the IBC mode is included in the current CTU or the left CTU and limited to a block in the already reconstructed area. For example, a value of the block vector may be limited such that the prediction block of the current block is positioned in an area of three 64×64 blocks preceding the 64×64 block to which the current block belongs in the coding/decoding order. By limiting the value of the block vector in this way, memory consumption and device complexity according to the IBC mode implementation may be reduced.

FIG. 8a is a flowchart illustrating an image encoding method according to an embodiment of the present invention, and FIG. 8b is a flowchart illustrating an image decoding method according to an embodiment of the present invention.

Hereinafter, an encoding/decoding method of the present disclosure will be described.

A single picture may be encoded using at least one of an intra prediction, inter prediction or intra block copy prediction method.

An intra block copy prediction based encoding/decoding method may be used when a luma component and a chroma component have independent block partitioning structures (that is, a dual tree structure) or when a luma component and a chroma component have the same block partitioning structure (that is, a single tree structure).

The intra block copy prediction method may mean a method of deriving a prediction block from an encoded/decoded region in the same picture (that is, intra) using a derived block vector.

When a block to be currently encoded/decoded (that is, a current block) is encoded/decoded using the intra block copy prediction method and a derived block vector is (x, y) , a block having the same size as the current block, which is separated from the current block by x pixels in a horizontal direction (that is, x rightward in the horizontal direction if x is a positive integer and $-x$ leftward in the horizontal direction if x is a negative integer) and y pixels in a vertical direction (that is, y downward in the vertical direction if y

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is a positive integer and $-y$ upward in the vertical direction if y is a negative integer), may be used as a prediction block of the current block. For example, FIG. 9 is a view illustrating a relation between a current block and a prediction block according to an embodiment of the present invention. Referring to FIG. 9, when x and y are negative integers and the top left sample position of the current block is (x_0, y_0) , the top left sample position of the prediction block of the current block is (x_0+x, y_0+y) .

If a current luma component block uses intra block copy prediction, the block may be encoded/decoded using one of the following methods.

Intra block copy based SKIP mode in which the block vector of a current block is derived from the block vector of a block encoded/decoded before the current block, similar to a skip mode in an intra prediction method, and a residual signal is not present

Intra block copy based MERGE mode in which the block vector of a current block is derived from the block vector of a block encoded/decoded before the current block, similar to a skip mode in a merge mode of an inter prediction method, and a residual signal is present

Intra block copy based AMVP mode in which a block vector is encoded similar to an AMVP mode of the inter prediction method

According to an embodiment, the encoding mode of the current luma component block in a decoder may be derived as follows.

At least one of the below-described encoding information may be used to determine the encoding mode of the current luma component block, and at least one of the encoding information may be included and transmitted in a bitstream.

The encoding information may include information indicating that the luma component block is in a SKIP mode (e.g., a SKIP mode identifier, a flag, an index, skip_flag, cu_skip_flag, etc.).

When information indicating the SKIP mode has a specific value, this indicates that the luma component block is in the SKIP mode. For example, if the identifier, the flag or the index has 1 which is a first value, this indicates that the luma component block is in the SKIP mode and, if the identifier, the flag or the index has 0 which is a second value, this indicates that the luma component block is not in the SKIP mode.

The encoding information may include the prediction mode information (e.g., an index, an identifier, a flag, etc.) of the luma component block. the prediction mode information may include an intra prediction mode, an inter prediction mode, an intra block copy

For example, the syntax indicating the prediction mode information having a first value of 0 indicates that an intra prediction mode is applied, the syntax having a second value of 1 indicates that an inter prediction mode is applied, and the syntax having a third value of 2 indicates that an intra block copy prediction mode is applied.

In addition, for example, first prediction mode information (e.g., index, flag, identifier, pred_mode_flag, etc.) may indicate an intra prediction mode. The first prediction mode information having a first value of 1 indicates that an intra prediction mode is applied and the first prediction mode information having a second value of 0 indicates that an intra prediction mode is not applied. If the intra prediction mode is not applied, second prediction mode information (e.g., index, flag, identifier, pred_mode_ibc_flag, etc.) may be received to indicate whether an inter prediction mode or an intra block copy prediction is applied. The second prediction mode information having a first value of 1 may indicate that

an intra block copy prediction mode is applied and the second prediction mode information having a second value of 0 may indicate that an inter prediction mode is applied.

The encoding information may include information indicating that the luma component block is in a Merge mode (e.g., a MERGE mode identifier, a flag, an index, merge_flag, etc.).

When the current luma component block is not in the SKIP mode but is in an intra block copy mode, the MERGE mode having a specific mode may indicate a MERGE mode. For example, the identifier, flag or index having a first value of 1 may indicate that a MERGE mode is applied, and the identifier, flag or index having a second value of 0 may indicate that the MERGE mode is not applied.

The encoding mode of the current luma component block may be determined using the encoding information as follows.

For example, when the luma component block is in the SKIP mode and a tile group, slice or tile is of Type I, the prediction mode information may not be received and the block may be determined as an intra block copy based SKIP mode. For example, a prediction mode applicable when a tile group, slice or tile is of Type I includes an intra prediction mode and an intra block copy prediction mode and the SKIP mode may not be present in the intra prediction mode. Accordingly, when the luma component block is in a SKIP mode and the corresponding tile group, slice or tile is of Type I, the corresponding block may be determined as an intra block copy based SKIP mode.

In addition, for example, when the luma component block is in the SKIP mode and the tile group, slice or tile is not of Type I, the prediction mode information may be received. At this time, upon determining that the luma component block is in an intra block copy prediction mode based on the prediction mode information, it may be determined that the luma block is in the intra block copy based SKIP mode.

In addition, for example, when the luma component block is not in the SKIP mode and it is determined that, based on the prediction mode information, the luma component block is in the intra block copy prediction mode, information indicating a MERGE mode may be received. Alternatively, for example, when the luma component block is not in the SKIP mode, information indicating the MERGE mode may be received. At this time, when the information indicating the MERGE mode indicates that the luma component block is in the MERGE mode, it may be determined that the luma component block is in an intra block copy based MERGE mode.

In addition, for example, when the luma component block is not in the SKIP mode and is not in the MERGE mode but is in the intra block copy prediction mode, it may be determined that the luma component block is in an intra block copy AMVP mode. For example, whether the prediction mode for the luma component block is in the intra block copy based MERGE mode or the intra block copy based AMVP mode may be determined, based on information indicating the MERGE mode. Here, the information indicating the MERGE mode may indicate whether the inter prediction encoding parameter of the current block is derived from the inter prediction encoding parameter of the neighbor block adjacent to the current block.

In a decoder, the current luma component block or the encoding mode of the current chroma component block may be derived as follows.

When the luma component and the chroma component has the same block partitioning structure (that is, single tree structure), the encoding mode may be determined as described below.

For example, the prediction mode (e.g., intra prediction, inter prediction or intra block copy prediction) of the chroma component block may be equal to the prediction mode of the luma component block corresponding thereto.

In addition, for example, when the corresponding luma component block is in an intra block copy based SKIP mode, a residual signal may not be encoded/decoded and transmitted in a chroma component block. At this time, information (e.g., cu_cbf, tu_cbf, etc.) indicating that the residual signal is not transmitted in a bitstream.

In addition, for example, when the luma component and the chroma component have the same block partitioning structure and a current chroma component block is in an intra block copy prediction mode (or when the luma component block corresponding to the current chroma component block is in an intra block copy prediction mode), information necessary to encode/decode the current chroma component block may be derived from the encoding/decoding information of the luma component block corresponding to the current chroma component block.

When the luma component and the chroma component have independent block partitioning structure (That is, dual tree structure), the prediction mode of the chroma component block may be determined from the prediction mode information of the chroma component block included and transmitted in the bitstream. The prediction mode of the chroma component block may include an intra prediction mode, an inter prediction mode, an intra block copy prediction mode, etc., as in the prediction mode of the luma component block.

For example, the syntax indicating the prediction mode information having a first value of 0 may indicate an intra prediction mode, the syntax having a second value of 1 may indicate an inter prediction mode, and the syntax having a third value of 2 may indicate an intra block copy prediction mode.

In addition, for example, the first prediction mode information (e.g., index, flag, identifier, pred_mode_flag, etc.) may indicate whether the prediction mode of the chroma component block is an intra prediction mode or an inter prediction mode. The first prediction mode information having a first value of 1 may indicate an intra prediction mode and the first prediction mode information having a second value of 0 may indicate an inter prediction mode. Additionally, the second prediction mode information (e.g., index, flag, identifier, pred_mode_ibc_flag, etc.) may be transmitted or derived, the second prediction mode information having a first value of 1 may indicate that an intra block copy mode is applied and the second prediction mode information having a second value of 0 may indicate that the prediction mode of the chroma component block is determined as an intra prediction mode or an inter prediction mode determined in the first prediction mode information.

In addition, for example, the second prediction mode information (e.g., index, flag, identifier, pred_mode_ibc_flag, etc.) may be transmitted or derived, the second prediction mode information having a first value of 1 may indicate an intra block copy mode and the second prediction mode information having a second value of 0 may indicate an intra prediction mode.

When the luma component and the chroma component have independent block partitioning structures and the current chroma component block is in an intra block copy

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prediction mode, information (e.g., a block vector, etc.) necessary to encode/decode the current chroma component block may be derived from the encoding/decoding information of the luma component block corresponding to the current chroma component block.

Hereinafter, step of deriving a block vector for intra block copy prediction will be described.

Step of deriving the block vector for intra block copy prediction may include at least one of step of deriving the block vector of the luma component block and step of deriving the block vector of the chroma component block.

Hereinafter, step of deriving the block vector of the luma component block will be described.

According to an embodiment, a method of deriving a block vector when the current block is a luma component block and is encoded in an intra block copy based SKIP mode or an intra block copy based MERGE mode will be described below.

In order to derive the block vector of the luma component block, a block vector candidate list may be constructed from block vector candidates of the luma component block encoded/decoded before the current block and one of the candidates in the constructed block vector candidate list may be used as the block vector of the current block. At this time, information (e.g., identifier, index, flag, merge_idx, etc.) for identifying the candidate in the block vector candidate list may be derived or encoded/decoded based on the encoding parameter transmitted in the bitstream.

The block vector candidate list may be composed of a maximum of N candidates. At this time, N may be a positive integer including 0. In addition, one or more candidates described below may be included in the block vector candidate list.

FIG. 10 is a view illustrating neighbor blocks adjacent to a current block according to an embodiment of the present invention. Referring to FIG. 10, a block vector may be derived from at least one of a block B1 adjacent to the upper end of the current block X, a block A1 adjacent to the left side of the current block, a block B0 located at the top corner of the current block, a block B2 located at the top left corner of the current block or a block A0 located at the bottom left corner of the current block, and the derived block vector may be determined as the block vector candidate of the current block.

For example, the block B2 located at the top left corner of the current block may not be used as the block vector candidate, if block vectors are present in all blocks A1, B1, B0 and A0.

In addition, for example, whether a block vector is present in each of the blocks A0, A1, B0, B1 and B2 according to predetermined order of priority (that is, whether the block is encoded/decoded using the intra block copy prediction method) may be determined and, if the block vector is present, the block vector of the block may be determined as the vector candidate. At this time, the predetermined order of priority of the block vector candidate list may be A1, B1, B0, A0 and B2.

The block vector candidate list may be constructed by the predetermined order of priority and redundancy check between the block vector candidate present in the block vector candidate list and a newly added block vector candidate may be performed.

For example, when the block vector candidate list is constructed in order of A1, B1, B0, A0 and B2, redundancy check between the block B1 and the block A1 may be performed and redundancy the block B0 and the block B1 may be performed. In addition, redundancy check between

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the block A0 and the block A1 may be performed and redundancy the block B2 and the blocks A1 and B1 may be performed. Redundancy check may be performed only when the block vector is present in the block.

In addition, for example, redundancy check may be performed between block vectors to be added and all block vectors present in the block vector candidate list.

When a block vector is present in at least one of the blocks A0, A1, B0, B1 and B2, whether the block vector of the block is available in the current block may be determined and, if available, the block vector of the neighbor block may be determined as a block vector candidate. If not available, the block vector of the neighbor block may not be used as a block vector candidate. At this time, whether the block vector is available may be determined depending on whether a reference block (block) at a position indicated by the block vector is available.

The block vectors of the blocks encoded/decoded before the current block may be stored in a buffer and at least one of the block vectors stored in the buffer may be determined as a block vector candidate. At this time, an intra block vector may be stored in a buffer having a specific size in order of encoding/decoding and, if the buffer is full, a first stored block vector may be deleted and a new (that is, a recently encoded/decoded) block vector may be stored. The order of priority may be changed when the block vector candidate list may be constructed in order of the block vectors stored in the buffer (e.g., from oldest to newest or from newest to oldest). For example, the block vectors may be included in the block vector candidate list from a block vector most recently stored in the buffer to a block vector first stored in the buffer or from a block vector first stored in the buffer to a block vector most recently stored in the buffer. The block vector candidate may be referred to as a history-based block vector candidate.

When the block vector candidate list is constructed using at least one of the history-based block vector candidates, whether the history-based block vector candidate is available in the current block, and, if available, the candidate may be added to the block vector candidate list. At this time, whether the history-based block vector is available may be determined depending on whether a reference sample (block) at a position indicated by the block vector is available.

When the block vector candidate list is constructed using at least one of the history-based block vector candidates, redundancy check may be performed between the history-based block vector candidate and the block vector candidates in the block vector candidate list and, if the same block vector is not present as the result of redundancy check, the history-based block vector candidate may be added to the block vector candidate list.

For example, when the block vector candidate list is constructed using at least one of the history-based block vector candidates, a predetermined candidate of the history-based block vector candidates may be added to the block vector candidate list without performing redundancy check with the block vector candidates in the block vector candidate list. For example, the predetermined candidate may mean a block vector candidate except for a first candidate of the history-based block vector candidates. Here, the first candidate may mean a block vector candidate which is first or most recently stored in the history-based block vector list composed of the history-based block vector candidates.

In addition, for example, when the block vector candidate list is constructed using at least one of the history-based block vector candidates, a predetermined candidate of the

history-based block vector candidates may be subjected to redundancy check with the block vector candidates in the block vector candidate list. When the same block vector is not present as the result of performing redundancy check, the predetermined candidate may be added to the block vector candidate list. For example, the predetermined candidate may mean the first candidate of the history-based block vector candidates. Here, the first candidate may mean a block vector candidate, which is most recently stored, in the history-based block vector list composed of the history-based block vector candidates.

The buffer including the history-based block vector candidates may be maintained during encoding/decoding in units of pictures, slices, tiles, CTUs, CTU rows or CTU columns and used in units of pictures, slices, tiles, CTUs, CTU rows or CTU columns. In addition, the buffer may include at least one of encoding information blocks encoded/decoded before the current block in units of pictures, slices, tiles, CTUs, CTU rows or CTU columns.

When a combined block vector candidate may be constructed using at least two of the block vector candidates in the block vector candidate list. At this time, when the combined block vector candidate is constructed, the history-based block vector candidates may not be used. At this time, when the combined block vector candidate is constructed, the block vector candidates of adjacent neighbor blocks may not be used. At this time, whether the combined block vector candidate composed of the block vector candidates is available may be determined and, only if available, the combined block vector candidate may be used. At this time, whether the block vector is available may be determined depending on whether the reference sample (block) at a position indicated by the block vector is available.

If the width of the current luma block is W and the height of the current luma block is H , $(-(W < n) + a, -(H < n) + b)$, $(-(W < n) + c, 0)$ or $(0, -(H < n) + d)$ may be included in the block vector candidate list as block vector candidates. At this time, n may be a positive integer greater than 0, and a , b , c and d may have an integer value. The block vector candidate may be referred to as a fixed base block vector candidate.

The block vector candidate list may be constructed in predetermined order using at least one of the block vectors of the adjacent neighbor block candidates, the history-based block vector candidates, the combined block vector candidates and the fixed base block vector candidates.

Here, the adjacent neighbor block may be at least one of A0 or A1 and at least one of B0, B1 or B2, referring to FIG. 10.

For example, the block vector candidate list may be constructed in order of the block vectors of the adjacent neighbor block candidates, the history-based block vector candidates, the combined block vector candidates and the fixed base block vector candidates.

In addition, for example, the fixed base block vector may be constructed in the following order until the number of candidates in the block vector candidate list satisfies a maximum number (or a maximum number of block candidates).

1. $(-(W < 1), 0)$
2. $(0, -(H < 1))$
3. $(-(W < 1) - 1, 0)$
4. $(0, -(H < 1) - 1)$
5. $(-(W < 1) - 2, 0)$
6. $(0, -(H < 1) - 2)$
7. $(-(W < 1) - 3, 0)$

8. $(0, -(H < 1) - 3)$
9. $(-(W < 1) - 4, 0)$
10. $(0, -(H < 1) - 4)$

Alternatively, the fixed base block vector may be a vector $(0, 0)$, and a block vector candidate list having a maximum number of candidates may be constructed by adding the fixed base block vector until the number of candidates in the block vector candidate list satisfies the maximum number. For example, the fixed base block vector may be added to the block vector candidate list until the maximum number of block vector candidates is satisfied, if the number of block vector candidates added to the block vector candidate list using the block vector of the adjacent neighbor block, the history-based block vector, the combined block vector, etc. is less than the maximum number N of block vector candidates. At this time, the fixed base block vector may be a vector $(0, 0)$.

When the block vector candidate list is constructed, the number of history-based block vector candidates which may be included in the block vector candidate list may be the maximum number N of block vector candidates or $(N - m)$. m may be an integer greater than 0.

When an upper block is partitioned and each block is encoded/decoded in an intra block copy based SKIP mode or an intra block copy based MERGE mode, if at least one of the blocks partitioned from the upper block is less than a predetermined threshold or a critical value, the partitioned blocks may share and use the block vector candidate list constructed in the upper block.

In determination as to whether the block vector candidate list constructed in the upper block is shared, the block vector candidate list constructed in the upper block may be used for the partitioned blocks when the following conditions are satisfied using the width and height of the upper block.

Quadtree partitioning: (the width of the upper block $\times$ the height of the upper block) / 4 $<$ threshold

Horizontal or vertical binary tree partitioning: (the width of the upper block $\times$ the height of the upper block) / 2 $<$ threshold

Ternary tree: (the width of the upper block $\times$ the height of the upper block) / 4 $<$ threshold

The threshold may be predetermined in the coder/decoder or signaled from the encoder to the decoder.

FIG. 11 is a view illustrating a current block partitioned when a predetermined threshold is 32 according to an embodiment of the present invention.

Referring to FIG. 11, if the area of a lower block of at least one of the quadtree partitioning, vertical or horizontal binary partitioning, ternary tree of the upper block is less than 32, the above condition is satisfied and each lower block is encoded/decoded in an intra block copy based SKIP mode or an intra block copy based MERGE mode, the lower blocks may be encoded/decoded using a block vector candidate list including history-based block vector candidates encoded/decoded before the upper block and the block vectors of the neighbor blocks (e.g., A1, B1, B0, A0 and B2) at the upper block position and stored in the buffer and fixed base block vectors derived from the width and height of the upper block.

When the width and height of the current block is equal to or less than a predetermined value, a predetermined vector candidate may not be allowed as a block vector candidate.

Referring to FIG. 11, when the width and height of the current is 4×4 (or when the product of the width and height of the current block is less than or equal to 16), the block vector of the neighbor block may not be allowed as a block vector candidate.

At this time, the block vector candidate list may be constructed using at least one of a history-based block vector, a combined block vector or a fixed basic block vector. For example, the block vector candidate list may be constructed using only the history-based block vector or using only the history-based block vector and the fixed basic block vector.

In addition, at this time, the update process of the history-based block vector candidate list may not be performed. For example, when the width and height of the current is 4×4 (or when the product of the width and height of the current block is less than or equal to 16), the encoded/decoded block vector of the current block may not be added to the history-based block vector candidate list.

In addition, the block vector candidate list may be constructed using the encoded/decoded history-based block vector candidates in a previous upper block.

When the block vector candidate list constructed in the upper block is shared and used, a prediction block indicated by the block vector of the lower block may not be located in the upper block.

For example, FIG. 12 is a view illustrating a process of sharing and using a block vector candidate list constructed in an upper block according to an embodiment of the present invention. Referring to FIG. 12, when the block vector candidate list at the upper block position is shared and used, it may be determined that the block vector BV in which at least one of the lower blocks is encoded/decoded in the intra prediction block copy mode is valid only when the prediction block is present in a region encoded/decoded before the upper block.

When the block vector candidate list constructed in the upper block is shared and used, at least one of the partitioned lower blocks may have one of an intra block copy based SKIP mode, an intra block copy based MERGE mode or an intra block copy based AMVP mode.

When the block vector candidate list constructed in the upper block is shared and used, at least one of the partitioned lower blocks may be encoded/decoded only in an intra block copy based SKIP mode, an intra block copy based MERGE mode, an intra block copy based AMVP mode or an AMVP mode using a motion vector.

When a merge candidate list constructed in the upper block is shared and used, the partitioned lower blocks may not be encoded/decoded in an intra block copy based SKIP and intra block copy based MERGE mode. Although the partitioned lower blocks may be encoded/decoded in an intra block AMVP mode, it may be determined that the block vector of the block is valid only when a prediction block acquired from the block vector is present in a region encoded/decoded before the upper block.

The merge candidate list may mean a list including a motion vector and a temporal motion vector, a history-based motion vector, a combined motion vector and a zero vector, not the block vectors of the adjacent neighbor blocks of the upper block.

When the width and/or height of the current block is equal to or less than a predetermined value, an intra prediction block copy SKIP mode and an intra prediction block MERGE mode may not be allowed.

For example, when the width and height of the current block is less than 8, intra block copy based SKIP mode and an intra prediction block MERGE mode may not be allowed.

If the value of the width W the height H of the current block is less than or equal to the threshold condition in which the merge candidate list constructed in the upper

block may be used in the lower block, the intra block copy based SKIP mode and the intra block copy based MERGE mode may not be allowed.

For example, if the threshold in which the merge candidate list constructed in the upper block may be used in the lower block is 32, the intra block copy based SKIP mode and the intra block copy based MERGE mode may be allowed only when the width the height H of the current block is greater than 32.

When the value of the width W the height H of the current block is less than or equal to the threshold condition in which the block vector candidate list constructed in the upper block may be used in the lower block, the SKIP mode and the MERGE mode may not be allowed. The SKIP mode and the MERGE mode may mean modes encoded/decoded with a motion vector rather than the block vector of the spatial/temporal neighbor block of the current block.

For example, if the threshold condition in which the block vector candidate list constructed in the upper block may be used in the lower block is 32, the SKIP mode and MERGE mode based on the motion vector rather than the block vector may be allowed only when the width the height H of the current block is greater than 32.

A combined merge candidate list may be composed of motion vectors and block vectors of the adjacent neighbor block of the upper block, and, when a value derived based on the width and/or height of at least one of the blocks partitioned from the upper block is less than a predetermined threshold, the combined merge candidate list constructed in the upper block may be shared and used.

A combined merge candidate list may be constructed using at least one of the neighbor block motion vector candidate of the upper block, the block vector candidate of the neighbor block, the temporal motion vector candidate, the history-based motion vector candidate, the history-based block vector candidate, the (0,0) motion vector candidate or the fixed base block vector and the combined merge candidate list may be shared and used.

For example, when the block is encoded/decoded in the intra prediction block copy SKIP mode or the intra prediction block MERGE mode, candidates corresponding to the block vector in the combined merge candidate list may be encoded/decoded. At this time, when encoding/decoding is performed in the intra prediction block copy SKIP mode or the intra prediction block MERGE mode, information for identifying the corresponding candidate in the combined merge candidate list may indicate only the block vector candidate.

In addition, for example, when the corresponding block is encoded/decoded in the SKIP mode or the MERGE mode, only candidates corresponding to the motion vector rather than the block vector in the combined merge candidate list may be encoded/decoded. At this time, when encoding/decoding is performed in the SKIP mode or the MERGE mode, information for identifying the corresponding candidate in the combined merge candidate list may indicate only a motion vector candidate.

According to an embodiment, when the current block is a luma component block and is encoded in an intra block copy based AMVP mode, a block vector derivation method is as follows.

Similar to the intra block copy based SKIP mode or MERGE mode, a prediction block vector candidate list may be constructed using a maximum of N prediction block vector candidates. One of the candidates included in the constructed prediction block vector candidate list may be used as the prediction block vector of the current block, and

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information (e.g., identifier, index, flag, mvp_l0_flag, etc.) for identifying the prediction candidate in the prediction block vector candidate list may be derived or encoded/decoded based on the encoding parameter transmitted in the bitstream.

A vector difference between the block vector of the current block and the prediction block vector may be calculated and the result of calculation may be entropy-encoded. The decoder may receive block vector difference information in the bitstream or derive the vector difference from the information transmitted in the bitstream and add the block vector difference and the prediction block vector of the current block to derive the block vector of the current block.

One or more candidates of the following may be included in the prediction block vector candidate list.

In FIG. 10, whether the block is encoded/decoded using the intra block copy prediction method in order of A0 and A1 may be determined and the block vector of the block encoded/decoded using the intra block copy prediction method may be determined as a prediction candidate A. Alternatively, whether the block corresponding to A1 is encoded/decoded using the intra block copy prediction method and, when encoding/decoding is performed using the intra block copy prediction method, the prediction candidate A may be determined.

In FIG. 10, whether the block is encoded/decoded using the intra block copy prediction method in order of B0, B1 and B2 may be determined, and the block vector of the block encoded/decoded using the intra block copy prediction method may be determined as a prediction candidate B. Alternatively, whether the block corresponding to B1 is encoded/decoded using the intra block copy prediction method may be determined and, when encoding/decoding is performed using the intra block copy prediction method, the prediction candidate B may be determined.

At this time, the predetermined order of the prediction block vector candidate list may be A and B.

The block vectors of the blocks encoded/decoded before the current block may be stored in a buffer and one or more of the block vectors stored in the buffer may be determined as the prediction block vector candidate. At this time, the block vector may be stored in a buffer having a specific size in order of encoding/decoding and, if the buffer is filled, the first stored block vector may be deleted and a new (that is, most recently encoded/decoded) block vector may be stored. Priority may be differentiated when the prediction block vector candidate list is constructed in order (e.g., in order from oldest to newest or in order from newest to oldest) of the block vectors stored in the buffer. For example, the block vector most recently stored in the buffer may be first included in the prediction block vector candidate list or the block vector first stored in the buffer may be first included in the prediction block vector candidate list. The block vector may be referred to as a history-based block vector candidate.

When the block vector candidate list is constructed using at least one of the history-based block vector candidates, whether the history-based block vector candidate is available in the current block may be determined and, if available, the candidate may be added to the block vector candidate list. At this time, whether the history-based block vector is available may be determined depending on whether the reference sample (block) at the position indicated by the block vector is available.

When the block vector candidate list is constructed using at least one of the history-based block vector candidates,

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redundancy check may be performed between the history-based block vector candidate and the block vector candidates present in the block vector candidate list, and, when the same block vector is not present as the result of redundancy check, the candidate may be added to the block vector candidate list.

For example, when the block vector candidate list is constructed using at least one of the history-based block vector candidates, redundancy check may not be performed between a predetermined candidate among the history-based block vector candidates and the block vector candidates present in the block vector candidate list and the candidate may be added to the block vector candidate list. For example, the predetermined candidate may mean a block vector candidate except for a first candidate among the history-based block vector candidates. Here, the first candidate may mean a first stored or most recently stored block vector candidate in the history-based block vector list composed of the history-based block vector candidates.

As another example, when the block vector candidate list is constructed using at least one of the history-based block vector candidates, redundancy check may be performed between a predetermined candidate among the history-based block vector candidates and the block vector candidates present in the block vector candidate list and the candidate may be added to the block vector candidate list. For example, the predetermined candidate may mean a first candidate among the history-based block vector candidates. Herein, the first candidate may mean a block vector candidate, which is most recently stored, in the history-based block vector list composed of the history-based block vector candidates.

The buffer including the history-based block vector candidates may be maintained during encoding/decoding in units of pictures, slices, tiles, CTUs, CTU rows, CTU columns and may be used in units of pictures, slices, tiles, CTUs, CTU rows, CTU columns. In addition, the buffer may include at least one of the encoding information of the block encoded/decoded before the current block in units of pictures, slices, tiles, CTUs, CTU rows, CTU columns.

The combined block vector candidate may be constructed using at least two of the block vector candidates present in the block vector candidate list. At this time, when the combined block vector candidate is constructed, the history-based block vector candidate may not be used. At this time, when the combined block vector candidate is constructed, the block vector candidates of the adjacent neighbor blocks may not be used. At this time, whether the combined block vector candidate composed of the block vector candidates is available in the current block may be determined and, only if available, this may be determined as the combined block vector candidate. At this time, whether the block vector is available may be determined depending on whether the reference sample (block) at a position indicated by the block vector is available.

When the width of the current luma block is W and the height of the current luma block is H , $(-(W < n) + a, -(H < n) + b)$, $(-(W < n) + c, 0)$ or $(0, -(H < n) + d)$ may be included in the block vector candidate list as the block vector candidate. At this time, n may be a positive integer greater than 0 and a, b, c and d may have an integer value. The block vector candidate may be referred to as a fixed base block vector candidate.

The block vector candidate list may be constructed in predetermined order using at least one of the block vector of

the adjacent neighbor block, the history-based block vector, the combined block vector or the fixed base block vector candidates.

For example, the block vector candidate list may be constructed in order of the block vector of the adjacent neighbor block, the history-based block vector, the combined block vector and the fixed base block vector candidate.

In addition, for example, the fixed base block vector may be constructed in the following order until the number of candidates in the block vector candidate list reaches a maximum number.

1. $(-(W < 1), 0)$
2. $(0, -(H < 1))$
3. $(-(W < 1) - 1, 0)$
4. $(0, -(H < 1) - 1)$
5. $(-(W < 1) - 2, 0)$
6. $(0, -(H < 1) - 2)$
7. $(-(W < 1) - 3, 0)$
8. $(0, -(H < 1) - 3)$
9. $(-(W < 1) - 4, 0)$
10. $(0, -(H < 1) - 4)$

Alternatively, the fixed base block vector may be a (0,0) vector, and the fixed base block vector may be added to construct the block vector candidate list having a maximum value of candidates until the number of candidates in the block vector candidate list reaches a maximum number. For example, when the number of block vector candidates added to the block vector candidate list using the block vector of the adjacent neighbor block, the history-based block vector, the combined block vector, etc. is less than the maximum number N of block vector candidates, the fixed base block vector may be added to the block vector candidate list until the maximum number of block vector candidates is reached. At this time, the fixed base block vector may be a (0,0) vector.

When the block vector candidate list is constructed, the number of history-based block vector candidates which may be included in the block vector candidate list may be the maximum number N of block vector candidates or (N-m). m may be a natural number greater than 0.

According to an embodiment, when the current block is a luma component block and is encoded/decoded in the intra block copy based SKIP mode, the intra block copy based MERGE mode or the intra block copy based AMVP mode, the block vector derivation method is as follows.

The block vector candidate list may be composed of up to N candidates. At this time, N may be a positive integer including 0. Meanwhile, the block vector candidate list may include one or more of the below-described candidates.

Referring to FIG. 10 again, a block vector may be derived from blocks corresponding to a block A1 adjacent to the left side of the current block X and a block B1 adjacent to the top of the current block, and the derived block vector may be determined as the block vector candidate for the current block.

Here, whether a block vector is present in each of the blocks included in the locations A1 and B1 (that is, whether the block has been encoded/decoded using an intra block copy prediction method) may be determined according to a predetermined priority and, when the block vector is present, the block vector of the block may be determined as a block vector candidate. At this time, the predetermined priority of the block vector candidate list may be A1 and B1.

When the block vector candidate list is constructed by the predetermined priority, redundancy check may be performed

between the block vector candidate present in the block vector candidate list and a newly added block vector candidate.

For example, when the block vector candidate list is constructed in order of A1 and B1, the block B1 may be subjected to redundancy check with the block A and, only when the block B1 has a block vector not equal to the block vector of the block A1, the block vector of the block B1 may be added to the list. The redundancy check may be performed only when the block vector is present in the block.

Meanwhile, when the number of candidates in the block vector candidate list is less than maximum number of allowable candidates in a predefined or derived list, at least one block vector candidate may be brought and used from a buffer in which the block vectors of the blocks encoded/decoded before the current block are stored.

Here, information indicating the maximum number of allowable block vectors in the block vector candidate list may be encoded/decoded at a high level such as a sequence parameter set (SPS), a picture parameter set (PPS), an adaptation parameter set (APS), a picture header, a sub picture, a slice header, a tile group, etc.

For example, in the slice header or SPS, information (e.g., `six_minus_max_num_merge_cand`) indicating the allowable maximum number of the merge candidate list of the inter prediction mode may be encoded/decoded and, as described below, the maximum number of allowable candidates may be derived based on the information on the allowable maximum number. In addition, the derived maximum number of allowable candidates may be defined as the maximum number of block vector candidates in the block vector list.

Maximum number `MaxNumMergeCand` of merge candidates = `6 - six_minus_max_num_merge_cand`

where, `MaxNumMergeCand` may have a value of 1 to 6.

Maximum number `MaxNumIBCCand` of block vector candidates = `maxNumMergeCand`

In addition, for example, in the slice header or SPS, information (e.g., `six_minus_max_num_merge_cand`) indicating the allowable maximum number of the merge list of the inter prediction mode may be encoded/decoded and, as described below, the maximum number `MaxNumMergeCand` of allowable candidates may be derived based on the information indicating the allowable maximum number. In addition, the maximum number of block vector candidates of the block vector list may be defined based on the derived number `MaxNumMergeCand`.

Maximum number `MaxNumMergeCand` of merge candidates = `6 - six_minus_max_num_merge_cand`

Maximum number `MaxNumIBCCand` of block vector candidates = `Max(N, MaxNumMergeCand)`, N is a positive integer greater than 0, and, for example, `N=2`.

In addition, for example, in the slice header or SPS, information (e.g., `six_minus_max_num_merge_cand`) indicating the allowable maximum number of the merge list of the inter prediction mode may be encoded/decoded and, as described above, the maximum number `MaxNumMergeCand` of allowable candidates may be derived based on the information indicating the allowable maximum number. In addition, the maximum number of block vector candidates in the block vector list may be defined based on the derived

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number MaxNumMergeCand and the encoding mode of the current block.

Maximum number MaxNumMergeCand of merge
candidates=6-six_minus_max_num_merge_cand

When the current block is in an intra block copy based SKIP mode or an intra block copy based MERGE mode,

Maximum number MaxNumIBCCand of block vec-
tor candidates=MaxNumMergeCand

Alternatively, when the current block is in an intra block copy based AMVP mode,

Maximum number MaxNumIBCCand of block vec-
tor candidates=Max(N, MaxNumMergeCand) *N*
being a positive integer greater than 0

For example, when $N=2$ and MaxNumMergeCand=1, the maximum number of candidates of the block vector candidate list in the intra block copy based SKIP mode or the intra block copy based MERGE mode may be derived as 1, and the maximum number of candidates of the block vector candidate list in the intra block copy based AMVP mode may be derived 2.

In addition, for example, in the slice header or SPS, information (e.g., six_minus_max_num_merge_cand) indicating the allowable maximum number of the merge list of the inter prediction mode may be encoded/decoded and, as described below, the maximum number MaxNumMergeCand of allowable candidates may be derived based on the information indicating the allowable maximum number. In addition, the maximum number of block vector candidates in the block vector list may be defined based on the derived number MaxNumMergeCand and the encoding mode of the current block.

Maximum number MaxNumMergeCand of merge
candidates=6-six_minus_max_num_merge_cand

When the current block is in the intra block copy based SKIP mode or the intra block copy based MERGE mode,

Maximum number MaxNumIBCCand of block
vectors=MaxNumMergeCand

When the current block is in the intra block copy based AMVP mode,

Maximum number MaxNumIBCCand of block vector
candidates= N , N being a positive integer greater than 0

For example, when the current block is in the intra block copy based AMVP mode, the maximum number of block vector candidates may be defined as 2.

In addition, for example, when MaxNumMergeCand=6, the maximum number of candidates of the block vector candidate list in the intra block copy based SKIP mode or the intra block copy based MERGE mode may be defined as 6 and the maximum number of candidates of the block vector candidate list in the intra block copy based AMVP mode may be defined as 2.

In addition, for example, in the slice header or SPS, information (e.g., six_minus_max_num_ibc_cand) indicating the allowable maximum number of the block vector candidate list may be separately encoded/decoded and, as described above, the maximum number of allowable block vector candidates of the block vector list may be derived based on the information indicating the allowable maximum number.

Maximum number (e.g., MaxNumIBCCand) of
allowable block vector candidates=6-six_mi-
nus_max_num_ibc_cand

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where, the maximum number (e.g., MaxNumIBCCand) of allowable block vector candidates may have a value of 0 to 6.

In addition, the information (e.g., six_minus_max_num_ibc_cand) indicating the maximum number of allowable block vector candidates may have a value of 0 to 5.

Here, the names of MaxNumIBCCand and six_minus_max_num_ibc_cand are arbitrarily described for convenience of description and signals having the other names may be configured.

In addition, for example, in the slice header or SPS, information (e.g., max_num_merge_cand_minus_max_num_ibc_cand) indicating the allowable maximum number of the block vector candidate list may be separately encoded/decoded and, as described below, the maximum number of allowable block vector candidates of the block vector list may be derived based on the information indicating the allowable maximum number.

Maximum number MaxNumIBCCand of allowable block
vector candidates=MaxNumMergeCand-max_num_
merge_cand_minus_max_num_ibc_cand where, MaxNu-
mIBCCand may have a value in a range of 2 to MaxNum-
MergeCand.

Meanwhile, information (e.g., six_minus_max_num_ibc_cand, max_num_merge_cand_minus_max_num_ibc_cand, etc.) indicating the allowable maximum number may be encoded/decoded only when an intra block copy (IBC) function is used at a high level such as an SPS, a PPS, an APS or a slice header. For example, the information indicating the allowable maximum number may be encoded/decoded only when information (e.g., sps_ibc_enabled_flag) indicating whether the intra block copy function decoded in the SPS is used. In addition, for example, the information indicating the allowable maximum number may be derived as 0 when information (e.g., sps_ibc_enabled_flag) indicating whether the intra block copy function decoded in the SPS is used is a first value of 0.

In addition, for example, information (e.g., pic_six_minus_max_num_ibc_merge_cand) indicating the allowable maximum number in the block vector candidate list in the picture header or SPS may be separately encoded/decoded and, as described below, the maximum number of block vector candidates included in the block vector candidate list may be derived based on the information indicating the allowable maximum number.

Maximum number MaxNumIBCCand of block vec-
tor candidates=6-pic_six_minus_max_
num_ibc_merge_cand

where, MaxNumIBCCand may has a value of 1 to 6. Alternatively, MaxNumIBCCand may have a value of 0 to 6. Meanwhile, if MaxNumIBCCand=0, this may mean that the IBC mode is not allowed in all slices related to the picture header.

In addition, for example, encoding/decoding of information (e.g., pic_six_minus_max_num_ibc_merge_cand) indicating the allowable maximum number of the block vector candidate list which may be signaled in the picture header may be omitted based on the encoding parameter value decoded at a high level such as a video parameter set, a sequence parameter set, a picture parameter set, etc., and the value of the information may be estimated based on the value encoded/decoded at the high level.

FIG. 13 is a view illustrating a process of deriving information indicating an allowable maximum number in a block vector candidate list according to an embodiment of the present invention.

Referring to FIG. 13, according to the value of information (e.g., `pps_six_minus_max_num_ibc_merge_cand_plus1`) decoded in the picture parameter set referred to by the picture header, information (e.g., `pic_six_minus_max_num_ibc_merge_cand`) indicating the maximum number of allowable block vector candidates of the block vector candidate list in the picture header may be encoded/decoded or estimated. In addition, when the information (e.g., `pic_six_minus_max_num_ibc_merge_cand`) is not present, the information may be estimated using information (e.g., `pps_six_minus_max_num_ibc_merge_cand_plus1`) encoded at a high level.

For example, when information (e.g., `pps_six_minus_max_num_ibc_merge_cand_plus1`) decoded in the picture parameter set has a value of 0, this may mean that information (e.g., `pic_six_minus_max_num_ibc_merge_cand`) indicating the maximum number of allowable block vector candidates of the block vector list is present in the picture header referring to the picture parameter set.

In addition, for information example, when (e.g., `pps_six_minus_max_num_ibc_merge_cand_plus1`) decoded in the picture parameter set has a value greater than 0, this may mean that information (e.g., `pic_six_minus_max_num_ibc_merge_cand`) indicating the maximum number of allowable block vector candidates of the block vector list is not present in the picture header referring to the picture parameter set. At this time, the information (e.g., `pic_six_minus_max_num_ibc_merge_cand`) may be estimated as the information (e.g., `pps_six_minus_max_num_ibc_merge_cand_plus1`)-1 decoded in the picture parameter set.

Referring to FIG. 13, information (e.g., `constant_slice_header_params_enabled_flag`) indicating presence/absence of predetermined information encoded/decoded in the picture header in the picture parameter may be encoded/decoded.

For example, information (e.g., `pps_six_minus_max_num_ibc_merge_cand_plus1`) decoded in the picture parameter set may be entropy-encoded/decoded when the other information (e.g., `constant_slice_header_params_enabled_flag`) decoded in the picture parameter set has a first value, and may be estimated as 0 without being entropy-encoded/decoded when the other information has a second value. Here, the first value may mean 1 and the second value may mean 0.

In addition, for example, information (e.g., `pps_six_minus_max_num_ibc_merge_cand_plus1`) decoded in the picture parameter set may be entropy-encoded/decoded when the following condition is satisfied.

Condition: information (e.g., `constant_slice_header_params_enabled_flag`) decoded in the picture parameter set has a first value and information (e.g., `sps_ibc_enabled_flag`) decoded in the sequence parameter set has a first value

In addition, for example, information (e.g., `pps_max_num_merge_cand_minus1_max_num_triangle_cand_plus1`) decoded in the picture parameter set may be entropy-encoded/decoded when the following condition is satisfied.

Condition: information (e.g., `constant_slice_header_params_enabled_flag`) decoded in the picture parameter set has a first value and information (e.g., `sps_triangle_enabled_flag`) decoded in the sequence parameter set has a first value

In addition, for example, information (e.g., `pps_collocated_from_l0_idc`) decoded in the picture parameter set may be entropy-encoded/decoded when the following condition is satisfied.

Condition: information (e.g., `constant_slice_header_params_enabled_flag`) decoded in the picture parameter set has a first value and information (e.g., `sps_temporal_mvp_enabled_flag`) decoded in the sequence parameter set has a first value

Meanwhile, the first value may mean 1 and the second value may mean 0.

Meanwhile, the maximum number of allowable block vector candidates of the block vector candidate list may be a fixed value N predefined in the encoder/decoder. Here, N may be a positive integer including 0.

For example, the maximum number of allowable block vector candidates of the block vector candidate list may be 2.

In addition, for example, the maximum number of allowable block vector candidates of the block vector candidate list may be 5.

In addition, for example, the maximum number of allowable block vector candidates of the block vector candidate list may be 6.

In addition, the maximum number of allowable block vector candidates of the block vector candidate list may be a fixed value N predefined in the encoder/decoder. Here, N may be a positive integer including 0, and the value N may vary according to the encoding mode of the current block.

For example, when the current block is in the intra block copy based SKIP mode or the intra block copy based MERGE mode, the predefined value N may be 6.

In addition, for example, when the current block is in the intra block copy based AMVP mode, the predefined value N may be 2.

When the number of block vectors in the block vector candidate list is less than the maximum number of allowable block vector candidates and one or more block vectors are present in a buffer in which the block vectors of the blocks encoded/decoded before the current block are stored, at least one block vector candidate may be brought and used from the buffer in which the block vectors of the blocks encoded/decoded before the current block are stored. For example, the at least one block vector candidate may be included in the block vector candidate list.

At this time, the intra block vector may be stored in a buffer having a specific size in order of encoding/decoding and, if the buffer is full, a first stored block vector may be deleted and a new (that is, a most recently encoded/decoded) block vector may be stored. The block vector candidate list may be constructed by differentiating priority according to the order of storage (e.g., from the oldest vector to the newest vector or from the newest vector to the oldest vector) of the block vectors stored in the buffer. For example, the vector most recently stored in the buffer may be first included in the block vector candidate list or the vector first stored in the buffer may be first included in the block vector candidate list. The block vector candidate may be referred to as a history-based block vector candidate.

When the number of block vectors in the block vector candidate list is less than the maximum number of allowable block vector candidates and one or more block vectors are present in a buffer in which the block vectors of the blocks encoded/decoded before the current block are stored, the history-based block vector candidate may be added to the block vector candidate list until the number of block vectors in the block vector candidate list becomes equal to the maximum number of allowable block vector candidates.

When the block vector candidate list is constructed using at least one of the history-based block vector candidates, redundancy check between the history-based block vector

candidate and the block vector candidates of the neighbor blocks of the current block may be performed and the candidate may be added to the block vector candidate list when the same block vector is not present as the result of performing redundancy check.

For example, referring to FIG. 10, redundancy check between the history-based block vector candidate and the block vector of the block A1 and/or the block B1, both of which are the neighbor block of the current block, may be performed.

In addition, for example, redundancy check with the block A1 and/or the block B1 may be performed only with respect to the first history-based block vector candidate.

In addition, for example, redundancy check with the block A1 and/or the block B1 may be performed only with respect to up to the second history-based block vector candidate.

In addition, for example, redundancy check with the block A1 and/or the block B1 may be performed with respect to all history-based block vector candidates.

Here, the first history-based block vector candidate may mean a most recently stored block vector candidate in the history-based block vector list composed of history-based block vector candidates.

In addition, the second history-based block vector candidate may mean a second block vector candidate stored after the first history-based block vector candidate in the history-based block vector list.

When the block vector candidate list is constructed using at least one of the history-based block vector candidates, a predetermined candidate among the history-based block vector candidates may be added to the block vector candidate list without performing redundancy check with the block vector candidates present in the block vector candidate list or the block vector candidates of the neighbor blocks.

When the number of block vector candidates in the list composed of the block vectors of the neighbor blocks and the history-based block vector candidates is less than the maximum number of allowable candidates of the predefined or derived list, a zero vector having horizontal and vertical block vector sizes of (0, 0) may be added until the number of block vector candidates in the list becomes equal to the maximum number of allowable candidates.

When the current block is in the intra block copy based AMVP mode and the number of candidates in the block vector list constructed in the predetermined order is less than the predefined number N (where, N being a positive integer greater than 0, for example, N=2) of AMVP candidates, a zero vector having horizontal and vertical block vector sizes of (0, 0) may be added until the number of block vector candidates in the block vector list becomes equal to the predefined number of AMVP candidates.

For example, when the current block is in the intra block copy based AMVP mode, the maximum number of candidates in the block vector list is defined in the same manner of the maximum number `MaxNumMergeCand` of merge candidates derived from information (e.g., `six_minus_max_num_merge_cand`) on the maximum number of merge candidates for the inter prediction merge mode decoded in the slice header or SPS, and the maximum number `MaxNumMergeCand` of merge candidates is 1, that is, when the number of candidates in the block vector candidate list is less than the predefined number 2 of AMVP candidates, a zero vector having horizontal and vertical block vector sizes of (0, 0) may be added to the block vector candidate list as a block vector candidate, such that the number 2 of AMVP candidates is satisfied.

Meanwhile, information (e.g., identifier, index, flag, etc.) for identifying the corresponding candidate in the block vector candidate list may be derived based on the encoding parameter transmitted in the bitstream.

For example, when the current block is in the intra block copy based SKIP mode or the intra block copy based MERGE mode, the candidate may be identified based on merge index (e.g., `merge_idx`) information decoded/estimated in the current block.

In addition, for example, when the current block is in the intra block copy based AMVP mode, the candidate may be identified based on L0 motion vector prediction flag (e.g., `mvp_l0_flag`) information decoded/inferred in the current block.

The L0 motion vector prediction flag (e.g., `mvp_l0_flag`) information may be inferred as 0 without performing entropy encoding/decoding, when the maximum number of candidates in the block vector list is 1.

For example, when information (e.g., `MaxNumIbcMergeCand`) indicating the maximum number of allowable candidate block vectors in the block vector list is 1 (that is, `MaxNumMergeCand=1`), information (e.g., `mvp_l0_flag`) indicating the index of the L0 motion vector predictor may be inferred as 0 without performing entropy encoding/decoding. That is, referring to FIG. 14, when `MaxNumMergeCand` is greater than 1, information (e.g., `mvp_l0_flag`) indicating the index of the L0 motion vector predictor may be entropy-encoded/decoded. Otherwise (when `MaxNumMergeCand` is equal to or less than 1), `mvp_l0_flag` may be inferred as 0 without performing entropy encoding/decoding. The information may be a flag.

Here, the L0 motion vector prediction flag may mean information indicating the index of the L0 motion vector predictor.

Here, the name of `MaxNumMergeCand` is arbitrarily described for convenience of description and signals having other names may be configured. That is, instead of `MaxNumMergeCand`, `MaxNumIbcMergeCand` may be used. For example, when `MaxNumIbcMergeCand` is greater than 1, information (e.g., `mvp_l0_flag`) indicating the index of the L0 motion vector predictor may be entropy-encoded/decoded. Otherwise (when `MaxNumIbcMergeCand` is equal to or less than 1), `mvp_l0_flag` may be inferred as 0 without performing entropy encoding/decoding. The information may be a flag.

Referring to FIG. 15, when the current block is in the intra block copy based SKIP mode or the intra block copy based MERGE mode and the allowable maximum number (e.g., `MaxNumIbcMergeCand`) of candidates in the block vector candidate list is greater than 1, merge index information (e.g., `merge_idx`) may be encoded/decoded.

When the current block is in the intra block copy based SKIP mode or the intra block copy based MERGE mode, the block vector candidate identified in the block vector list may be used as the block vector of the current block. Here, the block vector list may be constructed by the above-described methods.

When the current block is in the intra block copy based AMVP mode, a value generated by adding a decoded block vector difference (BVD) to the block vector candidate identified in the block vector list may be used as the block vector of the current block. Here, the block vector list may be constructed by the above-described methods.

Meanwhile, rounding may be performed with respect to the identified block vector candidate, according to the value of the information (e.g., `amvr_precision_flag`) indicating the precision of the decoded motion vector difference.

```

offset = (rightshift == 0)?0:(1<<(rightshift-1))
mvX[0] = ((mvX[0] + offset - (mvX[0] >= 0)) >> rightshift)<< leftshift
mvX[1] = ((mvX[0] + offset - (mvX[1] >= 0)) >> rightshift)<< leftshift

```

When `amvr_precision_flag` has a first value of 0 (the block vector has a 1-integer pixel unit size), `rightshift=4`, `leftshift=4`

When `amvr_precision_flag` has a second value of 1 (the block vector has a 4-integer pixel unit size), `rightshift=6`, `leftshift=6`

Hereinafter, step of deriving the block vector of the chroma component block will be described below.

According to an embodiment, when block partitioning of the luma component and the chroma component in the same CTU is the same (that is, the block partitioning structure of the luma block and the block partitioning structure of the chroma block are the same) and the current block is a chroma component block and is encoded using an intra block copy prediction method, the block vector of the chroma component block may be derived as follows.

The luma component block corresponding to the current chroma component block may be determined as follows.

When the top left sample position of the current chroma component block is (xc, yc), the width is Wc, and the height is Hc, the top left sample position of the luma component block corresponding to the current chroma component block may be (xc/K1, yc/K2), the width may be K1×Wc, the height may be K2×Hc. At this time, K1 and K2 may vary according to the chroma component format, and both K1 and K2 may be 2 when the chroma component format to be encoded is a 4:2:0 format and may be 1 when the chroma component format to be encoded is a 4:4:4 format. In addition, in the case of a 4:2:2 format, K1=2, K2=1.

Since the block partitioning structures of the chroma component block and the luma component block are the same, the luma component block corresponding to the current chroma component block may be composed of one luma component block.

When the block vector of the luma component block corresponding to the chroma component block is (MVL[0], MVL[1]), the block vector of the chroma component block may be (MVL[0]/K1, MVL[1]/K2). Both K1 and K2 may be 2 when the chroma component format of the picture to be encoded is a 4:2:0 format, and may be 1 when the chroma component format of the picture to be encoded is a 4:4:4 format. In addition, in the case of a 4:2:2 format, K1=2 and K2=1. Meanwhile, although it is assumed that the basic unit of MVL[0] and MVL[1] is 1 pixel, the basic unit may be 1/6 pixel or 1/N pixel, where N may be an arbitrary positive integer.

According to an embodiment, block partitioning of the luma component and the chroma component in the same CTU is independently performed (that is, the block partitioning structure of the luma component and the block partitioning structure of the chroma component are different) and the current block is a chroma component block and is encoded using an intra block copy prediction method, the block vector of the chroma component block may be derived as follows.

The luma component region corresponding to the current chroma component block may be determined as follows.

When the top left sample position of the current chroma component block is (xc, yc), the width is Wc and the height is Hc, the top left sample position of the luma component

block corresponding to the current chroma component block may be (xc/K1, yc/K2), the width may be K1×Wc, and the height may be K2×Hc. At this time, K1 and K2 may vary according to the chroma component format and both K1 and K2 may be 2 when the chroma component format of the picture to be encoded is a 4:2:0 format and may be 1 when the chroma component format of the picture to be encoded is a 4:4:4 format. In addition, in the case of a 4:2:2 format, K1=2 and K2=1.

At this time, the region of the luma component corresponding to the current chroma component block may include only a partitioned part of the luma component block. In addition, the luma component region corresponding to the chroma component block may be partitioned into at least one luma component block.

The current chroma component block may be partitioned in units of N×M subblocks and the subblock of the luma component region corresponding to the corresponding subblock may be partitioned in units of (N×K1)×(M×K2) subblocks. At this time, N and M may be integers of 1 or more.

FIG. 16 is a view illustrating a correspondence relation between a chroma component block and a luma component region according to another embodiment of the present invention.

Referring to FIG. 16, an example of a block partitioning structure in an arbitrary CTU having a chroma format of 4:2:0 is shown. The current chroma component block may be partitioned in units of N×M subblocks and the subblock of the luma component region corresponding to the corresponding subblock may be obtained by partitioning the luma component block in units of (N×K1)×(M×K2) subblocks. At this time, N and M may be integers of 1 or more. Alternatively, when the width of the current chroma component block is Wc and the height is Hc, partitioning may be performed in units of N×M subblocks in which the width is partitioned into P1 and the height is partitioned into P2. In this case, N=Wc/P1 and M=Hc/P2, where P1 and P2 may be integers of 1 or more.

There may be a luma component subblock corresponding to the subblock of the current chroma component block. The block vector of the current chroma component subblock may be derived from the block vector of the luma component subblock corresponding thereto. For example, when the block vector of the luma component subblock corresponding to the chroma component subblock is (MVL[0], MVL[1]), the block vector of the chroma component block may be (MVL[0]/K1, MVL[1]/K2). At this time, K1 and K2 may be the same as described above. Meanwhile, although it is assumed that the basic units of MVL[0] and MVL[1] are one pixel in the description, the basic unit may be a 1/6 pixel or a 1/N pixel, where N may be an arbitrary positive integer.

Meanwhile, all the samples located in the luma component subblock corresponding to the subblock of the current chroma component block may not be encoded/decoded using the same prediction method. For example, when the block partitioning structure of the luma component and the chroma component are independent, the luma component subblock corresponding to the chroma component subblock may not match the luma component prediction block or two or more luma component prediction blocks partitioned in the luma component subblock may be present. Here, the luma component prediction block may mean a block, to which the same prediction or transform encoding is applied when the luma component is encoded/decoded and may be determined by luma component block partitioning. Meanwhile, the luma component region corresponding to the chroma component block may not mean a prediction block deter-

mined by luma component block partitioning as in the example of FIG. 16 but may mean a luma component region corresponding to the position and size of the chroma component block.

Accordingly, the block vector of the luma component subblock corresponding to the subblock of the current chroma component block may be one of the followings.

1. The block vector of the luma component prediction block when the luma component prediction block including the top left sample of the luma component subblock corresponding to the subblock of the current chroma component block is encoded/decoded using the intra block copy prediction method

2. The block vector of the luma component prediction block when the luma component prediction block including the center position sample of the luma component subblock corresponding to the subblock of the current chroma component block is encoded/decoded using the intra block copy prediction method

3. FIG. 17 is a view illustrating a luma component subblock according to another embodiment of the present invention. The block vector of the luma component prediction block when the luma component prediction block including one of the shaded sample position in the luma component subblock corresponding to the subblock of the current chroma component block is encoded/decoded using the intra block copy prediction method as in the example of FIG. 17

4. The block vector of the luma component prediction block when the luma component prediction block occupying a largest region in the luma component subblock corresponding to the subblock of the current chroma component block is encoded/decoded using the intra block copy prediction method

There may be no block vector of the luma component subblock corresponding to the subblock of the current chroma component block in one of the following cases.

1. The case where the luma component prediction block including the top left sample of the luma component subblock corresponding to the subblock of the current chroma component block is not encoded/decoded using the intra block copy prediction method or is encoded/decoded using the intra prediction method

2. The case where the luma component prediction block including the center position sample of the luma component subblock corresponding to the subblock of the current chroma component block is not encoded/decoded using the intra block copy prediction method or is encoded/decoded using the intra prediction method

3. The case where the luma component prediction block including one of the shaded sample positions in the luma component subblock corresponding to the subblock of the current chroma component block is not encoded/decoded using the intra block copy prediction method or is encoded/decoded using the intra prediction method, as in the example of FIG. 17.

4. The case where the luma component prediction block occupying a largest region in the luma component subblock corresponding to the subblock of the current chroma component block is not encoded/decoded using the intra block copy prediction method or is encoded/decoded using the intra prediction method

When there is no block vector of the luma component subblock corresponding to the subblock (hereinafter referred to as a "current subblock") of the current chroma component

block, the block vector corresponding to the current chroma component block may be derived using one of the following methods.

The block vector of the current subblock may be set to (0, 0) or (D1, D2). At this time, D1 and D2 may be an integer of 0, ± 1 , ± 2 , ± 3 ,

The block vector of the current subblock may be set to (Wc+D1, D2) or (D1, Hc+D2). At this time, Wc may be the width of the current chroma component block, Hc may be the height of the current chroma component block, and D1 and D2 may be integers of 0, ± 1 , ± 2 , ± 3 ,

The block vector of the current subblock may be set to one of $-(Wc \ll n) + a$, $-(Hc \ll n) + b$, $-(Wc \ll n) + c$, 0 or $(0, -(Hc \ll n) + d)$. At this time, n may be a positive integer greater than or equal to 0, and a, b, c and d may have integer values.

The block vector of the neighbor subblock (e.g., at least one of the top subblock, the bottom subblock, the left subblock, the right subblock, the top left subblock, the top right subblock, the bottom left subblock or the bottom right subblock) of the current subblock may be used as the block vector of the current subblock.

The block vector of the current subblock may be derived using the statistical value of the block vector values of the subblocks, in which the block vector of the corresponding luma component subblock is present, among the subblocks of the current chroma component block.

For example, the block vector of the current subblock may be one of the mean, the media, the maximum value or the minimum value of the block vectors of the subblocks.

As another example, the block vector of the current subblock may be a block vector having a highest frequency of occurrence.

Meanwhile, when the corresponding luma component subblock is not encoded/decoded using the intra block copy prediction method or when there is at least one chroma component subblock in which the block vector of the luma component subblock is not present, the chroma component block may not be encoded/decoded using the intra block copy prediction method.

The prediction mode of the luma component subblock corresponding to the subblock of the current chroma component block may be set as follows. Here, the prediction mode may be intra prediction, inter prediction or intra block copy prediction and, more specifically, may be one of a skip mode, a merge mode, an AMVP mode or an affine skip mode among inter prediction modes or may be an intra block copy based SKIP mode, an intra block copy based MERGE mode or an intra block copy based AMVP mode among intra block copy prediction modes.

1. The prediction mode of the luma component prediction block including the top left sample of the luma component subblock corresponding to the subblock of the current chroma component block

2. The prediction mode of the luma component prediction block including the center position sample of the luma component subblock corresponding to the subblock of the current chroma component block

3. The prediction mode of the luma component prediction block including one of the shaded sample positions in the luma component subblock corresponding to the subblock of the current chroma component block as in the example of FIG. 17

4. The prediction mode of the luma component prediction block occupying a largest region in the luma component subblock corresponding to the subblock of the current chroma component block

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Hereinafter, step of deriving the prediction signal will be described.

Step of deriving the prediction signal for intra block copy prediction may include at least one of step of deriving the prediction signal of a luma component block and step of deriving the prediction signal of the chroma component block.

Hereinafter, step of deriving the prediction signal of the luma component block will be described.

A block separated from the current luma component block by the block vector of the derived luma component block may be referred to as a prediction block.

For example, when the top left sample position of the current luma component block is $(x0, y0)$, the width is WL, the height of HL and the block vector of the derived luma component is (xd, yd) , a block in which the sample position $(x0+xd, y0+yd)$ separated from the top left sample position of the current luma component block in the same picture by (xd, yd) is a top left sample position, the width is WL and the height is HL may be a prediction block. At this time, it may move from $(x0, y0)$ by xd leftward in the horizontal direction if xd is a negative integer, move from $(x0, y0)$ by xd rightward if xd is a positive integer, move from $(x0, y0)$ by yd upward in the vertical direction if yd is a negative integer and move from $(x0, y0)$ by yd downward if yd is a positive integer.

FIG. 18 is a view illustrating a relation between a current block and a prediction block according to another embodiment of the present invention.

Referring to FIG. 18, it is assumed that xd and yd are negative integers.

The sample value of the prediction block may be set to the prediction sample value of the current luma component block and the set value may be referred to as the prediction signal of the current luma component block.

Hereinafter, step of deriving the prediction signal of the chroma component block will be described.

According to an embodiment, when block partitioning of the luma component and the chroma component in the same CTU is equally performed (that is, the block partitioning structure of the luma component and the block partitioning structure of the chroma component are the same) and the current block is a chroma component block and is encoded using the intra block copy prediction method, the prediction signal of the chroma component block may be derived as follows.

A block separated from the current chroma component block by the block vector the derived chroma component block may be referred to as a prediction block.

For example, when the top left sample position of the current chroma component block is $(x0, y0)$, the width is Wc, the height of Hc and the block vector of the derived chroma component is (xd, yd) , a block in which the sample position $(x0+xd, y0+yd)$ separated from the top left sample position of the current chroma component block in the same picture by (xd, yd) is a top left sample position, the width is Wc and the height is Hc may be a prediction block. At this time, it may move from $x0$ by xd rightward in the horizontal direction if xd is a positive integer, move from $x0$ by xd leftward if xd is a negative integer, move from $y0$ by yd downward in the vertical direction if yd is a positive integer and move from $y0$ by yd upward if yd is a negative integer.

The sample value of the prediction block may be set as the prediction sample value of the current chroma component block, and the set value may be referred to as the prediction signal of the current chroma component block.

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According to an embodiment, when block partitioning of the luma component and the chroma component in the same CTU is independently performed (that is, the block partitioning structure of the luma component and the block partitioning structure of the chroma component are different) and the current block is a chroma component block and is encoded using the intra block copy prediction method, the prediction signal of the chroma component block may be derived as follows.

A subblock prediction signal may be derived in subblock units using the block vector derived in units of the subblocks of the current chroma component block.

A block separated from the subblock of the current chroma component block by the block vector of the subblock of the derived chroma component may be referred to as a prediction subblock.

For example, when the top left sample position of the subblock of the current chroma component block is $(sx0, sy0)$, the width is SWc, the height is SHc and the block vector of the derived chroma component is (Sxd, Syd) , $(sx0+Sxd, sy0+Syd)$ moved from the top left sample position of the subblock in the current chroma component block in the same picture by (xd, yd) may be a top left sample position and a block having a width of SWc and a height of SHc may be a prediction subblock.

The sample value of the prediction subblock may be set as the prediction sample value of the current chroma component subblock and the set value may be referred to as the prediction signal of the current chroma component subblock.

The prediction signal of the current chroma component block may be constructed using the subblock prediction signals of all subblocks included in the current chroma component block.

Hereinafter, step of deriving the residual signal will be described.

In general, when a residual signal is present, the residual signal may be transformed and encoded in an encoding process and may be included and transmitted in a bitstream, and the inverse process of the transform encoding process may be performed in a decoding process to derive the residual signal.

Identifier information indicating presence/absence of residual signal related signal (e.g., a quantization transform coefficient (or a quantized transform coefficient), etc.) included in the bitstream and transmitted to the decoder may include at least one of the following.

cu_cbf: When the luma component and the chroma component have the same block partitioning structure, this may mean information on presence/absence of the quantization transform coefficient of the residual signal of the luma component block and the quantization transform coefficient of the residual signal of the chroma component block in the coding block CU. When the luma component and the chroma component have independent block partitioning structures, this may mean information on presence/absence of the quantization transform coefficient of the residual signal of the luma component block (or the luma component block in the coding block CU) or the chroma component block (or the chroma component block in the coding block CU). The information on presence/absence of the quantization transform coefficient of the residual signal having a first value of 1 may mean that the quantization transform coefficient of the residual signal of the blocks is present and the information on presence/absence of the quantization transform coefficient of the residual signal having a second value of 0 may mean that the quantization transform coefficient of

the residual signal of the blocks is not present. When the luma component and the chroma component have the same block partitioning structure, if any one of the luma component block and the chroma component (Cb, Cr) block has a quantization transform coefficient of a residual signal, the information on presence/absence of the quantization transform coefficient of the residual signal may have the first value and, if there is no quantization transform coefficient of a residual signal for all components, the information on presence/absence of the quantization transform coefficient of the residual signal may have the second value. Here, the luma component block and the chroma component block may mean the luma component block in the coding block CU and the chroma component block in the coding block CU.

tu_cbf_luma: This may mean presence/absence of the quantization transform coefficient of the residual signal of the luma component block. Information on presence/absence of the quantization transform coefficient of the residual signal of the luma component block having a first value of 1 may mean that the quantization transform coefficient of the residual signal of the luma block is present and the information on presence/absence of the quantization transform coefficient of the residual signal of the luma component block having a second value of 0 may mean that the quantization transform coefficient of the residual signal of the luma block is not present. Here, the luma component block may mean the luma component block in the transform block TU.

tu_cbf_cr, tu_cbf_cb: This may mean presence/absence of the quantization transform coefficient of the residual signal of each of Cr and Cb of the chroma component. Information on presence/absence of the quantization transform coefficient of the residual signal of the chroma component (Cr or Cb) block having a first value of 1 may mean that the quantization transform coefficient of the residual signal of the chroma component (Cr or Cb) block is present and the information on presence/absence of the quantization transform coefficient of the residual signal of the chroma component (Cr or Cb) block having a second value of 0 may mean that the quantization transform coefficient of the residual signal of the chroma component (Cr or Cb) block is not present. Here, the chroma component block may mean the chroma component block in the transform block Tu.

Meanwhile, in the following embodiments, the luma component block and the chroma component block related to cu_cbf may mean the luma component block in the coding block CU and the chroma component block in the coding block CU, respectively. In addition, the luma component block related to tu_cbf_luma may mean the luma component block in the transform block TU. In addition, the chroma component block related to tu_cbf_cr and tu_cbf_cb may mean the chroma component block in the transform block TU.

In general, only when cu_cbf has a first value of 1, at least one of tu_cbf_luma, tu_cbf_cr or tu_cbf_cb may be further transmitted to indicate presence/absence of the quantization transform coefficients of the residual signals of the luma component, Cr of the chroma component and Cb of the chroma component.

When the luma component and the chroma component have independent block partitioning structures, cu_cbf may have the same information as tu_cbf_luma.

In the CTU to be currently encoded, when the block partitioning structures of the luma component and the chroma component are the same and the current luma component block is in an intra block copy based SKIP mode,

the residual signals of the current luma component block and the chroma component block may be derived as follows.

When the current luma component block is in an intra block copy based SKIP mode, there is no residual signal as in the SKIP mode of general intra prediction. In this case, all residual signal may be set to have a value of 0.

In the current chroma component block, when the corresponding luma component block is in an intra block copy based SKIP mode, there is no residual signal similar to the luma component block. In this case, all residual signal may be set to have a value of 0.

In the case of the intra block copy based SKIP mode, information for identifying presence of the residual signal (e.g., (an identifier, a flag, an index, cu_cbf, tu_cbf_luma, tu_cbf_cr, tu_cbf_cb, etc.) may not be transmitted in a bitstream.

For example, a cu_cbf value indicating whether all the quantization transform coefficients of the residual signals of the luma component, Cr of the chroma component and Cb of the chroma component are present may not be transmitted in the bitstream and may be set to a second value indicating that all the quantization transform coefficients of the residual signals of the luma component, Cr of the chroma component and Cb of the chroma component are not present in the decoding process. All tu_cbf_luma indicating presence/absence of the quantization transform coefficient of the residual signal of the luma component, tu_cbf_cr indicating presence/absence of the quantization transform coefficient of the residual signal of the Cr component of the chroma component, tu_cbf_cb indicating presence/absence of the quantization transform coefficient of the residual signal of the Cb component of the chroma component are not transmitted and are set to the second value in the decoding process to indicate that the quantization transform coefficients of the residual signals corresponding thereto are not present.

In CTU to be currently encoded, when the block partitioning structures of the luma component and the chroma component are the same and the current luma component block is in an intra block copy based MERGE mode, the residual signals of the current luma component block and the chroma component block may be derived as follows.

When the current luma component block is in an intra block copy based MERGE mode, the residual signal may always be present. In this case, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process.

In the current chroma component block, when the corresponding luma component block is in an intra block copy based MERGE mode, a residual signal may be present similar to the luma component block. In this case, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process.

In the case of the intra block copy based MERGE mode, information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence of the residual signal may not be transmitted in the bitstream. In the MERGE mode, since the residual signal is always present, a cu_cbf value indicating whether any one of the quantization transform coefficients of the residual signals of the luma component, Cr of the chroma component and Cb of the chroma component is present may also be set to the first value in the decoding process. In this

case, the information for identifying presence of the residual signal may not be transmitted in the bitstream and the quantization transform coefficient information of the residual signal may also be included and transmitted.

However, in the case of the intra block copy based MERGE mode, since there may be a component with the quantization transform coefficient of the residual among the luma component, Cr of the chroma component and Cb of the chroma component, identifiers (e.g., tu_cbf_luma in the case of the luma component and tu_cbf_cr and tu_cbf_cb in the case of the chroma component) indicating presence/absence of the quantization transform coefficient of the residual signal of each component may be included and transmitted in the bitstream.

In the CTU to be currently encoded, the block partitioning structures of the luma component and the chroma component are the same and the current luma component block is in an intra block copy based AMVP mode, the residual signals of the current luma component block and the chroma component block may be derived as follows.

When the current luma component block is in the intra block copy based AMVP mode, the residual signal may or may not be present. In this case, information for identifying presence/absence of the residual signal may always be transmitted in the bitstream. When the residual signal is present, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process. When the residual signal is not present, all residual signals may be set to have a value of 0.

In the current chroma component block, when the corresponding luma component block is in an intra block copy based AMVP mode, a residual signal may or may not be present similar to the luma component block. In this case, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process. When the residual signal is not present, all residual signals may be set to have a value of 0.

In the case of intra block copy based AMVP mode, since the quantization transform coefficients of the residual signals of the luma component, Cr of the chroma component and Cb of the chroma component may or may not be present, information for (e.g., an identifier, a flag, an index, cu_cbf, etc.) identifying presence/absence of the residual signal indicating whether the quantization transform coefficients of the residual signals of the luma component, Cr of the chroma component and Cb of the chroma component may always be included and transmitted in the bitstream.

In addition, when information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of a residual signal indicating whether the quantization transform coefficients of the residual signals of the luma component, Cr of the chroma component and Cb of the chroma component are present has a first value indicating presence of the residual signal, since there may be a component without the quantization transform coefficient of the residual among the luma component, Cr of the chroma component and Cb of the chroma component, identifies (e.g., tu_cbf_luma in the case of the luma component and tu_cbf_cr and tu_cbf_cb in the chroma component) indicating presence/absence of the quantization transform coefficient of the residual signal of each component may be included and transmitted in the bitstream.

In the CTU to be currently encoded, when block partitioning of the luma component and the chroma component is independent and the current luma component block is in an intra block copy based SKIP mode, the residual signal of the current luma component block may not be present as in the SKIP mode of general intra prediction. In this case, all residual signals may be set to have a value of 0 and information (e.g., an identifier, a flag, an index, cu_cbf, tu_cbf_luma, etc.) for identifying presence of the residual signal may not be transmitted in the bitstream.

When the luma component block is in an intra block copy based SKIP mode, cu_cbf transmitted for the luma component block in an independent partitioning structure may indicate whether the quantization transform coefficient of the residual signal of the luma component block is present. In this case, since the quantization transform coefficient of the residual signal of the luma component block is always not present, the information may not be transmitted in the bitstream and may be set to a second value of 0 in the decoding process.

In addition, identification information (e.g., tu_cbf_luma) indicating presence of the residual signal of the luma component may not be transmitted in the bitstream and may be set to a second value of 0 in the decoding process.

In the CTU to be current encoded, when block partitioning of the luma component and the chroma component is independent and the current luma component block is in an intra block copy based MERGE mode, the residual signal of the current luma component block may always be present as in MERGE mode of general intra prediction. In this case, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process.

When the luma component block is in an intra block copy based MERGE mode, in an independent partitioning structure, cu_cbf transmitted for the luma component block may indicate only whether the quantization transform coefficient of the residual signal of the luma component block is present. In this case, since the quantization transform coefficient of the residual signal of the luma component block is always present, the information may not be transmitted in the bitstream and may be set to a first value of 1 in the decoding process.

Meanwhile, identification information (e.g., tu_cbf_luma) indicating presence/absence of the residual signal of the luma component has the same value as cu_cbf transmitted for the luma component block in the independent partitioning structure and thus may not be transmitted in the bitstream and may be set to a first value of 1 in the decoding process.

In the CUT to be currently encoded, when block partitioning of the luma component and the chroma component is independent and the current luma component block is in an intra block copy based AMVP mode, the residual signal of the current luma component block may or may not be present as in the AMVP mode of general intra prediction. When the residual signal of the current luma component block is present, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process. When the residual signal is not present, all residual signals may be set to have a value of 0.

When the luma component block is in an intra block copy based AMVP mode, cu_cbf transmitted for the luma component block in an independent partitioning structure may indicate only whether the quantization transform coefficient of the residual signal of the luma component block is present. In this case, since the quantization transform coefficient of the residual signal of the luma component block may or may not be present, cu_cbf which is information for identifying whether the residual signal is present may always be transmitted in the bitstream.

Meanwhile, identification information (e.g., tu_cbf_luma) indicating presence/absence of the residual signal of the luma signal in the transform unit TU has the same value as cu_cbf transmitted for the luma component block in the coding block CU in the independent partitioning structure and thus may not be transmitted in the bitstream and may be set to the same value as cu_cbf in the decoding process. That is, it is possible to increase encoding/decoding efficiency by removing redundancy of signaling of both cu_cbf and tu_cbf_luma. For example, in the independent partitioning structure, when the prediction mode of the current luma component block is an intra block copy mode, tu_cbf_luma may not be signaled. At this time, a tu_cbf_luma value which is not signaled may be set to a cu_cbf value.

Whether a residual signal is present may be determined according to the value of cu_cbf or tu_cbf_luma. For example, when the value of cu_cbf or tu_cbf_luma has a first value, it may be determined that the residual signal is present.

When block partitioning of the luma component and the chroma component are independently performed in the same CTU (that is, the block partitioning structure of the luma component is different from that of the chroma component), and the current block is a chroma component block and is encoded using the intra block copy prediction method, the residual signal of the chroma component block may be derived as follows.

All subblocks included in the luma component block corresponding to the current chroma component block may have the same intra block copy prediction the encoding mode.

At this time, the intra block copy prediction the encoding mode may be an intra block copy based SKIP mode, an intra block copy based MERGE mode or an intra block copy based AMVP mode.

FIG. 19 is a view illustrating when prediction encoding modes of luma component subblocks corresponding to chroma component blocks according to an embodiment of the present invention are the same. For example, as in the example of FIG. 19, all prediction encoding modes of the luma component subblocks may be intra block copy based SKIP modes, intra block copy based MERGE modes or intra block copy based AMVP modes.

When all the subblocks included in the luma component block corresponding to the current chroma component block are in the same intra block copy prediction mode, whether the residual signal of the corresponding chroma component block is encoded/decoded may be determined based on the intra block copy prediction mode of the corresponding luma component blocks.

For example, when all the subblocks included in the luma component block corresponding to the current chroma component block are in the intra block copy based SKIP mode, as in the case where the luma component block is in an intra block copy based SKIP mode, the residual signal of the chroma component block may also be not encoded/decoded

and residual signal information may not be transmitted. In this case, all residual signals may be set to have a value of 0.

At this time, information (e.g., an identifier, a flag, cu_cbf, tu_cbf_cr/tu_cbf_cb, etc.) for identifying presence/absence of the residual signal of the block may not be sent. The information for identifying presence/absence of the residual signal, which has a first value, may indicate that the residual signal is present and the information having a second value may indicate that the residual signal is not present. When the chroma component block is an intra block copy based SKIP mode, the for identifying presence/absence of the residual signal of the block may be set.

As another example, when all subblocks included in the luma component block corresponding to the current chroma component block are in an intra block copy based MERGE mode, as in the case where the luma component block is in the intra block copy based MERGE mode, the chroma component block may always have a residual signal. In this case, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process.

In the independent partitioning structure, information (e.g., an identifier, a flag, cu_cbf, etc.) for identifying presence of the residual signal transmitted for the chroma component block may indicate whether the quantization transform coefficient of the residual signal is present in at least one of the Cb and Cr blocks of the chroma component.

In the case of the intra block copy based MERGE mode, information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of the residual signal may not be transmitted in the bitstream. In the MERGE mode, since the residual signal is always present, the cu_cbf value indicating whether any one of the quantization transform coefficients of the residual signals of Cr of the chroma component and Cb of the chroma component may always be set to a first value in the decoding process.

However, in the case of an intra block copy based MERGE mode, since there may be a component without the quantization transform coefficient of the residual signal between Cr of the chroma component and Cb of the chroma component, identifiers (e.g., tu_cbf_cr, tu_cbf_cb) indicating presence/absence of the quantization transform coefficient of the residual signal of each chroma component may be included and transmitted in the bitstream.

As another example, when all subblocks included in the luma component block corresponding to the current chroma component block are in an intra block copy based AMVP mode, as in the case where the luma component block is in an intra block copy based AMVP mode, the chroma component block may or may not have a residual signal. In this case, information for identifying presence/absence of the residual may always be transmitted in the bitstream. When the residual signal of the current luma component block is present, the quantization transform coefficient of the residual signal transformed and encoded in the encoding process may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process. When the residual signal is not present, all residual signals may be set to have a value of 0.

In the independent partitioning structure, information (e.g., an identifier, a flag, cu_cbf, etc.) for identifying presence of the residual signal transmitted for the chroma component block may indicate whether the quantization

transform coefficient of the residual signal is present in at least one of the Cb and Cr blocks of the chroma component.

In the case of the intra block copy based AMVP mode, since the quantization transform coefficients of the residual signals of Cr of the chroma component and Cb of the chroma component may or may not be present, information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of the residual signals of Cr of the chroma component and Cb of the chroma component may be transmitted in the bitstream.

In addition, when the information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of the residual signals of Cr of the chroma component and Cb of the chroma component has a first value indicating that the residual signal is present, since there may be a component without the quantization transform coefficient of the residual between Cr of the chroma component and Cb of the chroma component, identifiers (e.g., tu_cbf_cror tu_cbf_cb in the case of the chroma component) indicating presence/absence of the quantization transform coefficient of the residual signal of each component may be included and transmitted in the bitstream. When the identifier (e.g., tu_cbf_cr) indicating presence of the quantization transform coefficient of the residual signal of Cr of the chroma component has a first value, the quantization transform coefficient information of the residual signal for the Cr component may be included and transmitted in the bitstream. When the identifier (e.g., tu_cbf_cr) indicating presence of the quantization transform coefficient of the residual signal of Cb of the chroma component has a first value, the quantization transform coefficient information of the residual signal for the Cb component may be included and transmitted in the bitstream.

As another example, even if all subblocks included in the luma component block corresponding to the current chroma component block are in the same intra block copy prediction mode (e.g., an intra block copy based SKIP mode, an intra block copy based MERGE mode, an intra block copy based AMVP mode, etc.), since all samples included in the corresponding luma component block are not in the same intra block copy prediction mode, it may not be efficient to encode the residual signal according to the mode of the luma component block corresponding to the chroma component block.

Accordingly, when the chroma component block is in an intra block copy mode, regardless of the type of the mode of the subblock included in the corresponding luma component block, information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of the quantization transform coefficients of the residual signals of Cr of the chroma component and Cb of the chroma component may always be included and transmitted in the bitstream.

In addition, when the information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of the quantization transform coefficients of the residual signals of Cr of the chroma component and Cb of the chroma component has a first value indicating that the residual signal is present, since there may be a component without the quantization transform coefficient of the residual between Cr of the chroma component and Cb of the chroma component, identifiers (e.g., tu_cbf_cr or tu_cbf_cb in the case of the chroma component) indicating presence/absence of the quantization transform coefficient of the residual signal of each component may be included and transmitted in the bitstream. When the identifier (e.g., tu_cbf_cr) indicating that the quantization transform coefficient of the residual signal of Cr of the chroma component has a first

value, the quantization transform coefficient information of the residual signal of the Cr component may be included and transmitted in the bitstream. When the identifier (e.g., tu_cbf_cr) indicating presence of the quantization transform coefficient of the residual signal of Cb of the chroma component has a first value, the quantization transform coefficient information of the residual signal of the Cb component may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process. When absence of the residual signal of the chroma component block is identified, the quantization transform coefficient information of the residual signal of the chroma component block may not be transmitted and all residual signals may be set to have a value of 0.

As another example, when the current chroma component block is in an intra block copy prediction mode or when all subblocks included in the luma component block corresponding to the current chroma component block are in an intra block copy prediction mode, the subblocks of the luma component corresponding to the chroma component block may be in different intra block copy prediction encoding modes (e.g., an intra block copy based SKIP mode, an intra block copy based MERGE mode and an intra block copy based AMVP mode). FIG. 20 is a view illustrating when prediction encoding modes of luma component subblocks corresponding to chroma component blocks according to an embodiment of the present invention are different. For example, as in the example of FIG. 20, at least two of the intra block copy based SKIP mode, the intra block copy based MERGE mode and intra block copy based AMVP mode may be present in the corresponding luma component block.

At this time, regardless of the type of the mode of the subblock included in the corresponding luma component block, information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of the quantization transform coefficients of the residual signals of Cr of the chroma component and Cb of the chroma component may always be included and transmitted in the bitstream.

In addition, when the information (e.g., an identifier, a flag, an index, cu_cbf, etc.) for identifying presence/absence of the quantization transform coefficients of the residual signals of Cr of the chroma component and Cb of the chroma component has a first value indicating that the residual signal is present, since there may be a component without the quantization transform coefficient of the residual signal between Cr of the chroma component and Cb of the chroma component, identifiers (e.g., tu_cbf_cr, tu_cbf_cb in the case of the chroma component) indicating presence/absence of the quantization transform coefficient of the residual signal of each component may be included and transmitted in the bitstream. When the identifier (e.g., tu_cbf_cr) indicating presence of the quantization transform coefficient of the residual signal of Cr of the chroma component has a first value, the quantization transform coefficient information of the residual signal of the Cr component may be included and transmitted in the bitstream. When the identifier (e.g., tu_cbf_cr) indicating presence of the quantization transform coefficient of the residual signal of Cb of the chroma component has a first value, the quantization transform coefficient information of the residual signal of the Cb component may be included and transmitted in the bitstream and the residual signal may be derived by inverse transform encoding in the decoding process. When absence of the residual signal of the chroma component block is identified, the quantization transform coefficient information of the

residual signal of the chroma component block may not be transmitted and all residual signals may be set to have a value of 0.

Hereinafter, step of constructing the reconstructed signal will be described.

In the current luma component block, the reconstructed signal may be constructed by adding the residual signal of the luma component block to the prediction signal of the luma component block.

In the current chroma component block, the reconstructed signal may be constructed by adding the residual signal of the chroma component block to the prediction signal of the chroma component block.

Meanwhile, when the residual signal is not present, the prediction signal may be set as the reconstructed signal.

Hereinafter, step of entropy-encoding/decoding information on intra block copy prediction will be described.

The information on intra block copy prediction encoding may be entropy-encoded in a bitstream or may be entropy-decoded from the bitstream. Here, the information on intra block copy prediction encoding may include at least one of the following information.

- cu_skip_flag indicating whether a skip mode is used
- merge_flag indicating whether a merge mode is used
- merge_idx (merge index) indicating a merge candidate,
- pred_mode_flag indicating whether a prediction mode is intra prediction
- pre_mode_ibc_flag indicating whether the prediction mode is inter prediction or intra block copy prediction
- block vector candidate index (mvp_l0_flag)
- motion vector difference
- cu_cbf, tu_cbf_luma, tu_cbf_cb and tu_cbf_cr indicating presence/absence of the quantization transform coefficient of the residual signal

Here, the cu_skip_flag may mean whether a skip mode is used and may be entropy-encoded in units of at least one of encoding blocks or prediction blocks. For example, if information indicating whether the skip mode is used has a first value of 1, this may indicate that the skip mode is used and, when information indicating whether the skip mode is used has a second value of 0, this may indicate that the skip mode is not used.

The merge_flag may mean whether the merge mode is used and may be entropy-encoded in units of at least one of encoding blocks or prediction blocks. For example, if information indicating whether the merge mode is used has a first value of 1, this may indicate that the merge mode is used and, when information indicating whether the skip mode is used has a second value of 0, this may indicate that the merge mode is not used.

The merge_idx may mean information indicating a merge candidate in a merge candidate list and may be entropy-encoded in units of at least one of encoding blocks or prediction blocks. In addition, merge_idx may mean merge index information. In addition, merge_idx may indicate a block, in which a merge candidate is derived, among blocks reconstructed spatially adjacent to the current block. In addition, merge_idx may indicate at least one of the motion information of the merge candidate. For example, the merge index information having a first value of 0 may indicate a first merge candidate in the merge candidate list, the merge index information having a second value of 1 may indicate a second merge candidate in the merge candidate list, and the merge index information having a third value of 2 may indicate a third merge candidate in the merge candidate list. In addition, the merge index information having fourth to N-th value may indicate merge candidates corresponding to

the corresponding values in the order of candidates in the merge candidate list. Here, N may be a positive integer including 0.

The pred_mode_flag may mean whether an intra prediction mode is applied and may be entropy-encoded in units of at least one of encoding blocks or prediction blocks. For example, the information indicating whether the intra prediction mode is applied, which has a first value of 1, may indicate that the intra prediction mode is applied and the information indicating whether the intra prediction mode is applied, which has a second value of 0, may indicate that the intra prediction mode is not applied.

The pred_mode_ibc_flag may mean whether an intra block copy prediction mode is applied and may be entropy-encoded in units of at least one of encoding blocks, prediction blocks or encoding units. For example, the information indicating whether the intra block copy prediction mode is applied, which has a first value of 1, may indicate that the intra block copy prediction mode is applied and the information indicating whether the intra block copy prediction mode is applied, which has a second value of 0, may indicate that the intra block copy prediction mode is not applied.

The block vector candidate index mvp_l0_flag may indicate the block vector candidate used by the current block in the block vector candidate list of the intra block copy prediction mode and may be the block vector candidate index may be entropy-encoded/decoded. The prediction block of the current block may be derived using the block vector candidate index.

The motion vector difference may mean a difference between the block vector and a predicted block vector, and the prediction block of the current block may be derived using the block vector difference.

In cu_cbf, tu_cbf_luma, tu_cbf_cb and tu_cbf_cr indicating presence/absence of the quantization transform coefficient of the residual signal, cu_cbf may mean information on presence/absence of the quantization transform coefficient of the luma component block and the quantization transform coefficient of the chroma component block when the luma component and the chroma component has the same block partitioning structure, and may mean information on the quantization transform coefficient of the luma component block or the chroma component block when the luma component and the chroma component have independent block partitioning structures. The information indicating presence/absence of the quantization transform coefficient, which has a first value of 1, may mean that the quantization coefficient coefficients of the blocks are present and the information indicating presence/absence of the quantization transform coefficient, which has a second value of 0, may mean that the quantization coefficient coefficients of the blocks are not present. tu_cbf_luma may mean presence/absence of the quantization transform coefficient of the luma component block, tu_cbf_cr and tu_cbf_cb may respectively mean presence/absence of the quantization transform coefficients of Cr and Cb of the chroma component. The information on presence/absence of the quantization transform coefficient of the luma component block, which has a first value of 1, may mean that the quantization transform coefficient of the luma block is present and the information on presence/absence of the quantization transform coefficient of the luma component block, which has a second value of 0, may mean that the quantization transform coefficient of the luma block is not present. The information on presence/absence of the quantization transform coefficient of the chroma components Cb or Cr, which has a first value of 1, may mean that the quantization transform coefficient of

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the chroma block is present and the information on presence/absence of the quantization transform coefficient of the chroma components Cb or Cr, which has a second value of 0, may mean that the quantization transform coefficient of the chroma block is not present.

In addition, at least one of information on intra block copy prediction encoding may be entropy-encoded/decoded from at least one of a video parameter set, a sequence parameter set, a picture parameter set, an adaptation parameter set, a tile header, a tile group header, a slice header or a slice payload.

FIGS. 21 to 23 are views illustrating encoding information transmitted in association with intra block partitioning according to an embodiment of the present invention.

FIGS. 21a and 21b show an example of an encoding information transmission method for each block partitioning structure.

FIGS. 22 to 23 show an example of an encoding information transmission method of removing overlapping when cu_cbf and tu_cbf_luma information transmitted at the time of intra block partitioning prediction overlap.

In particular, FIGS. 22 to 23 show a method of redundant signaling of cu_cbf and tu_cbf_luma indicating the same information in an independent block partitioning structure and when the luma component block is intra block copy prediction. In addition, referring to FIGS. 22 to 23, in an independent block partitioning structure and when the luma component block is intra block copy prediction, tu_cbf_luma may not be transmitted and the cu_cbf value may be set to the tu_cbf_luma value.

Referring to FIG. 22, in the case of the luma component block of the independent block partitioning structure (that is, treeType is DUAL_TREE_LUMA), by signaling tu_cbf_luma only when the prediction mode is an intra prediction mode (that is, CurPredMode[x0][y0]=MODE_INTRA), tu_cbf_luma, tu_cbf_luma may not be transmitted in the case where the luma component block of the independent block partitioning structure is in an intra block copy prediction mode. In addition, the luma component block of the independent block partitioning structure (that is, treeType is DUAL_TREE_LUMA), tu_cbf_cb and tu_cbf_cr may not be signaled and, when tu_cbf_cb and tu_cbf_cr are not signaled, these may be set to 0 in the decoding process. Meanwhile, the tu_cbf_luma value which is not signaled may be set to a cu_cbf value.

Referring to FIG. 23, the luma component block of the independent block partitioning structure (that is, treeType is DUAL_TREE_LUMA), by signaling tu_cbf_luma only when the prediction mode is not an intra block copy prediction mode (that is, CurPredMode[x0][y0]=MODE_IBC), tu_cbf_luma may not be transmitted when the luma component block of the independent block partitioning structure is in an intra block copy prediction mode. In addition, in the case of the luma component block of the independent block partitioning structure (that is, treeType is DUAL_TREE_LUMA), tu_cbf_cb and tu_cbf_cr may not be signaled and, when tu_cbf_cb and tu_cbf_cr are not signaled, these may be set to 0 in the decoding process. Meanwhile, the tu_cbf_luma value which is not signaled may be set to a cu_cbf value.

According to the present invention, it is possible to provide a method of deriving a block vector of a chroma component block from a luma component block vector corresponding to a chroma component block in an independent block partitioning structure and a method and apparatus for minimizing residual signal encoding information of a

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chroma component block and a luma component block, when encoding is performed based on intra block copy.

When at least one of information on the motion compensation or region information is entropy-encoded/decoded, at least one of the following binarization methods may be used.

- Truncated Rice binarization method
- K-th order Exp_Golomb binarization method
- Restricted K-th order Exp_Golomb binarization method
- Fixed-length binarization method
- Unary binarization method
- Truncated Unary binarization method

When at least one of information on the motion compensation or region information is entropy-encoded/decoded, a context model may be determined using at least one of information on motion compensation of a neighbor block or region information of the neighbor block, at least one of information on motion compensation encoded/decoded before, region information encoded/decoded before, information on the depth of the current unit/block or information on the size of the current unit/block.

When at least one of information on the motion compensation or region information is entropy-encoded/decoded, entropy encoding/decoding may be performed using at least one of information on motion compensation of a neighbor block or region information of the neighbor block, at least one of information on motion compensation encoded/decoded before, region information encoded/decoded before, information on the depth of the current unit/block or information on the size of the current unit/block as a prediction value for the information on motion compensation of the current block or the region information.

The above embodiments may be performed in the same method in an encoder and a decoder.

At least one or a combination of the above embodiments may be used to encode/decode a video.

A sequence of applying to above embodiment may be different between an encoder and a decoder, or the sequence applying to above embodiment may be the same in the encoder and the decoder.

The above embodiment may be performed on each luma signal and chroma signal, or the above embodiment may be identically performed on luma and chroma signals.

A block form to which the above embodiments of the present invention are applied may have a square form or a non-square form.

At least one of the syntax elements (flags, indices, etc.) entropy-coded in the encoder and entropy-decoded in the decoder may use at least one of the following binarization, debinarization, entropy encoding/decoding methods.

- Method of binarization/debinarization of 0-th order Exp_Golomb having a sign (se(v))
- Method of binarization/debinarization of k-th order Exp_Golomb having a sign (sek(v))
- Method of binarization/debinarization of 0-th order Exp_Golomb of a positive integer without a sign (ue(v))
- Method of binarization/debinarization of k-th order Exp_Golomb of a positive integer without a sign (uek(v))
- Fixed-length binarization/debinarization method (f(n))
- Truncated Rice binarization/debinarization method or Truncated Unary binarization/debinarization method (tu(v))
- Truncated Binary binarization/debinarization method (tb(v))
- Context adaptive arithmetic encoding/decoding method (ae(v))

Byte-unit bit string (b(8))

Binarization/debinarization method of an integer having a sign (i(n))

Binarization/debinarization method of a positive integer without a sign (u(n))

Unary binarization/debinarization method

The above embodiment of the present invention may be applied depending on a size of at least one of a coding block, a prediction block, a transform block, a block, a current block, a coding unit, a prediction unit, a transform unit, a unit, and a current unit. Herein, the size may be defined as a minimum size or maximum size or both so that the above embodiments are applied, or may be defined as a fixed size to which the above embodiment is applied. In addition, in the above embodiments, a first embodiment may be applied to a first size, and a second embodiment may be applied to a second size. In other words, the above embodiments may be applied in combination depending on a size. In addition, the above embodiments may be applied when a size is equal to or greater than a minimum size and equal to or smaller than a maximum size. In other words, the above embodiments may be applied when a block size is included within a certain range.

For example, the above embodiments may be applied when a size of current block is 8x8 or greater. For example, the above embodiments may be applied when a size of current block is 4x4 only. For example, the above embodiments may be applied when a size of current block is 16x16 or smaller. For example, the above embodiments may be applied when a size of current block is equal to or greater than 16x16 and equal to or smaller than 64x64.

The above embodiments of the present invention may be applied depending on a temporal layer. In order to identify a temporal layer to which the above embodiments may be applied, a corresponding identifier may be signaled, and the above embodiments may be applied to a specified temporal layer identified by the corresponding identifier. Herein, the identifier may be defined as the lowest layer or the highest layer or both to which the above embodiment may be applied, or may be defined to indicate a specific layer to which the embodiment is applied. In addition, a fixed temporal layer to which the embodiment is applied may be defined.

For example, the above embodiments may be applied when a temporal layer of a current image is the lowest layer. For example, the above embodiments may be applied when a temporal layer identifier of a current image is 1. For example, the above embodiments may be applied when a temporal layer of a current image is the highest layer.

A slice type or a tile group type to which the above embodiments of the present invention are applied may be defined, and the above embodiments may be applied depending on the corresponding slice type or tile group type.

In the above-described embodiments, the methods are described based on the flowcharts with a series of steps or units, but the present invention is not limited to the order of the steps, and rather, some steps may be performed simultaneously or in different order with other steps. In addition, it should be appreciated by one of ordinary skill in the art that the steps in the flowcharts do not exclude each other and that other steps may be added to the flowcharts or some of the steps may be deleted from the flowcharts without influencing the scope of the present invention.

The embodiments include various aspects of examples. All possible combinations for various aspects may not be described, but those skilled in the art will be able to recognize different combinations. Accordingly, the present

invention may include all replacements, modifications, and changes within the scope of the claims.

The embodiments of the present invention may be implemented in a form of program instructions, which are executable by various computer components, and recorded in a computer-readable recording medium. The computer-readable recording medium may include stand-alone or a combination of program instructions, data files, data structures, etc. The program instructions recorded in the computer-readable recording medium may be specially designed and structured for the present invention, or well-known to a person of ordinary skilled in computer software technology field. Examples of the computer-readable recording medium include magnetic recording media such as hard disks, floppy disks, and magnetic tapes; optical data storage media such as CD-ROMs or DVD-ROMs; magneto-optimum media such as floptical disks; and hardware devices, such as read-only memory (ROM), random-access memory (RAM), flash memory, etc., which are particularly structured to store and implement the program instruction. Examples of the program instructions include not only a mechanical language code formatted by a compiler but also a high level language code that may be implemented by a computer using an interpreter. The hardware devices may be configured to be operated by one or more software modules or vice versa to conduct the processes according to the present invention.

Although the present invention has been described in terms of specific items such as detailed elements as well as the limited embodiments and the drawings, they are only provided to help more general understanding of the invention, and the present invention is not limited to the above embodiments. It will be appreciated by those skilled in the art to which the present invention pertains that various modifications and changes may be made from the above description.

Therefore, the spirit of the present invention shall not be limited to the above-described embodiments, and the entire scope of the appended claims and their equivalents will fall within the scope and spirit of the invention.

INDUSTRIAL APPLICABILITY

The present invention may be used to encode or decode an image.

The invention claimed is:

1. A method of decoding an image, the method comprising:

constructing a block vector candidate list of a current block;

deriving a block vector of the current block based on one of block vector candidates included in the block vector candidate list; and

obtaining prediction samples of the current block based on the block vector of the current block,

wherein constructing the block vector candidate list comprises,

in response to a number of block vector candidates included in the block vector candidate list being less than a maximum number, adding at least one of history-based block vector candidates included in a history-based block vector candidate list to the block vector candidate list; and

wherein whether a history-based block vector candidate is added to the block vector candidate list is determined by comparing the history-based block vector candidate with a block vector candidate already included in the block vector candidate list.

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2. The method of claim 1, wherein the history-based block vector candidates in the history-based block vector candidate list are derived from a block decoded prior to the current block.

3. The method of claim 2, wherein whether the history-based block vector candidate list is updated with the block vector of the current block is determined based on a size of the current block.

4. The method of claim 3, wherein when the size of the current block is 4×4, the history-based block vector candidate list is not updated with the block vector of the current block.

5. The method of claim 4, wherein when the size of the current block is greater than 4×4, the history-based block vector candidate list is updated with the block vector of the current block.

6. The method of claim 1, wherein the one of the block vector candidates is selected based on an L0 motion vector predictor flag, and

wherein when the maximum number of block vector candidates that can be included in the block vector candidate list is not greater than 1, decoding the L0 motion vector predictor flag is omitted, and a value of the L0 motion vector predictor flag is inferred to 0.

7. The method of claim 1, wherein when a size of the current block is 4×4, the history-based block vector candidate is added to the block vector candidate list without inserting a spatial block vector candidate to the block vector candidate list.

8. A method of encoding an image, the method comprising:

constructing a block vector candidate list of a current block;

deriving a block vector of the current block based on one of block vector candidates included in the block vector candidate list; and

obtaining prediction samples of the current block based on the block vector of the current block,

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wherein constructing the block vector candidate list comprises in response to a number of block vector candidates included in the block vector

candidate list being less than a maximum number, adding at least one of history-based block vector candidates included in a history-based block vector candidate list to the block vector candidate list; and

wherein whether a history-based block vector candidate is added to the block vector candidate list is determined by comparing the history-based block vector candidate with a block vector candidate already included in the block vector candidate list.

9. A non-transitory computer-readable recording medium storing a bitstream generated by the encoding method, wherein the encoding method comprising:

constructing a block vector candidate list of a current block;

deriving a block vector of the current block based on one of block vector candidates included in the block vector candidate list; and

obtaining prediction samples of the current block based on the block vector of the current block,

wherein constructing the block vector candidate list comprises

in response to a number of block vector candidates included in the block vector candidate list being less than a maximum number, adding at least one of history-based block vector candidates included in a history-based block vector candidate list to the block vector candidate list; and

wherein whether a history-based block vector candidate is added to the block vector candidate list is determined by comparing the history-based block vector candidate with a block vector candidate already included in the block vector candidate list.

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